State of California

State Water Resources Control Board

NOTICE OF INTENT

TO COMPLY WITH THE TERMS OF GENERAL PERMIT ORDER NO . 2000-___-DWQ
FOR THE DISCHARGE OF BIOSOLIDS TO LAND
FOR USE IN AGRICULTURAL, SILVICULTURAL, HORTICULTURAL AND LAND RECLAMATION ACTIVITIES

ATTACHMENT A

Mark Only One Item	1. ☒ New Discharge Under MODEL Permit 2. ☐ Change of Information-WDID # _____

I. Property Owner (Required)

Name				
Las Gallinas Valley Sanitary District				
Mailing Address				
300 Smith Ranch Road				
City	County	State	Zip	Phone
San Rafael	Marin	CA	94903	(415) 472-1734
Contact Person		(check one) Owner ___X___ Operator _____ Owner/operator _____		
Mike Prinz, General Manager				

II. Generator (Required . If more than one generator, attach the information and ensure that the signature block is copied, signed and attached.)

Name				
Same as Property Owner				
Mailing Address				
City	County	State	Zip	Phone
Contact Person				

III. Site Operator/Property Manager (if any)

Name				
Same as Property Owner				
Mailing Address				
City	County	State	Zip	Phone
Contact Person				

IV. Billing Address

Name				
Same as Property Owner				
Mailing Address				
City	County	State	Zip	Phone
Contact Person				

STATE USE ONLY

WDID: ☐☐☐☐☐☐☐☐☐☐☐	Regional Board Office: ☐☐	Date NOI Received: _____	Date NOI Processed: _____
	Fee Amount Received: $ _____	Check #: _____	

V. Site Operator

Name				
Same as Property Owner				
Mailing Address				
City	County	State	Zip	Phone
Contact Person				

VI. Hauler Information

Name				
Not Applicable				
Mailing Address				
City	County	State	Zip	Phone
Contact Person				
Type of Transportation				

VII. Site Location

Street (including address, if any) 300 Smith Ranch Road, San Rafael, CA 94903

Nearest Cross Street(s)

County: **Marin** | Total Size of Site (acres): 304

Township/Range/Section T **2 N**, R **6 W**, Section **2, 3, 10, 11, 15** B&M

Latitude/Longitude (From Center): _____ Deg. _____ Min. _____ Sec N. _____ Deg. _____ Min. _____ Sec. W

See Project Description /application area information **38.0255° N, 122.5195° W**

Attach a map of at least 1:24000 (1" = 2000") showing the proposed application site (e.g., USGS 7.5" topographic map). The map should also show run-on/runoff controls, storage areas, nearby surface waters, wells and residences, the application areas including setback and buffer zones .

VIII. Application Area Information **See attached Project Description and Application Area Information**

Subject	Value	Applicable Unit/ Type of Measure
Quantity of Biosolids to be Applied		dry tons per year
Total Biosolids Application Proposed		dry tons
Land Use Zone		
Adjacent Land Use Zones		
Application Area Size		acres
Proposed Nitrogen Loading		lb. Plant Available Nitrogen/acre
Proposed Crop, Use		crop type, human/animal/neither
Crop Nitrogen Usage		1b. Nitrogen/year
Nitrogen Usage Reference		
Depth of Root Zone for Crop Being Planted		inches
Will Setback Limits Be Met?		Yes or No
Distance to Nearest Inhabited Dwelling		feet/miles
Public Access Controls		Specify Type
Runoff Controls		Attach plans
Prevailing Wind Direction		
Minimum Depth to Ground Water		feet
How Minimum Depth to Ground Water is Determined		

Anticipated Average Daily Application Rate		dry tons/day
Source of Water for Crop		
Average Annual Precipitation		inches/year

Attach an anticipated annual time schedule for the field operations including anticipated biosolids applications windows, seeding operations, supplemental fertilization, and cultivation/harvest.

See attached Project Description and Application Area Information

IX. Soil Constituent Concentrations (Each Source)

Constituent	Concentration in Soil, mg/kg, dry weight
Arsenic	
Cadmium	
Copper	
Lead	
Mercury	
Molybdenum	
Nickel	
Selenium	
Zinc	
pH	
Estimated Permeability	cm/sec
Cation Exchange Capacity	meq/100g
Total Nitrogen	
Ammonia Nitrogen, as N	
Total Phosphorus, as P	
Total Potassium	

X Have any proposed biosolids application sites been fallow for more than one year? ☒YES ☐ NO

XI Are there existing agricultural, silvicultural, or horticultural operations at all the proposed application sites? ☒YES ☐ NO

XII Is it known whether any locations within the proposed land application site contain biologically unique or sensitive natural communities?
☒YES ☐ NO **No known biologically unique or sensitive natural communities present within proposed application areas**

If natural terrestrial habitats are present on the project site, a biological site assessment must be conducted to determine whether biologically unique or sensitive natural communities occur and whether they could be disturbed by the application of biosolids; this report must be forwarded to the appropriate regional office of DFG and the Endangered Species Unit of the USFWS in Sacramento for review and approval of the mitigation strategy, as necessary. If biologically unique or sensitive natural communities are present and more than 10% or 10 acres will be disturbed, whichever is less, the project will not be authorized under the GO unless the applicant submits a plan to mitigate for any significant impacts on biologically unique or sensitive natural communities and agrees to implement the mitigation.

XIII Biosolids Storage Plan (as required by Biosolids Storage and Transportations Spec. No, 8) **See Project Description**

A biosolids storage plan must be attached (if no *on-site* biosolids storage will be provided, a contingency plan for inclement weather operation must be provided). The biosolids' storage plan should include at a minimum:

If on-site storage will be provided:

a. Size of biosolids storage area
b. How frequently it will be used (emergency basis only or routine use)
c. Leachate controls
d. Erosion controls
e. Run-on/runoff controls

If no on-site storage will be provided:

a. Location of off-site storage facilities

b. Emergency storage plans

XIV Erosion Control Plan (if applicable) (as required by Discharge Specification No, 8) **Not Applicable**

Biosolids applied to ground surfaces having a 10 percent or greater slope requires an Erosion Control Plan. The Plan should outline conditions that justify application of biosolids to the 10 percent or greater slopes and specify the application and management practices to be used to assure containment of the biosolids on the application site.

XV. Spill Response and Traffic Plan (as required byBiosolids Storage and Transportation Spec. No. 14) **See Project Description**

a. The Spill Response Plan should include at a minimum:

1 Emergency contacts and notification procedures
2. Require personal protective equipment requirement
3. Response instructions for spill during biosolids transport
4. Response instructions for storage facility failure
5. Response instructions if hazardous or other unauthorized material is found

b. The Traffic Plan should include at a minimum:

1. The proposed route for all vehicles handling biosolids
2. Describe the anticipated maximum vehicle weight

XVI. Adverse Weather and Alternative Plan: (as required by Biosolids Storage and Transportation Spec. No. 8) **See Project Description**

Submit an Adverse Weather and Alternative Plan that details procedures to address times when biosolids cannot be applied to the site(s) due to adverse weather or other conditions (wind, precipitation, field preparation delays, access road limitations, etc.).

XVII. CERTIFICATION

"I certify under penalty of law that this document and all attachments were prepared under my direction and supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment." In addition, I certify that the provisions of the permit, including the criteria for eligibility, will be complied with.

Signature of Owner/Operator of Spreading Operations	Title
Printed or Typed Name	Date
Signature of Property Owner	Title **General Manager**
Printed or Typed Name **Mike Prinz**	Date
Signature of Site Operator/Manager (if any)	Title
Printed or Typed Name	Date

Las Gallinas Valley Sanitary District -- Biosolids Land Application on District Property

Project Description and Application Area Information

Las Gallinas Valley Sanitary District (District) is submitting a Notice of Intent to land apply District Class B biosolids to up to 304 acres of District property, 238 acres of which are currently under agricultural production (pasture hay crop) and irrigated with treated effluent, and 66 acres of which has been disked in the past for weed abatement and is not irrigated.

District biosolids would be applied in liquid form (~4% solids) using either mobile applicator vehicles and equipment or direct pumping from three existing sludge storage ponds located on District property to approved biosolids application areas; dewatered biosolids cake application may be done at some time in the future. Biosolids would be applied and incorporated into the soil during anytime between April and October, weather and field conditions permitting, followed by crop seeding or re-seeding, irrigation from May (as needed) to October using District treated effluent (WDR 92-064), and crop harvest between June – October annually.

District biosolids meets all Federal and state Biosolids General Order standards. Treated secondary effluent has and will be applied in the future to the 238-acre parcel per WDR 92-064, with nitrogen loading from effluent application accounted for in the proposed biosolids agronomic rate calculation. Supplemental irrigation may or may not be provided to the 66-acre parcel, depending on pasture hay mix crop response to biosolids application and response to natural rainfall.

District property where biosolids would be applied is located adjacent to San Pablo Bay to the east and primarily open space to the south and west. The closest residences are located approximately 0.3 miles to the west of APN 155-01-13. There is an existing network of groundwater monitoring wells surrounding a 9-acre sludge surface disposal unit on District property; semi-annual 2018 and 2019 monitoring data from the District indicate groundwater elevation ranges from 3 to 11 feet below ground surface. There are no drinking water wells in the area near District property, and the shallow groundwater in the area is unsuitable as a drinking water source because of high total dissolved solids related to soil type and proximity to San Pablo Bay.

The project would help transition the District from using the 9-acre permitted sludge surface disposal unit located on District property, to beneficial use, also on adjacent District property. On-site management of biosolids produced by the District is consistent with the goals of SB 1383 to divert organics to beneficial uses and reduce greenhouse gas emissions.

VIII. Application Area Information

Subject	Value	Applicable Unit – Type of Measure
Quantity of Biosolids to be Applied	200	dry tons per year
Total Biosolids Application Proposed (5-year plan)	1000	dry tons/5 years
Land Use Zone	A2- Agriculture, Limited	
Adjacent Land Use Zones	A2-Agriculture, Limited; PD (public)	
Application Area Size	304	acres
Anticipated Average Daily Application Rate	10-50	dry tons/day
Proposed Nitrogen Loading	200	lb. PAN/acre
Proposed Crop, Use	Animal	crop type, human / animal / neither
Crop Nitrogen Usage	200	lb. nitrogen/year
Nitrogen Usage Reference	UC Davis / Other	
Depth of Root Zone for Crop Being Planted	12-24	inches
Source of Water for Crop	Treated Effluent	
Will Setback Limits Be Met?	Yes	Yes or No
Distance to Nearest Inhabited Dwelling	~ 0. 3 miles	feet/miles
Public Access Controls	No Public Access	Specify type
Runoff Controls	Buffers	Attach plans
Prevailing Wind Direction	SW	
Minimum Depth to Ground Water	3	feet
How Minimum Depth to Ground Water is Determined	On-site wells	
Average Annual Precipitation	39	inches/year

Attach an anticipated annual time schedule for the field operations including anticipated biosolids applications windows, seeding operations, supplemental fertilization, and cultivation/harvest.

Schedule of Field Operations

Biosolids Application Window: April to October

Crop Seeding/Re-seeding: April to October

Supplemental Fertilization: Treated Secondary Effluent May to October Parcels 13 and 14

Harvest: June to October

IX. Biosolids and Soil Constituent Concentrations

Biosolids Constituent Concentrations – Las Gallinas Valley Sanitary District[1]

Constituent	Concentration (dry weight)	Units
Arsenic	<1	mg / kg
Cadmium	<0.5	mg / kg
Copper	230	mg / kg
Lead	<2.5	mg / kg
Mercury	<0.02	mg / kg
Molybdenum	<2.5	mg / kg
Nickel	<0.5	mg / kg
Selenium	<1	mg / kg
Zinc	1400	mg / kg
pH	7.9	
Total Nitrogen	81540	mg / kg
Ammonia Nitrogen, as N	40000	mg / kg
Total Phosphorus	9400	mg / kg
Total Potassium	10000	mg / kg

[1] Data included in Attachment A

Soil Constituent Concentrations (APN 155-01-33, 13, 14). Proposed Application Areas[2]

	APN 155-01-			
Constituent	**33**	**13**	**14**	**Units**
Arsenic	7	10	11	mg / kg
Cadmium	<0.4	<0.4	<0.4	mg / kg
Copper	25	30	30	mg / kg
Lead	31	13	13	mg / kg
Mercury	0.05	0.13	0.12	mg / kg
Molybdenum	<2.0	<2.0	<2.0	mg / kg
Nickel	79	48	48	mg / kg
Selenium	<5.0	<5.0	<5.0	mg / kg
Zinc	63	53	54	mg / kg
pH	7.1	6.3	6.5	mg / kg
Cation Exchange Capacity	32	26	28	meq / 100 g soil
Total Nitrogen	1527	1292	1450	mg / kg
Ammonia Nitrogen, as N	61	57	65	mg / kg
Total Phosphorus	287	502	435	mg / kg
Total Potassium	3500	2700	2767	mg / kg
Estimated Permeability[3]	0.06 to 2.0			in / hr

[2] Data included in Attachment B; values are the arithmetic average of 3 samples in APN 155-01-33, 5 samples in APN 155-01-13, and 6 samples in APN 155-01-14. 503/BGO Metals 6-inch sampling depth. All others 18-inch sampling depth.

[3]Soil Survey of Marin County, USDA-NRCS; Reyes Silty Clay Loam <2% slope.

XIII. Biosolids Storage Plan

Liquid biosolids generated by the District is stored in 3 existing sludge storage ponds within District property, known as Ponds A, B, and C (*See Application Area Figure, page 5*); total size is 3.5 million gallons. No in-field storage is proposed. Dewatering of District biosolids may be done in the future, with biosolids application performed immediately post-dewatering.

XIV. Erosion Control Plan

Not applicable, slopes <10%)

XV. Spill Response and Traffic Plan

a. **The Spill Response Plan.** Any spill would occur within District property and would be cleaned up immediately using District personnel and equipment.
b. **The Traffic Plan.** Not applicable to biosolids land application using District biosolids. Operations equipment would be trucked/transferred to the approved biosolids land application areas using designated access routes within District property boundaries. This may include biosolids pumping and applicator vehicle and support equipment, and farming equipment including harvesting and crop removal.

XVI. Adverse Weather and Alternative Plan

Biosolids land application is proposed to occur anytime between April and October, annually. No application of biosolids would occur when wind speeds exceed 25 mph.

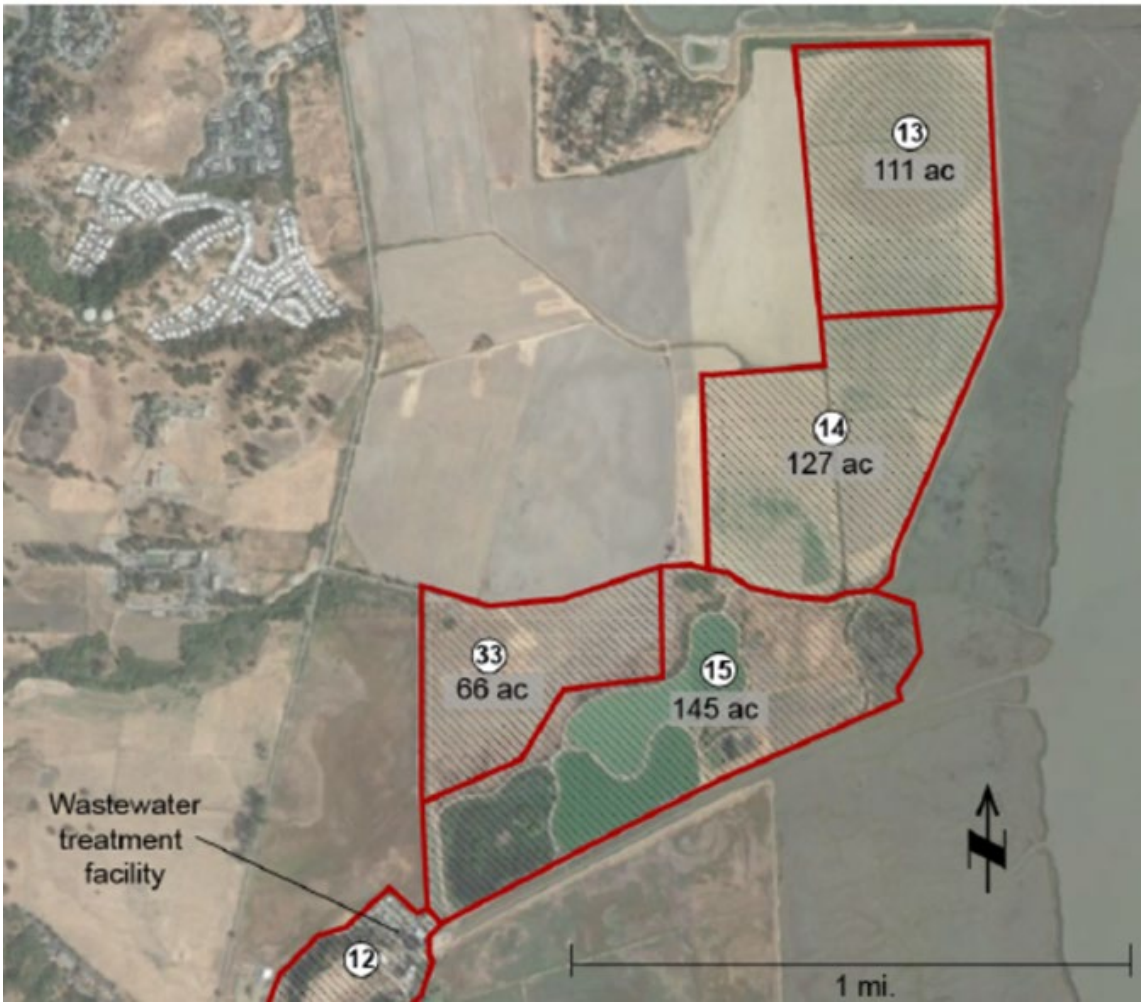
In the event of rainfall, no biosolids application will occur and would not commence until field conditions are safe and suitable for operations equipment including application and disking into soil surface.

In the event of an emergency such as wildfire or earthquake, biosolids application would cease. The District has a 3.5 million-gallon series of 3 sludge storage ponds, in the event of emergency or adverse weather conditions. In addition, the District uses a permitted 9-acre sludge surface disposal unit on District property.

Attach a map of at least 1:24000 (1” = 2000”) showing the proposed application site (e.g., USGS 7.5” topographic map).

The map should also show run-on/runoff controls, storage areas, nearby surface waters, wells and residences, the application areas including setback and buffer zones.

Application Area Map



Biosolids application is proposed on areas 13, 14, and 33.

Attachment A

Las Gallinas Valley Sanitary District Biosolids Constituent Analytical Report

Caltest
ANALYTICAL LABORATORY
ENVIRONMENTAL ANALYSES

Wednesday, December 02, 2020

Sahar Golshani
Las Gallinas Valley Sanitary
300 Smith Ranch Road
San Rafael, CA 94903

Re Lab Order: V110347 Collected By: SG
Project ID: BIOSOLIDS PO/Contract #:

Dear Sahar Golshani:

Enclosed are the analytical results for sample(s) received by the laboratory on Friday, November 06, 2020. Results reported herein conform to the most current NELAC standards, where applicable, unless otherwise narrated in the body of the report.

If you have any questions concerning this report, please feel free to contact me.

Enclosures

Project Manager: Greg Martindale

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(707) 258-4000 • Fax (707) 226-1001 • e-mail: info@caltestlabs.com

SAMPLE SUMMARY

Lab Order: V110347
Project ID: BIOSOLIDS

Lab ID	Sample ID	Matrix	Date Collected	Date Received
V110347001	**SLUDGE POND B**	Solid	11/06/2020 10:57	11/06/2020 15:30

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NARRATIVE

Lab Order: V110347
Project ID: BIOSOLIDS

General Qualifiers and Notes

Caltest authorizes this report to be reproduced only in its entirety. Results are specific to the sample(s) as submitted and only to the parameter(s) reported.

Caltest certifies that all test results for wastewater and hazardous waste analyses meet all applicable NELAC requirements; all microbiology and drinking water testing meet applicable ELAP requirements, unless stated otherwise.

All analyses performed by EPA Methods or Standard Methods.

Dilution Factors (DF) reported greater than '1' have been used to adjust the result, Reporting Limit (RL), and Method Detection Limit (MDL).

All Solid, sludge, and/or biosolids data is reported in Wet Weight, unless otherwise specified.

Filtrations performed at Caltest for dissolved metals (excluding mercury) and/or pH analysis are not performed within the 15 minute holding time as specified by 40CFR 136.3 table II.

Results Qualifiers: Report fields may contain codes and non-numeric data correlating to one or more of the following definitions:

ND - indicates analytical result has not been detected at or above the Reporting Limit (RL), or at above the Method Detection Limit (MDL) when it is included on the report and is not otherwise noted.

RL - Reporting Limit is the quantitation limit at which the laboratory is able to detect an analyte. An analyte not detected at or above the RL is reported as ND unless otherwise noted or qualified. For analyses pertaining to the State Implementation Plan of the California Toxics Rule, the Caltest Reporting Limit (RL) is equivalent to the Minimum Level (ML). A standard is always run at or below the ML. Where Reporting Limits are elevated due to dilution, the ML calibration criteria has been met.

MDL - The Method Detection Limit is defined as the minimum measured concentration of a substance that can be reported with 99% confidence that the measured concentration is distinguishable from method blank results.

J - reflects estimated analytical result value detected below the Reporting Limit (RL) and above the Method Detection Limit (MDL). The 'J' flag is equivalent to the DNQ Estimated Concentration flag.

B - indicates the analyte has been detected in the blank associated with the sample.

SS - compound is a Surrogate Spike used per laboratory quality assurance manual.

NOTE: This document represents a complete Analytical Report for the samples referenced herein and should be retained as a permanent record thereof.

Qualifiers and Compound Notes

1	Analysis Performed by BC Laboratories, Inc.; CA ELAP #1186. Some BC Labs sample and quality control results are input into Caltest's database to include in Caltest's report and EDD, as a convenience to our client. If quality control values in the two reports do not match, the BC reported value should be considered the correct value.
2	Detection and quantitation limits were raised due to matrix interference.
3	Surrogate is low due to matrix interference. Interference verified through second extraction/analysis.
4	The sample was low in solids, so it was analyzed as received.

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WET WEIGHT

ANALYTICAL RESULTS

Lab Order: V110347
Project ID: BIOSOLIDS

Solid results are reported on a wet weight basis.

Lab ID	**V110347001**	Date Collected	11/6/2020 10:57	Matrix	Solid
Sample ID	**SLUDGE POND B**	Date Received	11/6/2020 15:30	Results are expressed as wet weight values	

Parameters	Result	Units	R. L.	DF	Prepared	Batch	Analyzed	Batch	Qual
Calculation, Total Nitrogen	**Analytical Method:**		Calculated				**Analyzed by:**	CW	
Total Nitrogen	570	mg/kg	0.10	1			11/25/20 16:13	CALC	
Mercury Analysis	**Prep Method:**		EPA 7471A				**Prep by:**	UKS	
	Analytical Method:		EPA 7471A				**Analyzed by:**	UKS	
Mercury	ND	mg/kg	0.020	1	11/12/20 00:00	MPR 17653	11/13/20 18:47	MHG 6566	
TTLC Metals Analysis	**Prep Method:**		SW846 3050B				**Prep by:**	PJB	
	Analytical Method:		EPA 6010B				**Analyzed by:**	CW	
Arsenic	ND	mg/kg	0.99	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	1
Cadmium	ND	mg/kg	0.50	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	
Copper	1.6	mg/kg	0.99	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	
Lead	ND	mg/kg	2.5	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	
Molybdenum	ND	mg/kg	2.5	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	
Nickel	ND	mg/kg	0.50	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	
Potassium	73	mg/kg	50	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	
Selenium	ND	mg/kg	0.99	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	
Zinc	10	mg/kg	2.5	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	
Semivolatile Organic Analysis	**Prep Method:**		SW846 3540				**Prep by:**	PJB	
	Analytical Method:		SW846 8270C				**Analyzed by:**	CW	
Acenaphthene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	1
Acenaphthylene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Aniline	ND	mg/kg	0.20	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Anthracene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzidine	ND	mg/kg	3.0	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzo(a)anthracene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzo(a)pyrene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzo(b)fluoranthene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzo(g,h,i)perylene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzo(k)fluoranthene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzoic acid	ND	mg/kg	0.50	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzyl alcohol	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzyl butyl phthalate	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
4-Bromophenyl phenyl ether	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Carbazole	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
4-Chloroaniline	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
bis(2-Chloroethoxy) methane	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
bis(2-Chloroethyl) ether	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
bis(2-Chloroisopropyl) ether	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
4-Chloro-3-methylphenol	ND	mg/kg	0.20	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2-Chloronaphthalene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2-Chlorophenol	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
4-Chlorophenyl phenyl ether	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Chrysene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Dibenzo(a,h)anthracene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Dibenzofuran	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	

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ANALYTICAL RESULTS

WET WEIGHT

Lab Order: V110347
Project ID: BIOSOLIDS

Solid results are reported on a wet weight basis.

Lab ID	**V110347001**	Date Collected	11/6/2020 10:57	Matrix	Solid
Sample ID	**SLUDGE POND B**	Date Received	11/6/2020 15:30	Results are expressed as wet weight values	

Parameters	Result	Units	R. L.	DF	Prepared	Batch	Analyzed	Batch	Qual
1,2-Dichlorobenzene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
1,3-Dichlorobenzene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
1,4-Dichlorobenzene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
3,3'-Dichlorobenzidine	ND	mg/kg	0.20	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2,4-Dichlorophenol	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Diethylphthalate	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2,4-Dimethylphenol	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Dimethylphthalate	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Di-n-butylphthalate	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2,4-Dinitrophenol	ND	mg/kg	0.50	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2,4-Dinitrotoluene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2,6-Dinitrotoluene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Di-n-octylphthalate	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
1,2Diphenylhydrazine/Azobenzen	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
bis(2-Ethylhexyl) phthalate	ND	mg/kg	0.20	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Fluoranthene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Fluorene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Hexachlorobenzene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Hexachlorobutadiene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Hexachlorocyclo pentadiene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Hexachloroethane	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Indeno(1,2,3-cd)pyrene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Isophorone	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2-Methyl-4,6-dinitrophenol	ND	mg/kg	0.50	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2-Methylnaphthalene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2-Methylphenol (o-Cresol)	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
3 & 4-Methylphenol(m&p Cresol)	ND	mg/kg	0.20	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Naphthalene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2-Nitroaniline	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
3-Nitroaniline	ND	mg/kg	0.20	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
4-Nitroaniline	ND	mg/kg	0.20	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Nitrobenzene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2-Nitrophenol	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
4-Nitrophenol	ND	mg/kg	0.20	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
N-Nitrosodimethylamine	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
N-Nitroso-di-n-propylamine	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
N-Nitrosodiphenylamine	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Pentachlorophenol	ND	mg/kg	0.20	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Phenanthrene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Phenol	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Pyrene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Pyridine	ND	mg/kg	0.50	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
1,2,4-Trichlorobenzene	ND	mg/kg	0.10	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2,4,5-Trichlorophenol	ND	mg/kg	0.20	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2,4,6-Trichlorophenol	ND	mg/kg	0.20	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2-Fluorobiphenyl (SS)	46	%	30-140	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	

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ANALYTICAL RESULTS

WET WEIGHT

Lab Order: V110347
Project ID: BIOSOLIDS

Solid results are reported on a wet weight basis.

Lab ID	**V110347001**	Date Collected	11/6/2020 10:57	Matrix	Solid
Sample ID	**SLUDGE POND B**	Date Received	11/6/2020 15:30	Results are expressed as wet weight values	

Parameters	Result	Units	R. L.	DF	Prepared	Batch	Analyzed	Batch	Qual
2-Fluorophenol (SS)	65	%	20-130	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Nitrobenzene-d5 (SS)	77	%	30-130	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Phenol-d5 (SS)	74	%	30-130	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2,4,6-Tribromophenol (SS)	109	%	20-150	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Semivolatile Organic Analysis	**Prep Method:**		SW846 3541				**Prep by:**	PJB	
	Analytical Method:		SW846 8081				**Analyzed by:**	CW	
Aldrin	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	1
alpha-BHC	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
beta-BHC	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
delta-BHC	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
gamma-BHC (Lindane)	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Chlordane	ND	mg/kg	0.28	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
4,4'-DDD	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
4,4'-DDE	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
4,4'-DDT	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Dieldrin	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Endosulfan I	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Endosulfan II	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Endosulfan sulfate	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Endrin	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Endrin aldehyde	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Endrin ketone	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Heptachlor	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Heptachlor epoxide	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Methoxychlor	ND	mg/kg	0.0028	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Toxaphene	ND	mg/kg	0.28	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Decachlorobiphenyl (SS)	35	%	40-130	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	3
Tetrachloro-m-xylene (SS)	54	%	20-130	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Chlorinated Pesticides Analysis	**Prep Method:**		SW846 3541				**Prep by:**	PJB	
	Analytical Method:		SW846 8082				**Analyzed by:**	CW	
PCB 1016	ND	mg/kg	0.57	56.604	11/12/20 10:40	REFD 4354	11/14/20 03:45	REFD 4359	1,2
PCB 1221	ND	mg/kg	0.57	56.604	11/12/20 10:40	REFD 4354	11/14/20 03:45	REFD 4359	2
PCB 1232	ND	mg/kg	0.57	56.604	11/12/20 10:40	REFD 4354	11/14/20 03:45	REFD 4359	2
PCB 1242	ND	mg/kg	0.57	56.604	11/12/20 10:40	REFD 4354	11/14/20 03:45	REFD 4359	2
PCB 1248	ND	mg/kg	0.57	56.604	11/12/20 10:40	REFD 4354	11/14/20 03:45	REFD 4359	2
PCB 1254	ND	mg/kg	0.57	56.604	11/12/20 10:40	REFD 4354	11/14/20 03:45	REFD 4359	2
PCB 1260	ND	mg/kg	0.57	56.604	11/12/20 10:40	REFD 4354	11/14/20 03:45	REFD 4359	2
Decachlorobiphenyl (SS)	66	%	40-120	56.604	11/12/20 10:40	REFD 4354	11/14/20 03:45	REFD 4359	2
Wet Chem Analysis	**Analytical Method:**		EPA 350.1				**Analyzed by:**	CW	
Ammonia (as N)	280	mg/kg	19	0.943			11/19/20 14:00	REFD 4361	1
Wet Chem Analysis	**Analytical Method:**		EPA 351.2				**Analyzed by:**	CW	
Total Kjeldahl Nitrogen	570	mg/kg	40	1			11/25/20 16:13	REFD 4366	1

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ANALYTICAL RESULTS

WET WEIGHT

Lab Order: V110347
Project ID: BIOSOLIDS

Solid results are reported on a wet weight basis.

Lab ID	**V110347001**	Date Collected	11/6/2020 10:57	Matrix	Solid
Sample ID	**SLUDGE POND B**	Date Received	11/6/2020 15:30	Results are expressed as wet weight values	

Parameters	Result	Units	R. L.	DF	Prepared	Batch	Analyzed	Batch	Qual
Wet Chem Analysis	**Analytical Method:**		EPA 365.4				**Analyzed by:**	CW	
Total Phosphorus as P	66	mg/kg	10	1			11/24/20 09:50	REFD 4365	1
Wet Chem Analysis	**Analytical Method:**		EPA 300.0				**Analyzed by:**	CW	
Nitrogen, Nitrate (as N)	ND	mg/kg	1.0	1			11/18/20 12:43	REFD 4363	1
Wet Chem Analysis	**Analytical Method:**		EPA 353.2				**Analyzed by:**	CW	
Nitrogen, Nitrite (as N)	3.8	mg/kg	0.50	1			11/18/20 12:18	REFD 4362	1
pH, Soil and Waste Analysis	**Analytical Method:**		SW846 9045				**Analyzed by:**	DR	
pH	7.9	pH Units		1			11/10/20 14:27	WET 10682	
Electrical Conductance Analysis	**Analytical Method:**		SM 2510 B-97/-11				**Analyzed by:**	JH	
Conductivity	4500	umhos/cm	10	1			11/13/20 16:44	WET 10693	4
Percent Solids Analysis	**Analytical Method:**		SM 2540 G-11				**Analyzed by:**	JH	
Solids, Percent	0.7	%	0.1	1			11/11/20 17:42	WGR 7767	

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ANALYTICAL RESULTS

DRY WEIGHT

Lab Order: V110347
Project ID: BIOSOLIDS

Solid results are reported on a dry weight basis.

Lab ID	**V110347001**	Date Collected	11/6/2020 10:57	Matrix	Solid
Sample ID	**SLUDGE POND B**	Date Received	11/6/2020 15:30	Results are expressed as dry weight values	

Parameters	Result	Units	R. L.	DF	Prepared	Batch	Analyzed	Batch	Qual
Calculation, Total Nitrogen	**Analytical Method:**		Calculated				**Analyzed by:**	CW	
Total Nitrogen	81000	mg/kg	14	1			11/25/20 16:13	CALC	
Mercury Analysis	**Prep Method:**		EPA 7471A				**Prep by:**	UKS	
	Analytical Method:		EPA 7471A				**Analyzed by:**	UKS	
Mercury	ND	mg/kg	2.9	1	11/12/20 00:00	MPR 17653	11/13/20 18:47	MHG 6566	
TTLC Metals Analysis	**Prep Method:**		SW846 3050B				**Prep by:**	PJB	
	Analytical Method:		EPA 6010B				**Analyzed by:**	CW	
Arsenic	ND	mg/kg	140	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	1
Cadmium	ND	mg/kg	71	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	
Copper	230	mg/kg	140	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	
Lead	ND	mg/kg	350	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	
Molybdenum	ND	mg/kg	350	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	
Nickel	ND	mg/kg	71	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	
Potassium	10000	mg/kg	7100	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	
Selenium	ND	mg/kg	140	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	
Zinc	1400	mg/kg	350	0.99	11/12/20 07:30	REFD 4352	11/13/20 20:43	REFD 4357	
Semivolatile Organic Analysis	**Prep Method:**		SW846 3540				**Prep by:**	PJB	
	Analytical Method:		SW846 8270C				**Analyzed by:**	CW	
Acenaphthene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	1
Acenaphthylene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Aniline	ND	mg/kg	29	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Anthracene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzidine	ND	mg/kg	430	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzo(a)anthracene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzo(a)pyrene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzo(b)fluoranthene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzo(g,h,i)perylene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzo(k)fluoranthene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzoic acid	ND	mg/kg	71	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzyl alcohol	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Benzyl butyl phthalate	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
4-Bromophenyl phenyl ether	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Carbazole	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
4-Chloroaniline	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
bis(2-Chloroethoxy) methane	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
bis(2-Chloroethyl) ether	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
bis(2-Chloroisopropyl) ether	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
4-Chloro-3-methylphenol	ND	mg/kg	29	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2-Chloronaphthalene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2-Chlorophenol	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
4-Chlorophenyl phenyl ether	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Chrysene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Dibenzo(a,h)anthracene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Dibenzofuran	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	

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ANALYTICAL RESULTS

DRY WEIGHT

Lab Order: V110347
Project ID: BIOSOLIDS

Solid results are reported on a dry weight basis.

Lab ID	**V110347001**	Date Collected	11/6/2020 10:57	Matrix	Solid
Sample ID	**SLUDGE POND B**	Date Received	11/6/2020 15:30	Results are expressed as dry weight values	

Parameters	Result	Units	R. L.	DF	Prepared	Batch	Analyzed	Batch	Qual
1,2-Dichlorobenzene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
1,3-Dichlorobenzene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
1,4-Dichlorobenzene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
3,3'-Dichlorobenzidine	ND	mg/kg	29	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2,4-Dichlorophenol	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Diethylphthalate	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2,4-Dimethylphenol	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Dimethylphthalate	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Di-n-butylphthalate	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2,4-Dinitrophenol	ND	mg/kg	71	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2,4-Dinitrotoluene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2,6-Dinitrotoluene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Di-n-octylphthalate	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
1,2Diphenylhydrazine/Azobenzen	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
bis(2-Ethylhexyl) phthalate	ND	mg/kg	29	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Fluoranthene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Fluorene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Hexachlorobenzene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Hexachlorobutadiene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Hexachlorocyclo pentadiene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Hexachloroethane	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Indeno(1,2,3-cd)pyrene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Isophorone	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2-Methyl-4,6-dinitrophenol	ND	mg/kg	71	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2-Methylnaphthalene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2-Methylphenol (o-Cresol)	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
3 & 4-Methylphenol(m&p Cresol)	ND	mg/kg	29	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Naphthalene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2-Nitroaniline	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
3-Nitroaniline	ND	mg/kg	29	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
4-Nitroaniline	ND	mg/kg	29	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Nitrobenzene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2-Nitrophenol	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
4-Nitrophenol	ND	mg/kg	29	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
N-Nitrosodimethylamine	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
N-Nitroso-di-n-propylamine	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
N-Nitrosodiphenylamine	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Pentachlorophenol	ND	mg/kg	29	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Phenanthrene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Phenol	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Pyrene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Pyridine	ND	mg/kg	71	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
1,2,4-Trichlorobenzene	ND	mg/kg	14	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2,4,5-Trichlorophenol	ND	mg/kg	29	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2,4,6-Trichlorophenol	ND	mg/kg	29	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2-Fluorobiphenyl (SS)	46	%	30-140	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	

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ANALYTICAL RESULTS

DRY WEIGHT

Lab Order: V110347
Project ID: BIOSOLIDS

Solid results are reported on a dry weight basis.

Lab ID	**V110347001**	Date Collected	11/6/2020 10:57	Matrix	Solid
Sample ID	**SLUDGE POND B**	Date Received	11/6/2020 15:30	Results are expressed as dry weight values	

Parameters	Result	Units	R. L.	DF	Prepared	Batch	Analyzed	Batch	Qual
2-Fluorophenol (SS)	65	%	20-130	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Nitrobenzene-d5 (SS)	77	%	30-130	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Phenol-d5 (SS)	74	%	30-130	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
2,4,6-Tribromophenol (SS)	109	%	20-150	4.831	11/12/20 11:30	REFD 4356	11/14/20 00:53	REFD 4360	
Semivolatile Organic Analysis	**Prep Method:**		SW846 3541				**Prep by:**	PJB	
	Analytical Method:		SW846 8081				**Analyzed by:**	CW	
Aldrin	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	1
alpha-BHC	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
beta-BHC	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
delta-BHC	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
gamma-BHC (Lindane)	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Chlordane	ND	mg/kg	40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
4,4'-DDD	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
4,4'-DDE	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
4,4'-DDT	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Dieldrin	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Endosulfan I	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Endosulfan II	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Endosulfan sulfate	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Endrin	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Endrin aldehyde	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Endrin ketone	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Heptachlor	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Heptachlor epoxide	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Methoxychlor	ND	mg/kg	0.40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Toxaphene	ND	mg/kg	40	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Decachlorobiphenyl (SS)	35	%	40-130	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	3
Tetrachloro-m-xylene (SS)	54	%	20-130	5.556	11/11/20 18:30	REFD 4353	11/13/20 04:17	REFD 4358	
Chlorinated Pesticides Analysis	**Prep Method:**		SW846 3541				**Prep by:**	PJB	
	Analytical Method:		SW846 8082				**Analyzed by:**	CW	
PCB 1016	ND	mg/kg	81	56.604	11/12/20 10:40	REFD 4354	11/14/20 03:45	REFD 4359	1,2
PCB 1221	ND	mg/kg	81	56.604	11/12/20 10:40	REFD 4354	11/14/20 03:45	REFD 4359	2
PCB 1232	ND	mg/kg	81	56.604	11/12/20 10:40	REFD 4354	11/14/20 03:45	REFD 4359	2
PCB 1242	ND	mg/kg	81	56.604	11/12/20 10:40	REFD 4354	11/14/20 03:45	REFD 4359	2
PCB 1248	ND	mg/kg	81	56.604	11/12/20 10:40	REFD 4354	11/14/20 03:45	REFD 4359	2
PCB 1254	ND	mg/kg	81	56.604	11/12/20 10:40	REFD 4354	11/14/20 03:45	REFD 4359	2
PCB 1260	ND	mg/kg	81	56.604	11/12/20 10:40	REFD 4354	11/14/20 03:45	REFD 4359	2
Decachlorobiphenyl (SS)	66	%	40-120	56.604	11/12/20 10:40	REFD 4354	11/14/20 03:45	REFD 4359	2
Wet Chem Analysis	**Analytical Method:**		EPA 350.1				**Analyzed by:**	CW	
Ammonia (as N)	40000	mg/kg	2700	0.943			11/19/20 14:00	REFD 4361	1
Wet Chem Analysis	**Analytical Method:**		EPA 351.2				**Analyzed by:**	CW	
Total Kjeldahl Nitrogen	81000	mg/kg	5700	1			11/25/20 16:13	REFD 4366	1

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DRY WEIGHT

ANALYTICAL RESULTS

Lab Order: V110347
Project ID: BIOSOLIDS

Solid results are reported on a dry weight basis.

Lab ID	**V110347001**	Date Collected	11/6/2020 10:57	Matrix	Solid
Sample ID	**SLUDGE POND B**	Date Received	11/6/2020 15:30	Results are expressed as dry weight values	

Parameters	Result	Units	R. L.	DF	Prepared	Batch	Analyzed	Batch	Qual
Wet Chem Analysis	**Analytical Method:**		EPA 365.4				**Analyzed by:**	CW	
Total Phosphorus as P	9400	mg/kg	1400	1			11/24/20 09:50	REFD 4365	1
Wet Chem Analysis	**Analytical Method:**		EPA 300.0				**Analyzed by:**	CW	
Nitrogen, Nitrate (as N)	ND	mg/kg	140	1			11/18/20 12:43	REFD 4363	1
Wet Chem Analysis	**Analytical Method:**		EPA 353.2				**Analyzed by:**	CW	
Nitrogen, Nitrite (as N)	540	mg/kg	71	1			11/18/20 12:18	REFD 4362	1
pH, Soil and Waste Analysis	**Analytical Method:**		SW846 9045				**Analyzed by:**	DR	
pH	7.9	pH Units		1			11/10/20 14:27	WET 10682	
Electrical Conductance Analysis	**Analytical Method:**		SM 2510 B-97/-11				**Analyzed by:**	JH	
Conductivity	4500	umhos/cm	10	1			11/13/20 16:44	WET 10693	4
Percent Solids Analysis	**Analytical Method:**		SM 2540 G-11				**Analyzed by:**	JH	
Solids, Percent	0.7	%	0.1	1			11/11/20 17:42	WGR 7767	

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QUALITY CONTROL DATA

Lab Order: V110347
Project ID: BIOSOLIDS

Analysis Description:	Mercury Analysis	**QC Batch:**	MPR/17653
Analysis Method:	EPA 7471A	**QC Batch Method:**	EPA 7471A

METHOD BLANK: 975336

Parameter	Blank Result	Reporting Limit	Units	Qualifiers
Mercury	ND	0.020	mg/kg	

LABORATORY CONTROL SAMPLE: 975337

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% REC Limits	Qualifier
Mercury	mg/kg	0.1	0.108	105	75-125	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 975339 975340

Parameter	Units	V110272001 Result	Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limit	RPD	Max RPD	Qualifiers
Mercury	mg/kg	0.19	0.1	0.265	0.262	72	68	75-125	1.1	35	7

Analysis Description:	TTLC Metals Analysis	**QC Batch:**	REFD/4352
Analysis Method:	EPA 6010B	**QC Batch Method:**	SW846 3050B

METHOD BLANK: 977537

Parameter	Blank Result	Reporting Limit	Units	Qualifiers
Arsenic	ND	1.0	mg/kg	
Cadmium	ND	0.50	mg/kg	
Copper	ND	1.0	mg/kg	
Lead	ND	2.5	mg/kg	
Molybdenum	ND	2.5	mg/kg	
Nickel	ND	0.50	mg/kg	
Selenium	ND	1.0	mg/kg	
Zinc	ND	2.5	mg/kg	

METHOD BLANK: 977537

Parameter	Blank Result	Reporting Limit	Units	Qualifiers
Potassium	ND	50	mg/kg	1

LABORATORY CONTROL SAMPLE: 977538

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% REC Limits	Qualifier
Arsenic	mg/kg	20	18.651	93	75-125	

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QUALITY CONTROL DATA

Lab Order: V110347
Project ID: BIOSOLIDS

Analysis Description:	TTLC Metals Analysis	**QC Batch:**	REFD/4352
Analysis Method:	EPA 6010B	**QC Batch Method:**	SW846 3050B

LABORATORY CONTROL SAMPLE: 977538

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% REC Limits	Qualifier
Cadmium	mg/kg	10	9.9687	100	75-125	
Copper	mg/kg	100	103.41	103	75-125	
Lead	mg/kg	100	102.67	103	75-125	
Molybdenum	mg/kg	100	104.6	105	75-125	
Nickel	mg/kg	100	106.64	107	75-125	
Selenium	mg/kg	20	19.978	100	75-125	
Zinc	mg/kg	100	103.08	103	75-125	

LABORATORY CONTROL SAMPLE: 977538

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% REC Limits	Qualifier
Potassium	mg/kg	500	486.81	97	75-125	1

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 977539 977540

Parameter	Units	2033159 Result	Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limit	RPD	Max RPD	Qualifiers
Arsenic	mg/kg	0	20	17.601	17.509	88	88	75-125	0.5	20	
Cadmium	mg/kg	0.07	10	10.493	10.047	104	100	75-125	4.3	20	
Copper	mg/kg	0.99	100	112.19	108.03	111	107	75-125	3.8	20	
Lead	mg/kg	0.58	100	98.335	94.971	98	94	75-125	3.5	20	
Molybdenum	mg/kg	0.21	100	105.45	100.7	105	100	75-125	4.6	20	
Nickel	mg/kg	0.32	100	109.18	103.06	109	103	75-125	5.8	20	
Selenium	mg/kg	0	20	23.168	22.948	116	115	75-125	1	20	
Zinc	mg/kg	61	100	144.14	125.82	83	65	75-125	14	20	8

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 977539 977540

Parameter	Units	2033159 Result	Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limit	RPD	Max RPD	Qualifiers
Potassium	mg/kg	0	500	548.27	512.44	110	102	75-125	6.8	20	1

SAMPLE DUPLICATE: 977541

Parameter	Units	2033159 Result	DUP Result	RPD	Max RPD	Reporting Limit	MDL	Qualifiers
Potassium	mg/kg	ND	0	0	20	50	5.0	1

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QUALITY CONTROL DATA

Lab Order: V110347
Project ID: BIOSOLIDS

Analysis Description:	TTLC Metals Analysis	**QC Batch:**	REFD/4352
Analysis Method:	EPA 6010B	**QC Batch Method:**	SW846 3050B

SAMPLE DUPLICATE: 977541

Parameter	Units	2033159 Result	DUP Result	RPD	Max RPD	Reporting Limit	MDL	Qualifiers
Arsenic	mg/kg	ND	0	0	20	1.0	0.40	
Cadmium	mg/kg	0.067085	0.052039	25	20	0.50	0.05	10
Copper	mg/kg	0.99458	1.051	5.5	20	1.0	0.05	
Lead	mg/kg	0.58127	0	200	20	2.5	0.41	
Molybdenum	mg/kg	0.21263	0.13234	47	20	2.5	0.05	10
Nickel	mg/kg	0.3182	0.3736	16	20	0.50	0.15	
Selenium	mg/kg	ND	0	0	20	1.0	0.98	
Zinc	mg/kg	60.827	15.998	117	20	2.5	0.09	9

Analysis Description:	Semivolatile Organic Analysis	**QC Batch:**	REFD/4353
Analysis Method:	SW846 8081	**QC Batch Method:**	SW846 3541

METHOD BLANK: 977543

Parameter	Blank Result	Reporting Limit	Units	Qualifiers
Aldrin	ND	0.0005	mg/kg	1
alpha-BHC	ND	0.0005	mg/kg	
beta-BHC	ND	0.0005	mg/kg	
delta-BHC	ND	0.0005	mg/kg	
gamma-BHC (Lindane)	ND	0.0005	mg/kg	
Chlordane	ND	0.050	mg/kg	
4,4'-DDD	ND	0.0005	mg/kg	
4,4'-DDE	ND	0.0005	mg/kg	
4,4'-DDT	ND	0.0005	mg/kg	
Dieldrin	ND	0.0005	mg/kg	
Endosulfan I	ND	0.0005	mg/kg	
Endosulfan II	ND	0.0005	mg/kg	
Endosulfan sulfate	ND	0.0005	mg/kg	
Endrin	ND	0.0005	mg/kg	
Endrin aldehyde	ND	0.0005	mg/kg	
Endrin ketone	ND	0.0005	mg/kg	
Heptachlor	ND	0.0005	mg/kg	
Heptachlor epoxide	ND	0.0005	mg/kg	
Methoxychlor	ND	0.0005	mg/kg	
Toxaphene	ND	0.050	mg/kg	
Decachlorobiphenyl (SS)	68	40-130	%	
Tetrachloro-m-xylene (SS)	81	20-130	%	

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QUALITY CONTROL DATA

Lab Order: V110347
Project ID: BIOSOLIDS

Analysis Description:	Semivolatile Organic Analysis	**QC Batch:**	REFD/4353
Analysis Method:	SW846 8081	**QC Batch Method:**	SW846 3541

LABORATORY CONTROL SAMPLE: 977544

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% REC Limits	Qualifier
Aldrin	mg/kg	0.0049	0.0043319	88	70-130	1
gamma-BHC (Lindane)	mg/kg	0.0049	0.0051023	103	60-140	
4,4'-DDT	mg/kg	0.0049	0.0045461	92	60-140	
Dieldrin	mg/kg	0.0049	0.0047885	97	70-130	
Endrin	mg/kg	0.0049	0.0039289	80	60-140	
Heptachlor	mg/kg	0.0049	0.0046309	94	60-140	
Decachlorobiphenyl (SS)	%			72	40-130	
Tetrachloro-m-xylene (SS)	%			90	20-130	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 977545 977546

Parameter	Units	2033159 Result	Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limit	RPD	Max RPD	Qualifiers
Aldrin	mg/kg	0	0.0051	0.003786	0.003961	75	78	50-140	4.5	30	1
gamma-BHC (Lindane)	mg/kg	0	0.0051	0.005080	0.005318	101	105	50-140	4.6	30	
4,4'-DDT	mg/kg	0	0.0051	0.003880	0.004484	77	88	50-140	14	30	
Dieldrin	mg/kg	0	0.0051	0.004742	0.005045	94	100	40-140	6.2	30	
Endrin	mg/kg	0	0.0051	0.003316	0.004005	66	79	50-150	19	30	
Heptachlor	mg/kg	0	0.0051	0.003925	0.004002	78	79	60-140	1.9	30	
Decachlorobiphenyl (SS)	%					53	61	40-130	15		
Tetrachloro-m-xylene (SS)	%					81	83	20-130	2.9		

Analysis Description:	Chlorinated Pesticides Analysis	**QC Batch:**	REFD/4354
Analysis Method:	SW846 8082	**QC Batch Method:**	SW846 3541

METHOD BLANK: 977547

Parameter	Blank Result	Reporting Limit	Units	Qualifiers
PCB 1016	ND	0.010	mg/kg	1
PCB 1221	ND	0.010	mg/kg	
PCB 1232	ND	0.010	mg/kg	
PCB 1242	ND	0.010	mg/kg	
PCB 1248	ND	0.010	mg/kg	
PCB 1254	ND	0.010	mg/kg	
PCB 1260	ND	0.010	mg/kg	
Decachlorobiphenyl (SS)	87	40-120	%	

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QUALITY CONTROL DATA

Lab Order: V110347
Project ID: BIOSOLIDS

Analysis Description:	Chlorinated Pesticides Analysis	**QC Batch:**	REFD/4354
Analysis Method:	SW846 8082	**QC Batch Method:**	SW846 3541

LABORATORY CONTROL SAMPLE: 977548

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% REC Limits	Qualifier
PCB 1016	mg/kg	0.083	0.067333	81	60-120	1
PCB 1260	mg/kg	0.083	0.062	74	60-120	
Decachlorobiphenyl (SS)	%			70	40-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 977549 977550

Parameter	Units	2033159 Result	Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limit	RPD	Max RPD	Qualifiers
PCB 1016	mg/kg	0	0.082	0.077632	0.075	94	91	60-120	3.4	30	1
PCB 1260	mg/kg	0	0.082	0.081908	0.074013	100	90	60-120	10	30	
Decachlorobiphenyl (SS)	%					95	87	40-120	9.2		

Analysis Description:	Semivolatile Organic Analysis	**QC Batch:**	REFD/4356
Analysis Method:	SW846 8270C	**QC Batch Method:**	SW846 3540

METHOD BLANK: 977555

Parameter	Blank Result	Reporting Limit	Units	Qualifiers
Acenaphthene	ND	0.10	mg/kg	1
Acenaphthylene	ND	0.10	mg/kg	
Aniline	ND	0.20	mg/kg	
Anthracene	ND	0.10	mg/kg	
Benzidine	ND	3.0	mg/kg	
Benzo(a)anthracene	ND	0.10	mg/kg	
Benzo(a)pyrene	ND	0.10	mg/kg	
Benzo(b)fluoranthene	ND	0.10	mg/kg	
Benzo(g,h,i)perylene	ND	0.10	mg/kg	
Benzo(k)fluoranthene	ND	0.10	mg/kg	
Benzoic acid	ND	0.50	mg/kg	
Benzyl alcohol	ND	0.10	mg/kg	
Benzyl butyl phthalate	ND	0.10	mg/kg	
4-Bromophenyl phenyl ether	ND	0.10	mg/kg	
Carbazole	ND	0.10	mg/kg	
4-Chloroaniline	ND	0.10	mg/kg	
bis(2-Chloroethoxy) methane	ND	0.10	mg/kg	
bis(2-Chloroethyl) ether	ND	0.10	mg/kg	
bis(2-Chloroisopropyl) ether	ND	0.10	mg/kg	
4-Chloro-3-methylphenol	ND	0.20	mg/kg	
2-Chloronaphthalene	ND	0.10	mg/kg	
2-Chlorophenol	ND	0.10	mg/kg	
4-Chlorophenyl phenyl ether	ND	0.10	mg/kg	

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QUALITY CONTROL DATA

Lab Order: V110347
Project ID: BIOSOLIDS

Analysis Description:	Semivolatile Organic Analysis	**QC Batch:**	REFD/4356
Analysis Method:	SW846 8270C	**QC Batch Method:**	SW846 3540

Parameter	Blank Result	Reporting Limit	Units	Qualifiers
Chrysene	ND	0.10	mg/kg	
Dibenzo(a,h)anthracene	ND	0.10	mg/kg	
Dibenzofuran	ND	0.10	mg/kg	
1,2-Dichlorobenzene	ND	0.10	mg/kg	
1,3-Dichlorobenzene	ND	0.10	mg/kg	
1,4-Dichlorobenzene	ND	0.10	mg/kg	
3,3'-Dichlorobenzidine	ND	0.20	mg/kg	
2,4-Dichlorophenol	ND	0.10	mg/kg	
Diethylphthalate	ND	0.10	mg/kg	
2,4-Dimethylphenol	ND	0.10	mg/kg	
Dimethylphthalate	ND	0.10	mg/kg	
Di-n-butylphthalate	ND	0.10	mg/kg	
2,4-Dinitrophenol	ND	0.50	mg/kg	
2,4-Dinitrotoluene	ND	0.10	mg/kg	
2,6-Dinitrotoluene	ND	0.10	mg/kg	
Di-n-octylphthalate	ND	0.10	mg/kg	
1,2Diphenylhydrazine/Azobenzen	ND	0.10	mg/kg	
bis(2-Ethylhexyl) phthalate	ND	0.20	mg/kg	
Fluoranthene	ND	0.10	mg/kg	
Fluorene	ND	0.10	mg/kg	
Hexachlorobenzene	ND	0.10	mg/kg	
Hexachlorobutadiene	ND	0.10	mg/kg	
Hexachlorocyclo pentadiene	ND	0.10	mg/kg	
Hexachloroethane	ND	0.10	mg/kg	
Indeno(1,2,3-cd)pyrene	ND	0.10	mg/kg	
Isophorone	ND	0.10	mg/kg	
2-Methyl-4,6-dinitrophenol	ND	0.50	mg/kg	
2-Methylnaphthalene	ND	0.10	mg/kg	
2-Methylphenol (o-Cresol)	ND	0.10	mg/kg	
3 & 4-Methylphenol(m&p Cresol)	ND	0.20	mg/kg	
Naphthalene	ND	0.10	mg/kg	
2-Nitroaniline	ND	0.10	mg/kg	
3-Nitroaniline	ND	0.20	mg/kg	
4-Nitroaniline	ND	0.20	mg/kg	
Nitrobenzene	ND	0.10	mg/kg	
2-Nitrophenol	ND	0.10	mg/kg	
4-Nitrophenol	ND	0.20	mg/kg	
N-Nitrosodimethylamine	ND	0.10	mg/kg	
N-Nitroso-di-n-propylamine	ND	0.10	mg/kg	
N-Nitrosodiphenylamine	ND	0.10	mg/kg	
Pentachlorophenol	ND	0.20	mg/kg	
Phenanthrene	ND	0.10	mg/kg	
Phenol	ND	0.10	mg/kg	
Pyrene	ND	0.10	mg/kg	
Pyridine	ND	0.50	mg/kg	
1,2,4-Trichlorobenzene	ND	0.10	mg/kg	
2,4,5-Trichlorophenol	ND	0.20	mg/kg	

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QUALITY CONTROL DATA

Lab Order: V110347
Project ID: BIOSOLIDS

Analysis Description:	Semivolatile Organic Analysis	**QC Batch:**	REFD/4356
Analysis Method:	SW846 8270C	**QC Batch Method:**	SW846 3540

Parameter	Blank Result	Reporting Limit	Units	Qualifiers
2,4,6-Trichlorophenol	ND	0.20	mg/kg	
2-Fluorobiphenyl (SS)	91	30-140	%	
2-Fluorophenol (SS)	73	20-130	%	
Nitrobenzene-d5 (SS)	101	30-130	%	
Phenol-d5 (SS)	84	30-130	%	
2,4,6-Tribromophenol (SS)	116	20-150	%	

LABORATORY CONTROL SAMPLE: 977556

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% REC Limits	Qualifier
Acenaphthene	mg/kg	1.7	1.2558	75	50-130	1
4-Chloro-3-methylphenol	mg/kg	1.7	1.4086	84	50-130	
2-Chlorophenol	mg/kg	1.7	1.1921	71	50-130	
1,4-Dichlorobenzene	mg/kg	1.7	1.4859	88	50-130	
2,4-Dinitrotoluene	mg/kg	1.7	1.5783	94	50-130	
Hexachlorobenzene	mg/kg	1.7	1.6963	101	40-130	
Hexachlorobutadiene	mg/kg	1.7	1.695	101	50-130	
Hexachloroethane	mg/kg	1.7	1.6465	98	50-130	
2-Methylphenol (o-Cresol)	mg/kg	1.7	1.3	77	50-130	
3 & 4-Methylphenol(m&p Cresol)	mg/kg	3.4	2.7695	82	50-130	
Nitrobenzene	mg/kg	1.7	1.772	105	50-130	
4-Nitrophenol	mg/kg	1.7	1.4827	88	30-130	
N-Nitroso-di-n-propylamine	mg/kg	1.7	1.5373	91	40-120	
Pentachlorophenol	mg/kg	1.7	0.90861	54	20-130	
Phenol	mg/kg	1.7	1.3081	78	40-120	
Pyrene	mg/kg	1.7	1.2829	76	40-150	
Pyridine	mg/kg	1.7	0	0	10-110	11
1,2,4-Trichlorobenzene	mg/kg	1.7	1.6113	96	50-120	
2,4,6-Trichlorophenol	mg/kg	1.7	1.2965	77	50-130	
2-Fluorobiphenyl (SS)	%			78	30-140	
2-Fluorophenol (SS)	%			68	20-130	
Nitrobenzene-d5 (SS)	%			94	30-130	
Phenol-d5 (SS)	%			78	30-130	
2,4,6-Tribromophenol (SS)	%			104	20-150	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 977557 977558

Parameter	Units	2033159 Result	Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limit	RPD	Max RPD	Qualifiers
Acenaphthene	mg/kg	0	1.6	1.2869	1.1714	78	70	30-140	9.4	30	1
4-Chloro-3-methylphenol	mg/kg	0	1.6	1.4697	1.3202	89	79	50-130	11	30	
2-Chlorophenol	mg/kg	0	1.6	1.1878	1.0814	72	64	50-130	9.4	30	
1,4-Dichlorobenzene	mg/kg	0	1.6	1.4844	1.3534	90	81	50-130	9.2	30	

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QUALITY CONTROL DATA

Lab Order: V110347
Project ID: BIOSOLIDS

Analysis Description:	Semivolatile Organic Analysis	**QC Batch:**	REFD/4356
Analysis Method:	SW846 8270C	**QC Batch Method:**	SW846 3540

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 977557 977558

Parameter	Units	2033159 Result	Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limit	RPD	Max RPD	Qualifiers
2,4-Dinitrotoluene	mg/kg	0	1.6	1.5928	1.4179	97	85	50-130	12	30	
Hexachlorobenzene	mg/kg	0	1.6	1.7791	1.6229	108	97	50-130	9.2	30	
Hexachlorobutadiene	mg/kg	0	1.6	1.7812	1.6245	108	97	50-130	9.2	30	
Hexachloroethane	mg/kg	0	1.6	1.6431	1.494	100	89	50-130	9.5	30	
2-Methylphenol (o-Cresol)	mg/kg	0	1.6	1.3194	1.195	80	71	50-130	9.9	30	
3 & 4-Methylphenol(m&p Cresol)	mg/kg	0	3.3	2.8059	2.5348	85	76	50-130	10	30	
Nitrobenzene	mg/kg	0	1.6	1.8034	1.6308	110	97	30-120	10	30	
4-Nitrophenol	mg/kg	0	1.6	1.5122	1.2694	92	76	30-140	17	30	
N-Nitroso-di-n-propylamine	mg/kg	0	1.6	1.5025	1.374	91	82	20-130	8.9	30	
Pentachlorophenol	mg/kg	0	1.6	0.94719	0.84682	58	50	30-130	11	30	
Phenol	mg/kg	0	1.6	1.2875	1.175	78	70	40-150	9.1	30	
Pyrene	mg/kg	0	1.6	1.2409	1.1757	75	70	40-140	5.4	30	
Pyridine	mg/kg	0	1.6	0.13562	0	8.2	0	10-110	200	30	8
1,2,4-Trichlorobenzene	mg/kg	0	1.6	1.6509	1.5089	100	90	50-130	9	30	
2,4,6-Trichlorophenol	mg/kg	0	1.6	1.3369	1.2046	81	72	50-130	10	30	
2-Fluorobiphenyl (SS)	%					83	75	30-140	7.7		
2-Fluorophenol (SS)	%					71	64	20-130	7.6		
Nitrobenzene-d5 (SS)	%					97	87	30-130	8.6		
Phenol-d5 (SS)	%					80	72	30-130	8.7		
2,4,6-Tribromophenol (SS)	%					115	102	20-150	10		

Analysis Description:	Wet Chem Analysis	**QC Batch:**	REFD/4361
Analysis Method:	EPA 350.1	**QC Batch Method:**	EPA 350.1

METHOD BLANK: 978028

Parameter	Blank Result	Reporting Limit	Units	Qualifiers
Ammonia (as N)	ND	20	mg/kg	1

LABORATORY CONTROL SAMPLE: 978029

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% REC Limits	Qualifier
Ammonia (as N)	mg/kg	200	194.63	99	80-120	1

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QUALITY CONTROL DATA

Lab Order: V110347
Project ID: BIOSOLIDS

Analysis Description:	Wet Chem Analysis	**QC Batch:**	REFD/4361
Analysis Method:	EPA 350.1	**QC Batch Method:**	EPA 350.1

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 978030 978031

Parameter	Units	0201201 Result	Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limit	RPD	Max RPD	Qualifiers
Ammonia (as N)	mg/kg	1800	200	1896.4	1913.5	71	82	80-120	0.9	20	1,12,13

SAMPLE DUPLICATE: 978032

Parameter	Units	0201201 Result	DUP Result	RPD	Max RPD	Reporting Limit	MDL	Qualifiers
Ammonia (as N)	mg/kg	1756.3	1625.6	7.7	20	98	25	1

Analysis Description:	Wet Chem Analysis	**QC Batch:**	REFD/4362
Analysis Method:	EPA 353.2	**QC Batch Method:**	EPA 353.2

METHOD BLANK: 978033

Parameter	Blank Result	Reporting Limit	Units	Qualifiers
Nitrogen, Nitrite (as N)	ND	0.50	mg/kg	1

LABORATORY CONTROL SAMPLE: 978034

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% REC Limits	Qualifier
Nitrogen, Nitrite (as N)	mg/kg	0.53	0.5341	101	90-110	1

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 978035 978036

Parameter	Units	0201201 Result	Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limit	RPD	Max RPD	Qualifiers
Nitrogen, Nitrite (as N)	mg/kg	0.15	5.3	5.2101	5.1711	96	95	80-120	0.8	20	1

SAMPLE DUPLICATE: 978037

Parameter	Units	0201201 Result	DUP Result	RPD	Max RPD	Reporting Limit	MDL	Qualifiers
Nitrogen, Nitrite (as N)	mg/kg	0.14756	0.15629	5.7	20	0.50	0.10	1

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QUALITY CONTROL DATA

Lab Order: V110347
Project ID: BIOSOLIDS

Analysis Description:	Wet Chem Analysis	**QC Batch:**	REFD/4363
Analysis Method:	EPA 300.0	**QC Batch Method:**	EPA 300.0

METHOD BLANK: 978038

Parameter	Blank Result	Reporting Limit	Units	Qualifiers
Nitrogen, Nitrate (as N)	ND	1.0	mg/kg	1

LABORATORY CONTROL SAMPLE: 978039

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% REC Limits	Qualifier
Nitrogen, Nitrate (as N)	mg/kg	5	5.126	103	90-110	1

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 978040 978041

Parameter	Units	0201201 Result	Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limit	RPD	Max RPD	Qualifiers
Nitrogen, Nitrate (as N)	mg/kg	1.6	51	52.697	52.545	101	101	80-120	0.3	20	1

SAMPLE DUPLICATE: 978042

Parameter	Units	0201201 Result	DUP Result	RPD	Max RPD	Reporting Limit	MDL	Qualifiers
Nitrogen, Nitrate (as N)	mg/kg	1.57	1.37	14	20	1.0	0.26	1

Analysis Description:	Wet Chem Analysis	**QC Batch:**	REFD/4365
Analysis Method:	EPA 365.4	**QC Batch Method:**	EPA 365.4

METHOD BLANK: 978045

Parameter	Blank Result	Reporting Limit	Units	Qualifiers
Total Phosphorus as P	ND	10	mg/kg	1

LABORATORY CONTROL SAMPLE: 978046

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% REC Limits	Qualifier
Total Phosphorus as P	mg/kg	200	202.86	101	85-115	1

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QUALITY CONTROL DATA

Lab Order: V110347
Project ID: BIOSOLIDS

Analysis Description:	Wet Chem Analysis	**QC Batch:**	REFD/4365
Analysis Method:	EPA 365.4	**QC Batch Method:**	EPA 365.4

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 978047 978048

Parameter	Units	0201201 Result	Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limit	RPD	Max RPD	Qualifiers
Total Phosphorus as P	mg/kg	6800	200	7396.4	7579.6	309	401	80-120	2.4	20	12

SAMPLE DUPLICATE: 978049

Parameter	Units	0201201 Result	DUP Result	RPD	Max RPD	Reporting Limit	MDL	Qualifiers
Total Phosphorus as P	mg/kg	6778.4	6570.8	3.1	20	200	64	1

Analysis Description:	Wet Chem Analysis	**QC Batch:**	REFD/4366
Analysis Method:	EPA 351.2	**QC Batch Method:**	EPA 351.2

METHOD BLANK: 978050

Parameter	Blank Result	Reporting Limit	Units	Qualifiers
Total Kjeldahl Nitrogen	ND	40	mg/kg	1

LABORATORY CONTROL SAMPLE: 978051

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% REC Limits	Qualifier
Total Kjeldahl Nitrogen	mg/kg	400	404.16	101	90-110	1

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 978052 978053

Parameter	Units	0201201 Result	Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limit	RPD	Max RPD	Qualifiers
Total Kjeldahl Nitrogen	mg/kg	14000	400	16057	15768	473	400	90-110	1.8	20	1,12

SAMPLE DUPLICATE: 978054

Parameter	Units	0201201 Result	DUP Result	RPD	Max RPD	Reporting Limit	MDL	Qualifiers
Total Kjeldahl Nitrogen	mg/kg	14167	13940	1.6	20	800	260	1

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QUALITY CONTROL DATA

Lab Order: V110347
Project ID: BIOSOLIDS

Analysis Description:	pH, Soil and Waste Analysis	**QC Batch:**	WET/10682
Analysis Method:	SW846 9045	**QC Batch Method:**	SW846 9045

SAMPLE DUPLICATE: 974960

Parameter	Units	V101037001 Result	DUP Result	RPD	Max RPD	Reporting Limit	MDL	Qualifiers
pH	pH Units	6.83	6.79	0.6	1.5			

Analysis Description:	Electrical Conductance Analysis	**QC Batch:**	WET/10693
Analysis Method:	SM 2510 B-97/-11	**QC Batch Method:**	SM 2510 B-97/-11

METHOD BLANK: 975676

Parameter	Blank Result	Reporting Limit	Units	Qualifiers
Conductivity	ND	10	umhos/c	

LABORATORY CONTROL SAMPLE: 975677

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% REC Limits	Qualifier
Conductivity	umhos/c	1400	1410	100	80-120	

SAMPLE DUPLICATE: 975678

Parameter	Units	V110472001 Result	DUP Result	RPD	Max RPD	Reporting Limit	MDL	Qualifiers
Conductivity	umhos/c	480	478	0.4	20	10	10	

Analysis Description:	Percent Solids Analysis	**QC Batch:**	WGR/7767
Analysis Method:	SM 2540 G-11	**QC Batch Method:**	SM 2540 G-11

METHOD BLANK: 975269

Parameter	Blank Result	Reporting Limit	Units	Qualifiers
Solids, Percent	ND	0.1	%	

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QUALITY CONTROL DATA

Lab Order: V110347
Project ID: BIOSOLIDS

Analysis Description:	Percent Solids Analysis	**QC Batch:**	WGR/7767
Analysis Method:	SM 2540 G-11	**QC Batch Method:**	SM 2540 G-11

SAMPLE DUPLICATE: 975270

Parameter	Units	V110347001 Result	DUP Result	RPD	Max RPD	Reporting Limit	MDL	Qualifiers
Solids, Percent	%	0.7	0.6	15	20	0.1	0.1	

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QUALITY CONTROL DATA QUALIFIERS

Lab Order: V110347
Project ID: BIOSOLIDS

QUALITY CONTROL PARAMETER QUALIFIERS

Results Qualifiers: Report fields may contain codes and non-numeric data correlating to one or more of the following definitions:

NS - means not spiked and will not have recoveries reported for Analyte Spike Amounts

QC Codes Keys: These descriptors are used to help identify the specific QC samples and clarify the report.

MB - Method Blank

Method Blanks are reported to the same Method Detection Limits (MDLs) or Reporting Limits (RLs) as the analytical samples in the corresponding QC batch.

LCS/LCSD - Laboratory Control Spike / Laboratory Control Spike Duplicate

DUP - Duplicate of Original Sample Matrix

MS/MSD - Matrix Spike / Matrix Spike Duplicate

RPD - Relative Percent Difference

%Recovery - Spike Recovery stated as a percentage

1	Analysis Performed by BC Laboratories, Inc.; CA ELAP #1186. Some BC Labs sample and quality control results are input into Caltest's database to include in Caltest's report and EDD, as a convenience to our client. If quality control values in the two reports do not match, the BC reported value should be considered the correct value.
7	Low Matrix Spike recovery(ies) due to possible matrix interferences in the QC sample. QC batch accepted based on LCS and RPD results.
8	Matrix spike recovery(s) is(are) not within the control limits.
9	Sample precision is not within the control limits.
10	The difference between duplicate readings is less than the quantitation limit.
11	The Laboratory Control Sample Soil (LCSS) recovery is not within laboratory established control limits.
12	The sample concentration is more than 4 times the spike level.
13	Detection and quantitation limits were raised due to sample dilution caused by high analyte concentration or matrix interference.

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Lab Order: V110347
Project ID: BIOSOLIDS

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
V110347001	**SLUDGE POND B**	Calculated	CALC/	Calculated	CALC/
V110347001	**SLUDGE POND B**	EPA 7471A	MPR/17653	EPA 7471A	MHG/6566
V110347001	**SLUDGE POND B**	SW846 3050B	REFD/4352	EPA 6010B	REFD/4357
V110347001	**SLUDGE POND B**	SW846 3541	REFD/4353	SW846 8081	REFD/4358
V110347001	**SLUDGE POND B**	SW846 3541	REFD/4354	SW846 8082	REFD/4359
V110347001	**SLUDGE POND B**	SW846 3540	REFD/4356	SW846 8270C	REFD/4360
V110347001	**SLUDGE POND B**	EPA 350.1	REFD/4361		
V110347001	**SLUDGE POND B**	EPA 353.2	REFD/4362		
V110347001	**SLUDGE POND B**	EPA 300.0	REFD/4363		
V110347001	**SLUDGE POND B**	EPA 365.4	REFD/4365		
V110347001	**SLUDGE POND B**	EPA 351.2	REFD/4366		
V110347001	**SLUDGE POND B**	SW846 9045	WET/10682		
V110347001	**SLUDGE POND B**	SM 2510 B-97/-11	WET/10693		
V110347001	**SLUDGE POND B**	SM 2540 G-11	WGR/7767		

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1885 North Kelly Road • Napa, California 94558
(707) 258-4000 • Fax (707) 226-1001 • e-mail: info@caltestlabs.com

Date of Report: 11/30/2020

Carol Battaglia

Caltest Analytical
1885 North Kelly Road
Napa, CA 94558

Client Project:	V110347
BCL Project:	Solid Analysis
BCL Work Order:	2033164
Invoice ID:	B399162

Enclosed are the results of analyses for samples received by the laboratory on 11/11/2020. If you have any questions concerning this report, please feel free to contact me.

Sincerely,

Contact Person: Vanessa Sandoval
Client Service Rep

Authorized Signature

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Chain of Custody and Cooler Receipt Form for 2033164 Page 1 of 2

Caltest ANALYTICAL LABORATORY
ENVIRONMENTAL ANALYSES

SUBCONTRACT CHAIN OF CUSTODY
PROJECT ID / PO NUMBER - V110347

Page 1 of 1

RUSH! PM Review: GM

Sending Laboratory
Caltest Analytical Laboratory
1885 N. Kelly Rd.
Napa, CA 94558
Phone: (707) 258-4000
Fax: (707) 226-1001
Report To: Client Services
Email: pmgr@caltestlabs.com

Receiving Laboratory
BC Laboratories, Inc.
4100 Atlas Ct
Bakersfield, CA 93308
Phone: 800-878-4911
Fax: 661-327-1918

Due Date: 11/25/20 (contact if due date cannot be met)

20-33164

Sampled By: SG

Caltest Lab ID (ref. only)	Client Sample ID	Matrix	Sample Date	Container Type	Presevative	Requested Analyses	Comments
V110347001	SLUDGE POND B	S	11/06/2020 10:57			8081 Analysis	NEED CUSTOM CALTEST EDD FORMAT
						8082 Analysis	NEED CUSTOM CALTEST EDD FORMAT
						8270 Analysis	NEED CUSTOM CALTEST EDD FORMAT
						Ammonia as N, EPA 350.1	
						Metals by 6010 (see notes)	REPORT - As, Cd, Cu, Pb, Mo, Ni, Se & Zn & K
						Nitrate as N, EPA 300.0	
						Nitrite as N, EPA 353.2	
						Ref.Lab Analysis - BCL	REPORT EC / SALINITY
						TKN, EPA 351.2	
						Total Phosphorus as P, EPA 365.4	

CHK BY [signature]
DISTRIBUTION [illegible]
SUB-OUT ☐

Relinquished By	Date / Time	Received By	Date / Time
[signature]	11/10/20 1600	GSO	
		[signature]	11-11-20 1000

Tuesday, November 10 2020 11:36AM

Matrix: W – Water DW - Drinking Water (Regulatory) S - Soil/Sludge/Solid ML - Final Effluent (Low Level/Minimum Level RLs)

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Chain of Custody and Cooler Receipt Form for 2033164 Page 2 of 2

BC LABORATORIES INC. COOLER RECEIPT FORM Page 1 Of 1

Submission #: 20-33164

SHIPPING INFORMATION
Fed Ex ☐ UPS ☐ Ontrac ☐ Hand Delivery ☐
BC Lab Field Service ☐ Other ☒ (Specify) GLS

SHIPPING CONTAINER
Ice Chest ☒ None ☐ Box ☐
Other ☐ (Specify)

FREE LIQUID
YES ☐ NO ☐
W / S

Refrigerant: Ice ☒ Blue Ice ☐ None ☐ Other ☐ Comments:

Custody Seals Ice Chest ☐ Intact? Yes ☐ No ☐ Containers ☐ Intact? Yes ☐ No ☐ None ☒ Comments:

All samples received? Yes ☐ No ☐ All samples containers intact? Yes ☐ No ☐ Description(s) match COC? Yes ☐ No ☐

COC Received ☒ YES ☐ NO

Emissivity: 97 Container: PE Thermometer ID: 274 Date/Time 11-11-20 1000
Temperature: (A) 1.3 °C / (C) 1.1 °C Analyst Init TKJ

SAMPLE CONTAINERS	SAMPLE NUMBERS 1	2	3	4	5	6	7	8	9	10
QT PE UNPRES										
4oz / 8oz / 16oz PE UNPRES										
2oz Cr^{+6}										
QT INORGANIC CHEMICAL METALS										
INORGANIC CHEMICAL METALS 4oz / 8oz / 16oz										
PT CYANIDE										
PT NITROGEN FORMS										
PT TOTAL SULFIDE										
2oz. NITRATE / NITRITE										
PT TOTAL ORGANIC CARBON										
PT CHEMICAL OXYGEN DEMAND										
PtA PHENOLICS										
40ml VOA VIAL TRAVEL BLANK										
40ml VOA VIAL										
QT EPA 1664										
PT ODOR										
RADIOLOGICAL										
BACTERIOLOGICAL										
40 ml VOA VIAL- 504										
QT EPA 508/608/8080										
QT EPA 515.1/8150										
QT EPA 525										
QT EPA 525 TRAVEL BLANK										
40ml EPA 547										
40ml EPA 531.1										
8oz EPA 548										
QT EPA 549										
QT EPA 8015M										
QT EPA 8270										
8oz / 16oz / 32oz AMBER										
8oz / 16oz / 32oz JAR	AB									
FOIL SLEEVE										
PCB VIAL										
PLASTIC BAG										
PEDLAR BAG										
FERROUS IRON										
ENCORE										
SMART KIT										
SUMMA CANISTER										

Comments:
Sample Numbering Completed By: SJM Date/Time: 11/11/20 11:40
A = Actual / C = Corrected

Rev 21 05/23/2016
IS:\WP\Doc\WordPerfect\LAB_DOCS\FORMS\SAMREC rev 20

Caltest Analytical 1885 North Kelly Road Napa, CA 94558	**Reported:** 11/30/2020 13:14 Project: Solid Analysis Project Number: V110347 Project Manager: Carol Battaglia

Laboratory / Client Sample Cross Reference

Laboratory	Client Sample Information			
2033164-01	**COC Number:**	---	**Receive Date:**	11/11/2020 10:00
	Project Number:	---	**Sampling Date:**	11/06/2020 10:57
	Sampling Location:	---	**Sample Depth:**	---
	Sampling Point:	V110347001 Sludge Pond B	**Lab Matrix:**	Solids
	Sampled By:	SG	**Sample Type:**	Soil

Caltest Analytical
1885 North Kelly Road
Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Organochlorine Pesticides (EPA Method 8081A)

BCL Sample ID: 2033164-01 | **Client Sample Name:** V110347001 Sludge Pond B, 11/6/2020 10:57:00AM, SG

Constituent	Dry Basis Result	As Recvd Result	Units	As Received PQL	As Received MDL	Method	TTLC Limits	Lab Quals	Run #
Aldrin		ND	mg/kg	0.0028	0.00010	EPA-8081A	1.4		1
alpha-BHC		ND	mg/kg	0.0028	0.00021	EPA-8081A			1
beta-BHC		ND	mg/kg	0.0028	0.00027	EPA-8081A			1
delta-BHC		ND	mg/kg	0.0028	0.00021	EPA-8081A			1
gamma-BHC (Lindane)		ND	mg/kg	0.0028	0.00010	EPA-8081A	4.0		1
Chlordane (Technical)		ND	mg/kg	0.28	0.0056	EPA-8081A	2.5		1
4,4'-DDD		ND	mg/kg	0.0028	0.00036	EPA-8081A	1.0		1
4,4'-DDE		ND	mg/kg	0.0028	0.00053	EPA-8081A	1.0		1
4,4'-DDT		ND	mg/kg	0.0028	0.00022	EPA-8081A	1.0		1
Dieldrin		ND	mg/kg	0.0028	0.00020	EPA-8081A	8.0		1
Endosulfan I		ND	mg/kg	0.0028	0.00011	EPA-8081A			1
Endosulfan II		ND	mg/kg	0.0028	0.00019	EPA-8081A			1
Endosulfan sulfate		ND	mg/kg	0.0028	0.00014	EPA-8081A			1
Endrin		ND	mg/kg	0.0028	0.00036	EPA-8081A	0.2		1
Endrin aldehyde		ND	mg/kg	0.0028	0.00010	EPA-8081A			1
Endrin ketone		ND	mg/kg	0.0028	0.00036	EPA-8081A			1
Heptachlor		ND	mg/kg	0.0028	0.00048	EPA-8081A	4.7		1
Heptachlor epoxide		ND	mg/kg	0.0028	0.000094	EPA-8081A			1
Methoxychlor		ND	mg/kg	0.0028	0.00052	EPA-8081A	100		1
Toxaphene		ND	mg/kg	0.28	0.0078	EPA-8081A	5		1
TCMX (Surrogate)		54.4	%	20 - 130 (LCL - UCL)		EPA-8081A			1
Decachlorobiphenyl (Surrogate)		34.8	%	40 - 130 (LCL - UCL)		EPA-8081A		A20	1

Run #	Method	Prep Date	Run Date/Time	Analyst	Instrument	Dilution	QC Batch ID	Prep Method
1	EPA-8081A	11/11/20 18:30	11/13/20 04:17	HKS	GC-17	5.556	B092719	EPA 3550B

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Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

PCB Analysis (EPA Method 8082)

BCL Sample ID: 2033164-01 | **Client Sample Name:** V110347001 Sludge Pond B, 11/6/2020 10:57:00AM, SG

Constituent	Dry Basis Result	As Recvd Result	Units	As Received PQL	As Received MDL	Method	TTLC Limits	Lab Quals	Run #
PCB-1016		ND	mg/kg	0.57	0.13	EPA-8082	50	A10	1
PCB-1221		ND	mg/kg	0.57	0.19	EPA-8082	50	A10	1
PCB-1232		ND	mg/kg	0.57	0.15	EPA-8082	50	A10	1
PCB-1242		ND	mg/kg	0.57	0.085	EPA-8082	50	A10	1
PCB-1248		ND	mg/kg	0.57	0.12	EPA-8082	50	A10	1
PCB-1254		ND	mg/kg	0.57	0.091	EPA-8082	50	A10	1
PCB-1260		ND	mg/kg	0.57	0.10	EPA-8082	50	A10	1
Total PCB's (Summation)		ND	mg/kg	0.57	0.28	EPA-8082	50	A10	1
Decachlorobiphenyl (Surrogate)		66.7	%	40 - 120 (LCL - UCL)		EPA-8082		A10	1

Run #	Method	Prep Date	Run Date/Time	Analyst	Instrument	Dilution	QC Batch ID	Prep Method
1	EPA-8082	11/12/20 10:40	11/14/20 03:45	HKS	GC-14	56.604	B092779	EPA 3550B

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Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Base Neutral and Acid Extractables Organic Analysis (EPA Method 8270C)

BCL Sample ID: 2033164-01 | **Client Sample Name:** V110347001 Sludge Pond B, 11/6/2020 10:57:00AM, SG

Constituent	Dry Basis Result	As Recvd Result	Units	As Received PQL	As Received MDL	Method	TTLC Limits	Lab Quals	Run #
Acenaphthene		ND	mg/kg	0.48	0.032	EPA-8270C			1
Acenaphthylene		ND	mg/kg	0.48	0.032	EPA-8270C			1
Aniline		ND	mg/kg	0.97	0.053	EPA-8270C			1
Anthracene		ND	mg/kg	0.48	0.032	EPA-8270C			1
Benzidine		ND	mg/kg	14	0.045	EPA-8270C			1
Benzo[a]anthracene		ND	mg/kg	0.48	0.037	EPA-8270C			1
Benzo[b]fluoranthene		ND	mg/kg	0.48	0.032	EPA-8270C			1
Benzo[k]fluoranthene		ND	mg/kg	0.48	0.040	EPA-8270C			1
Benzo[a]pyrene		ND	mg/kg	0.48	0.032	EPA-8270C			1
Benzo[g,h,i]perylene		ND	mg/kg	0.48	0.063	EPA-8270C			1
Benzoic acid		ND	mg/kg	2.4	0.068	EPA-8270C			1
Benzyl alcohol		ND	mg/kg	0.48	0.032	EPA-8270C			1
Benzyl butyl phthalate		ND	mg/kg	0.48	0.032	EPA-8270C			1
alpha-BHC		ND	mg/kg	0.48	0.032	EPA-8270C			1
beta-BHC		ND	mg/kg	0.48	0.036	EPA-8270C			1
delta-BHC		ND	mg/kg	0.48	0.032	EPA-8270C			1
gamma-BHC (Lindane)		ND	mg/kg	0.48	0.032	EPA-8270C	4.0		1
bis(2-Chloroethoxy)methane		ND	mg/kg	0.48	0.032	EPA-8270C			1
bis(2-Chloroethyl) ether		ND	mg/kg	0.48	0.047	EPA-8270C			1
bis(2-Chloroisopropyl)ether		ND	mg/kg	0.48	0.032	EPA-8270C			1
bis(2-Ethylhexyl)phthalate		ND	mg/kg	0.97	0.032	EPA-8270C			1
4-Bromophenyl phenyl ether		ND	mg/kg	0.48	0.032	EPA-8270C			1
4-Chloroaniline		ND	mg/kg	0.48	0.072	EPA-8270C			1
2-Chloronaphthalene		ND	mg/kg	0.48	0.032	EPA-8270C			1
4-Chlorophenyl phenyl ether		ND	mg/kg	0.48	0.032	EPA-8270C			1
Chrysene		ND	mg/kg	0.48	0.032	EPA-8270C			1
Dibenzo[a,h]anthracene		ND	mg/kg	0.48	0.032	EPA-8270C			1
Dibenzofuran		ND	mg/kg	0.48	0.032	EPA-8270C			1
1,2-Dichlorobenzene		ND	mg/kg	0.48	0.032	EPA-8270C			1
1,3-Dichlorobenzene		ND	mg/kg	0.48	0.032	EPA-8270C			1
1,4-Dichlorobenzene		ND	mg/kg	0.48	0.032	EPA-8270C			1
3,3-Dichlorobenzidine		ND	mg/kg	0.97	0.032	EPA-8270C			1

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1885 North Kelly Road
Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Base Neutral and Acid Extractables Organic Analysis (EPA Method 8270C)

BCL Sample ID: 2033164-01 | **Client Sample Name:** V110347001 Sludge Pond B, 11/6/2020 10:57:00AM, SG

Constituent	Dry Basis Result	As Recvd Result	Units	As Received PQL	As Received MDL	Method	TTLC Limits	Lab Quals	Run #
Diethyl phthalate		ND	mg/kg	0.48	0.032	EPA-8270C			1
Dimethyl phthalate		ND	mg/kg	0.48	0.032	EPA-8270C			1
Di-n-butyl phthalate		ND	mg/kg	0.48	0.032	EPA-8270C			1
2,4-Dinitrotoluene		ND	mg/kg	0.48	0.041	EPA-8270C			1
2,6-Dinitrotoluene		ND	mg/kg	0.48	0.032	EPA-8270C			1
Di-n-octyl phthalate		ND	mg/kg	0.48	0.032	EPA-8270C			1
1,2-Diphenylhydrazine		ND	mg/kg	0.48	0.032	EPA-8270C			1
Fluoranthene		ND	mg/kg	0.48	0.032	EPA-8270C			1
Fluorene		ND	mg/kg	0.48	0.032	EPA-8270C			1
Hexachlorobenzene		ND	mg/kg	0.48	0.032	EPA-8270C			1
Hexachlorobutadiene		ND	mg/kg	0.48	0.032	EPA-8270C			1
Hexachlorocyclopentadiene		ND	mg/kg	0.48	0.072	EPA-8270C			1
Hexachloroethane		ND	mg/kg	0.48	0.032	EPA-8270C			1
Indeno[1,2,3-cd]pyrene		ND	mg/kg	0.48	0.033	EPA-8270C			1
Isophorone		ND	mg/kg	0.48	0.032	EPA-8270C			1
2-Methylnaphthalene		ND	mg/kg	0.48	0.032	EPA-8270C			1
Naphthalene		ND	mg/kg	0.48	0.032	EPA-8270C			1
2-Nitroaniline		ND	mg/kg	0.48	0.032	EPA-8270C			1
3-Nitroaniline		ND	mg/kg	0.97	0.032	EPA-8270C			1
4-Nitroaniline		ND	mg/kg	0.97	0.053	EPA-8270C			1
Nitrobenzene		ND	mg/kg	0.48	0.032	EPA-8270C			1
N-Nitrosodimethylamine		ND	mg/kg	0.48	0.19	EPA-8270C			1
N-Nitrosodi-N-propylamine		ND	mg/kg	0.48	0.032	EPA-8270C			1
N-Nitrosodiphenylamine		ND	mg/kg	0.48	0.032	EPA-8270C			1
Phenanthrene		ND	mg/kg	0.48	0.032	EPA-8270C			1
Pyrene		ND	mg/kg	0.48	0.032	EPA-8270C			1
1,2,4-Trichlorobenzene		ND	mg/kg	0.48	0.032	EPA-8270C			1
4-Chloro-3-methylphenol		ND	mg/kg	0.97	0.032	EPA-8270C			1
2-Chlorophenol		ND	mg/kg	0.48	0.032	EPA-8270C			1
2,4-Dichlorophenol		ND	mg/kg	0.48	0.032	EPA-8270C			1
2,4-Dimethylphenol		ND	mg/kg	0.48	0.032	EPA-8270C			1
4,6-Dinitro-2-methylphenol		ND	mg/kg	2.4	0.032	EPA-8270C			1

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Caltest Analytical
1885 North Kelly Road
Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Base Neutral and Acid Extractables Organic Analysis (EPA Method 8270C)

BCL Sample ID: 2033164-01 | **Client Sample Name:** V110347001 Sludge Pond B, 11/6/2020 10:57:00AM, SG

Constituent	Dry Basis Result	As Recvd Result	Units	As Received PQL	As Received MDL	Method	TTLC Limits	Lab Quals	Run #
2,4-Dinitrophenol		ND	mg/kg	2.4	0.032	EPA-8270C			1
2-Methylphenol		ND	mg/kg	0.48	0.032	EPA-8270C			1
3- & 4-Methylphenol		ND	mg/kg	0.97	0.068	EPA-8270C			1
Total Methylphenol		ND	mg/kg	0.97	0.10	EPA-8270C			1
2-Nitrophenol		ND	mg/kg	0.48	0.032	EPA-8270C			1
4-Nitrophenol		ND	mg/kg	0.97	0.087	EPA-8270C			1
Pentachlorophenol		ND	mg/kg	0.97	0.082	EPA-8270C	17		1
Phenol		ND	mg/kg	0.48	0.032	EPA-8270C			1
2,4,5-Trichlorophenol		ND	mg/kg	0.97	0.053	EPA-8270C			1
2,4,6-Trichlorophenol		ND	mg/kg	0.97	0.032	EPA-8270C			1
Carbazole		ND	mg/kg	0.48	0.032	EPA-8270C			1
Pentachlorobenzene		ND	mg/kg	2.4	0.033	EPA-8270C			1
Pyridine		ND	mg/kg	2.4	0.31	EPA-8270C			1
1,2,4,5-Tetrachlorobenzene		ND	mg/kg	2.4	0.036	EPA-8270C			1
2-Fluorophenol (Surrogate)		64.9	%	20 - 130 (LCL - UCL)		EPA-8270C			1
Phenol-d5 (Surrogate)		73.1	%	30 - 130 (LCL - UCL)		EPA-8270C			1
Nitrobenzene-d5 (Surrogate)		77.4	%	30 - 130 (LCL - UCL)		EPA-8270C			1
2-Fluorobiphenyl (Surrogate)		46.2	%	30 - 140 (LCL - UCL)		EPA-8270C			1
2,4,6-Tribromophenol (Surrogate)		108	%	20 - 150 (LCL - UCL)		EPA-8270C			1
p-Terphenyl-d14 (Surrogate)		37.0	%	30 - 150 (LCL - UCL)		EPA-8270C			1

Run #	Method	Prep Date	Run Date/Time	Analyst	Instrument	Dilution	QC Batch ID	Prep Method
1	EPA-8270C	11/12/20 11:30	11/14/20 00:53	MK1	MS-B8	4.831	B092737	EPA 3550B

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Caltest Analytical
1885 North Kelly Road
Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Chemical Analysis

BCL Sample ID: 2033164-01 | **Client Sample Name:** V110347001 Sludge Pond B, 11/6/2020 10:57:00AM, SG

Constituent	Dry Basis Result	As Recvd Result	Units	As Received PQL	As Received MDL	Method	MB Bias	Lab Quals	Run #
Nitrate as N		0.85	**mg/kg**	**1.0**	**0.26**	**EPA-300.0**	ND	**J**	1
Nitrite as N		3.8	**mg/kg**	**0.50**	**0.10**	**EPA-353.2**	ND		2
Total Kjeldahl Nitrogen		570	**mg/kg**	**40**	**13**	**EPA-351.2**	ND		3
Ammonia as N		280	**mg/kg**	**20**	**5.0**	**EPA-350.1**	ND		4
Total Phosphorus		66	**mg/kg**	**10**	**3.2**	**EPA-365.4**	ND		5

Run #	Method	Prep Date	Run Date/Time	Analyst	Instrument	Dilution	QC Batch ID	Prep Method
1	EPA-300.0	11/18/20 07:45	11/18/20 12:43	SAV	IC5	1	B092977	Water Extract X10
2	EPA-353.2	11/18/20 07:45	11/18/20 12:18	MC1	KONE-1	1	B092978	Water Extract X10
3	EPA-351.2	11/23/20 10:38	11/25/20 16:13	JMH2	SC-1	1	B093433	EPA 351.2
4	EPA-350.1	11/17/20 09:05	11/19/20 14:00	JMH2	SC-1	0.943	B092891	EPA 350.1
5	EPA-365.4	11/23/20 07:30	11/24/20 09:50	JMH2	SC-2	1	B093434	EPA 365.4

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Caltest Analytical
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Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Total Concentrations (TTLC)

BCL Sample ID: 2033164-01 | **Client Sample Name:** V110347001 Sludge Pond B, 11/6/2020 10:57:00AM, SG

Constituent	Dry Basis Result	As Recvd Result	Units	As Received PQL	MDL	Method	TTLC Limits	Lab Quals	Run #
Arsenic		ND	mg/kg	1.0	0.40	EPA-6010B	500		1
Cadmium		ND	mg/kg	0.50	0.052	EPA-6010B	100		1
Copper		1.6	**mg/kg**	**1.0**	**0.050**	**EPA-6010B**	2500		1
Lead		ND	mg/kg	2.5	0.41	EPA-6010B	1000		1
Molybdenum		ND	mg/kg	2.5	0.050	EPA-6010B	3500		1
Nickel		0.26	mg/kg	0.50	0.15	EPA-6010B	2000	J	1
Selenium		ND	mg/kg	1.0	0.98	EPA-6010B	100		1
Zinc		10	**mg/kg**	**2.5**	**0.087**	**EPA-6010B**	5000		1
Potassium		73	**mg/kg**	**50**	**5.0**	**EPA-6010B**			1

Run #	Method	Prep Date	Run Date/Time	Analyst	Instrument	Dilution	QC Batch ID	Prep Method
1	EPA-6010B	11/12/20 07:30	11/13/20 20:43	JRG	PE-OP4	0.990	B092496	EPA 3050B

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Caltest Analytical
1885 North Kelly Road
Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Organochlorine Pesticides (EPA Method 8081A)

Quality Control Report - Method Blank Analysis

Constituent	QC Sample ID	MB Result	Units	PQL	MDL	Lab Quals
QC Batch ID: B092719						
Aldrin	B092719-BLK1	ND	mg/kg	0.00050	0.000018	
alpha-BHC	B092719-BLK1	ND	mg/kg	0.00050	0.000038	
beta-BHC	B092719-BLK1	ND	mg/kg	0.00050	0.000048	
delta-BHC	B092719-BLK1	ND	mg/kg	0.00050	0.000037	
gamma-BHC (Lindane)	B092719-BLK1	ND	mg/kg	0.00050	0.000018	
Chlordane (Technical)	B092719-BLK1	ND	mg/kg	0.050	0.0010	
4,4'-DDD	B092719-BLK1	ND	mg/kg	0.00050	0.000064	
4,4'-DDE	B092719-BLK1	ND	mg/kg	0.00050	0.000095	
4,4'-DDT	B092719-BLK1	ND	mg/kg	0.00050	0.000040	
Dieldrin	B092719-BLK1	ND	mg/kg	0.00050	0.000036	
Endosulfan I	B092719-BLK1	ND	mg/kg	0.00050	0.000020	
Endosulfan II	B092719-BLK1	ND	mg/kg	0.00050	0.000034	
Endosulfan sulfate	B092719-BLK1	ND	mg/kg	0.00050	0.000026	
Endrin	B092719-BLK1	ND	mg/kg	0.00050	0.000065	
Endrin aldehyde	B092719-BLK1	ND	mg/kg	0.00050	0.000018	
Endrin ketone	B092719-BLK1	ND	mg/kg	0.00050	0.000065	
Heptachlor	B092719-BLK1	ND	mg/kg	0.00050	0.000086	
Heptachlor epoxide	B092719-BLK1	ND	mg/kg	0.00050	0.000017	
Methoxychlor	B092719-BLK1	ND	mg/kg	0.00050	0.000094	
Toxaphene	B092719-BLK1	ND	mg/kg	0.050	0.0014	
TCMX (Surrogate)	**B092719-BLK1**	**81.2**	%	**20 - 130 (LCL - UCL)**		
Decachlorobiphenyl (Surrogate)	**B092719-BLK1**	**68.4**	%	**40 - 130 (LCL - UCL)**		

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Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Organochlorine Pesticides (EPA Method 8081A)

Quality Control Report - Laboratory Control Sample

Constituent	QC Sample ID	Type	Result	Spike Level	Units	Percent Recovery	RPD	Control Limits Percent Recovery	Control Limits RPD	Lab Quals
QC Batch ID: B092719										
Aldrin	B092719-BS1	LCS	0.0043319	0.0049342	mg/kg	87.8		70 - 130		
gamma-BHC (Lindane)	B092719-BS1	LCS	0.0051023	0.0049342	mg/kg	103		60 - 140		
4,4'-DDT	B092719-BS1	LCS	0.0045461	0.0049342	mg/kg	92.1		60 - 140		
Dieldrin	B092719-BS1	LCS	0.0047885	0.0049342	mg/kg	97.0		70 - 130		
Endrin	B092719-BS1	LCS	0.0039289	0.0049342	mg/kg	79.6		60 - 140		
Heptachlor	B092719-BS1	LCS	0.0046309	0.0049342	mg/kg	93.9		60 - 140		
TCMX (Surrogate)	B092719-BS1	LCS	0.0089079	0.0098684	mg/kg	90.3		20 - 130		
Decachlorobiphenyl (Surrogate)	B092719-BS1	LCS	0.014153	0.019737	mg/kg	71.7		40 - 130		

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Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Organochlorine Pesticides (EPA Method 8081A)

Quality Control Report - Precision & Accuracy

Constituent	Type	Source Sample ID	Source Result	Result	Spike Added	Units	RPD	Percent Recovery	Control Limits RPD	Control Limits Percent Recovery	Lab Quals
QC Batch ID: B092719	Used client sample: N										
Aldrin	MS	2032744-40	ND	0.0037869	0.0050505	mg/kg		75.0		50 - 140	
	MSD	2032744-40	ND	0.0039611	0.0050676	mg/kg	4.5	78.2	30	50 - 140	
gamma-BHC (Lindane)	MS	2032744-40	ND	0.0050808	0.0050505	mg/kg		101		50 - 140	
	MSD	2032744-40	ND	0.0053186	0.0050676	mg/kg	4.6	105	30	50 - 140	
4,4'-DDT	MS	2032744-40	ND	0.0038808	0.0050505	mg/kg		76.8		50 - 140	
	MSD	2032744-40	ND	0.0044848	0.0050676	mg/kg	14.4	88.5	30	50 - 140	
Dieldrin	MS	2032744-40	ND	0.0047424	0.0050505	mg/kg		93.9		40 - 140	
	MSD	2032744-40	ND	0.0050453	0.0050676	mg/kg	6.2	99.6	30	40 - 140	
Endrin	MS	2032744-40	ND	0.0033165	0.0050505	mg/kg		65.7		50 - 150	
	MSD	2032744-40	ND	0.0040054	0.0050676	mg/kg	18.8	79.0	30	50 - 150	
Heptachlor	MS	2032744-40	ND	0.0039256	0.0050505	mg/kg		77.7		60 - 140	
	MSD	2032744-40	ND	0.0040027	0.0050676	mg/kg	1.9	79.0	30	60 - 140	
TCMX (Surrogate)	MS	2032744-40	ND	0.0081559	0.010101	mg/kg		80.7		20 - 130	
	MSD	2032744-40	ND	0.0083956	0.010135	mg/kg	2.9	82.8		20 - 130	
Decachlorobiphenyl (Surrogate)	MS	2032744-40	ND	0.010684	0.020202	mg/kg		52.9		40 - 130	
	MSD	2032744-40	ND	0.012414	0.020270	mg/kg	15.0	61.2		40 - 130	

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1885 North Kelly Road
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Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

PCB Analysis (EPA Method 8082)

Quality Control Report - Method Blank Analysis

Constituent	QC Sample ID	MB Result	Units	PQL	MDL	Lab Quals
QC Batch ID: B092779						
PCB-1016	B092779-BLK1	ND	mg/kg	0.010	0.0023	
PCB-1221	B092779-BLK1	ND	mg/kg	0.010	0.0033	
PCB-1232	B092779-BLK1	ND	mg/kg	0.010	0.0026	
PCB-1242	B092779-BLK1	ND	mg/kg	0.010	0.0015	
PCB-1248	B092779-BLK1	ND	mg/kg	0.010	0.0021	
PCB-1254	B092779-BLK1	ND	mg/kg	0.010	0.0016	
PCB-1260	B092779-BLK1	ND	mg/kg	0.010	0.0018	
Total PCB's (Summation)	B092779-BLK1	ND	mg/kg	0.010	0.0050	
Decachlorobiphenyl (Surrogate)	**B092779-BLK1**	**86.7**	%	**40 - 120 (LCL - UCL)**		

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1885 North Kelly Road
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Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

PCB Analysis (EPA Method 8082)

Quality Control Report - Laboratory Control Sample

Constituent	QC Sample ID	Type	Result	Spike Level	Units	Percent Recovery	RPD	Control Limits Percent Recovery	Control Limits RPD	Lab Quals
QC Batch ID: B092779										
PCB-1016	B092779-BS1	LCS	0.067333	0.083333	mg/kg	80.8		60 - 120		
PCB-1260	B092779-BS1	LCS	0.062000	0.083333	mg/kg	74.4		60 - 120		
Decachlorobiphenyl (Surrogate)	B092779-BS1	LCS	0.014000	0.020000	mg/kg	70.0		40 - 120		

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Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

PCB Analysis (EPA Method 8082)

Quality Control Report - Precision & Accuracy

Constituent	Type	Source Sample ID	Source Result	Result	Spike Added	Units	RPD	Percent Recovery	Control Limits RPD	Control Limits Percent Recovery	Lab Quals
QC Batch ID: B092779	Used client sample: N										
PCB-1016	MS	2030363-89	ND	0.077632	0.082237	mg/kg		94.4		60 - 120	
	MSD	2030363-89	ND	0.075000	0.082237	mg/kg	3.4	91.2	30	60 - 120	
PCB-1260	MS	2030363-89	ND	0.081908	0.082237	mg/kg		99.6		60 - 120	
	MSD	2030363-89	ND	0.074013	0.082237	mg/kg	10.1	90.0	30	60 - 120	
Decachlorobiphenyl (Surrogate)	MS	2030363-89	ND	0.018750	0.019737	mg/kg		95.0		40 - 120	
	MSD	2030363-89	ND	0.017105	0.019737	mg/kg	9.2	86.7		40 - 120	

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1885 North Kelly Road
Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Base Neutral and Acid Extractables Organic Analysis (EPA Method 8270C)

Quality Control Report - Method Blank Analysis

Constituent	QC Sample ID	MB Result	Units	PQL	MDL	Lab Quals
QC Batch ID: B092737						
Acenaphthene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Acenaphthylene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Aniline	B092737-BLK1	ND	mg/kg	0.20	0.011	
Anthracene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Benzidine	B092737-BLK1	ND	mg/kg	3.0	0.0093	
Benzo[a]anthracene	B092737-BLK1	ND	mg/kg	0.10	0.0077	
Benzo[b]fluoranthene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Benzo[k]fluoranthene	B092737-BLK1	ND	mg/kg	0.10	0.0082	
Benzo[a]pyrene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Benzo[g,h,i]perylene	B092737-BLK1	ND	mg/kg	0.10	0.013	
Benzoic acid	B092737-BLK1	ND	mg/kg	0.50	0.014	
Benzyl alcohol	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Benzyl butyl phthalate	B092737-BLK1	ND	mg/kg	0.10	0.0067	
alpha-BHC	B092737-BLK1	ND	mg/kg	0.10	0.0067	
beta-BHC	B092737-BLK1	ND	mg/kg	0.10	0.0075	
delta-BHC	B092737-BLK1	ND	mg/kg	0.10	0.0067	
gamma-BHC (Lindane)	B092737-BLK1	ND	mg/kg	0.10	0.0067	
bis(2-Chloroethoxy)methane	B092737-BLK1	ND	mg/kg	0.10	0.0067	
bis(2-Chloroethyl) ether	B092737-BLK1	ND	mg/kg	0.10	0.0097	
bis(2-Chloroisopropyl)ether	B092737-BLK1	ND	mg/kg	0.10	0.0067	
bis(2-Ethylhexyl)phthalate	B092737-BLK1	ND	mg/kg	0.20	0.0067	
4-Bromophenyl phenyl ether	B092737-BLK1	ND	mg/kg	0.10	0.0067	
4-Chloroaniline	B092737-BLK1	ND	mg/kg	0.10	0.015	
2-Chloronaphthalene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
4-Chlorophenyl phenyl ether	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Chrysene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Dibenzo[a,h]anthracene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Dibenzofuran	B092737-BLK1	ND	mg/kg	0.10	0.0067	
1,2-Dichlorobenzene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
1,3-Dichlorobenzene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
1,4-Dichlorobenzene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
3,3-Dichlorobenzidine	B092737-BLK1	ND	mg/kg	0.20	0.0067	
Diethyl phthalate	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Dimethyl phthalate	B092737-BLK1	ND	mg/kg	0.10	0.0067	

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Caltest Analytical
1885 North Kelly Road
Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Base Neutral and Acid Extractables Organic Analysis (EPA Method 8270C)

Quality Control Report - Method Blank Analysis

Constituent	QC Sample ID	MB Result	Units	PQL	MDL	Lab Quals
QC Batch ID: B092737						
Di-n-butyl phthalate	B092737-BLK1	ND	mg/kg	0.10	0.0067	
2,4-Dinitrotoluene	B092737-BLK1	ND	mg/kg	0.10	0.0085	
2,6-Dinitrotoluene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Di-n-octyl phthalate	B092737-BLK1	ND	mg/kg	0.10	0.0067	
1,2-Diphenylhydrazine	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Fluoranthene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Fluorene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Hexachlorobenzene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Hexachlorobutadiene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Hexachlorocyclopentadiene	B092737-BLK1	ND	mg/kg	0.10	0.015	
Hexachloroethane	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Indeno[1,2,3-cd]pyrene	B092737-BLK1	ND	mg/kg	0.10	0.0069	
Isophorone	B092737-BLK1	ND	mg/kg	0.10	0.0067	
2-Methylnaphthalene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Naphthalene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
2-Nitroaniline	B092737-BLK1	ND	mg/kg	0.10	0.0067	
3-Nitroaniline	B092737-BLK1	ND	mg/kg	0.20	0.0067	
4-Nitroaniline	B092737-BLK1	ND	mg/kg	0.20	0.011	
Nitrobenzene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
N-Nitrosodimethylamine	B092737-BLK1	ND	mg/kg	0.10	0.040	
N-Nitrosodi-N-propylamine	B092737-BLK1	ND	mg/kg	0.10	0.0067	
N-Nitrosodiphenylamine	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Phenanthrene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Pyrene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
1,2,4-Trichlorobenzene	B092737-BLK1	ND	mg/kg	0.10	0.0067	
4-Chloro-3-methylphenol	B092737-BLK1	ND	mg/kg	0.20	0.0067	
2-Chlorophenol	B092737-BLK1	ND	mg/kg	0.10	0.0067	
2,4-Dichlorophenol	B092737-BLK1	ND	mg/kg	0.10	0.0067	
2,4-Dimethylphenol	B092737-BLK1	ND	mg/kg	0.10	0.0067	
4,6-Dinitro-2-methylphenol	B092737-BLK1	ND	mg/kg	0.50	0.0067	
2,4-Dinitrophenol	B092737-BLK1	ND	mg/kg	0.50	0.0067	
2-Methylphenol	B092737-BLK1	ND	mg/kg	0.10	0.0067	
3- & 4-Methylphenol	B092737-BLK1	ND	mg/kg	0.20	0.014	
Total Methylphenol	B092737-BLK1	ND	mg/kg	0.20	0.021	

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Caltest Analytical
1885 North Kelly Road
Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Base Neutral and Acid Extractables Organic Analysis (EPA Method 8270C)

Quality Control Report - Method Blank Analysis

Constituent	QC Sample ID	MB Result	Units	PQL	MDL	Lab Quals
QC Batch ID: B092737						
2-Nitrophenol	B092737-BLK1	ND	mg/kg	0.10	0.0067	
4-Nitrophenol	B092737-BLK1	ND	mg/kg	0.20	0.018	
Pentachlorophenol	B092737-BLK1	ND	mg/kg	0.20	0.017	
Phenol	B092737-BLK1	ND	mg/kg	0.10	0.0067	
2,4,5-Trichlorophenol	B092737-BLK1	ND	mg/kg	0.20	0.011	
2,4,6-Trichlorophenol	B092737-BLK1	ND	mg/kg	0.20	0.0067	
Carbazole	B092737-BLK1	ND	mg/kg	0.10	0.0067	
Pentachlorobenzene	B092737-BLK1	ND	mg/kg	0.50	0.0069	
Pyridine	B092737-BLK1	ND	mg/kg	0.50	0.065	
1,2,4,5-Tetrachlorobenzene	B092737-BLK1	ND	mg/kg	0.50	0.0075	
2-Fluorophenol (Surrogate)	**B092737-BLK1**	**72.9**	%	**20 - 130 (LCL - UCL)**		
Phenol-d5 (Surrogate)	**B092737-BLK1**	**84.4**	%	**30 - 130 (LCL - UCL)**		
Nitrobenzene-d5 (Surrogate)	**B092737-BLK1**	**101**	%	**30 - 130 (LCL - UCL)**		
2-Fluorobiphenyl (Surrogate)	**B092737-BLK1**	**90.8**	%	**30 - 140 (LCL - UCL)**		
2,4,6-Tribromophenol (Surrogate)	**B092737-BLK1**	**116**	%	**20 - 150 (LCL - UCL)**		
p-Terphenyl-d14 (Surrogate)	**B092737-BLK1**	**76.0**	%	**30 - 150 (LCL - UCL)**		

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Caltest Analytical
1885 North Kelly Road
Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Base Neutral and Acid Extractables Organic Analysis (EPA Method 8270C)

Quality Control Report - Laboratory Control Sample

Constituent	QC Sample ID	Type	Result	Spike Level	Units	Percent Recovery	RPD	Control Limits Percent Recovery	Control Limits RPD	Lab Quals
QC Batch ID: B092737										
Acenaphthene	B092737-BS1	LCS	1.2558	1.6835	mg/kg	74.6		50 - 130		
1,4-Dichlorobenzene	B092737-BS1	LCS	1.4859	1.6835	mg/kg	88.3		50 - 130		
2,4-Dinitrotoluene	B092737-BS1	LCS	1.5783	1.6835	mg/kg	93.8		50 - 130		
Hexachlorobenzene	B092737-BS1	LCS	1.6963	1.6835	mg/kg	101		40 - 130		
Hexachlorobutadiene	B092737-BS1	LCS	1.6950	1.6835	mg/kg	101		50 - 130		
Hexachloroethane	B092737-BS1	LCS	1.6465	1.6835	mg/kg	97.8		50 - 130		
Nitrobenzene	B092737-BS1	LCS	1.7720	1.6835	mg/kg	105		50 - 130		
N-Nitrosodi-N-propylamine	B092737-BS1	LCS	1.5373	1.6835	mg/kg	91.3		40 - 120		
Pyrene	B092737-BS1	LCS	1.2829	1.6835	mg/kg	76.2		40 - 150		
1,2,4-Trichlorobenzene	B092737-BS1	LCS	1.6113	1.6835	mg/kg	95.7		50 - 120		
4-Chloro-3-methylphenol	B092737-BS1	LCS	1.4086	1.6835	mg/kg	83.7		50 - 130		
2-Chlorophenol	B092737-BS1	LCS	1.1921	1.6835	mg/kg	70.8		50 - 130		
2-Methylphenol	B092737-BS1	LCS	1.3000	1.6835	mg/kg	77.2		50 - 130		
3- & 4-Methylphenol	B092737-BS1	LCS	2.7695	3.3670	mg/kg	82.3		50 - 130		
4-Nitrophenol	B092737-BS1	LCS	1.4827	1.6835	mg/kg	88.1		30 - 130		
Pentachlorophenol	B092737-BS1	LCS	0.90861	1.6835	mg/kg	54.0		20 - 130		
Phenol	B092737-BS1	LCS	1.3081	1.6835	mg/kg	77.7		40 - 120		
2,4,6-Trichlorophenol	B092737-BS1	LCS	1.2965	1.6835	mg/kg	77.0		50 - 130		
Pyridine	B092737-BS1	LCS	ND	1.6835	mg/kg	3.5		10 - 110		L21
2-Fluorophenol (Surrogate)	B092737-BS1	LCS	0.91442	1.3468	mg/kg	67.9		20 - 130		
Phenol-d5 (Surrogate)	B092737-BS1	LCS	1.0457	1.3468	mg/kg	77.6		30 - 130		
Nitrobenzene-d5 (Surrogate)	B092737-BS1	LCS	1.2667	1.3468	mg/kg	94.1		30 - 130		
2-Fluorobiphenyl (Surrogate)	B092737-BS1	LCS	1.0531	1.3468	mg/kg	78.2		30 - 140		
2,4,6-Tribromophenol (Surrogate)	B092737-BS1	LCS	1.3947	1.3468	mg/kg	104		20 - 150		
p-Terphenyl-d14 (Surrogate)	B092737-BS1	LCS	0.43442	0.67340	mg/kg	64.5		30 - 150		

Caltest Analytical
1885 North Kelly Road
Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Base Neutral and Acid Extractables Organic Analysis (EPA Method 8270C)

Quality Control Report - Precision & Accuracy

Constituent	Type	Source Sample ID	Source Result	Result	Spike Added	Units	RPD	Percent Recovery	Control Limits RPD	Control Limits Percent Recovery	Lab Quals
QC Batch ID: B092737	Used client sample: N										
Acenaphthene	MS	2030363-83	ND	1.2869	1.6447	mg/kg		78.2		30 - 140	
	MSD	2030363-83	ND	1.1714	1.6779	mg/kg	9.4	69.8	30	30 - 140	
1,4-Dichlorobenzene	MS	2030363-83	ND	1.4844	1.6447	mg/kg		90.3		50 - 130	
	MSD	2030363-83	ND	1.3534	1.6779	mg/kg	9.2	80.7	30	50 - 130	
2,4-Dinitrotoluene	MS	2030363-83	ND	1.5928	1.6447	mg/kg		96.8		50 - 130	
	MSD	2030363-83	ND	1.4179	1.6779	mg/kg	11.6	84.5	30	50 - 130	
Hexachlorobenzene	MS	2030363-83	ND	1.7791	1.6447	mg/kg		108		50 - 130	
	MSD	2030363-83	ND	1.6229	1.6779	mg/kg	9.2	96.7	30	50 - 130	
Hexachlorobutadiene	MS	2030363-83	ND	1.7812	1.6447	mg/kg		108		50 - 130	
	MSD	2030363-83	ND	1.6245	1.6779	mg/kg	9.2	96.8	30	50 - 130	
Hexachloroethane	MS	2030363-83	ND	1.6431	1.6447	mg/kg		99.9		50 - 130	
	MSD	2030363-83	ND	1.4940	1.6779	mg/kg	9.5	89.0	30	50 - 130	
Nitrobenzene	MS	2030363-83	ND	1.8034	1.6447	mg/kg		110		30 - 120	
	MSD	2030363-83	ND	1.6308	1.6779	mg/kg	10.1	97.2	30	30 - 120	
N-Nitrosodi-N-propylamine	MS	2030363-83	ND	1.5025	1.6447	mg/kg		91.4		20 - 130	
	MSD	2030363-83	ND	1.3740	1.6779	mg/kg	8.9	81.9	30	20 - 130	
Pyrene	MS	2030363-83	ND	1.2409	1.6447	mg/kg		75.4		40 - 140	
	MSD	2030363-83	ND	1.1757	1.6779	mg/kg	5.4	70.1	30	40 - 140	
1,2,4-Trichlorobenzene	MS	2030363-83	ND	1.6509	1.6447	mg/kg		100		50 - 130	
	MSD	2030363-83	ND	1.5089	1.6779	mg/kg	9.0	89.9	30	50 - 130	
4-Chloro-3-methylphenol	MS	2030363-83	ND	1.4697	1.6447	mg/kg		89.4		50 - 130	
	MSD	2030363-83	ND	1.3202	1.6779	mg/kg	10.7	78.7	30	50 - 130	
2-Chlorophenol	MS	2030363-83	ND	1.1878	1.6447	mg/kg		72.2		50 - 130	
	MSD	2030363-83	ND	1.0814	1.6779	mg/kg	9.4	64.4	30	50 - 130	
2-Methylphenol	MS	2030363-83	ND	1.3194	1.6447	mg/kg		80.2		50 - 130	
	MSD	2030363-83	ND	1.1950	1.6779	mg/kg	9.9	71.2	30	50 - 130	
3- & 4-Methylphenol	MS	2030363-83	ND	2.8059	3.2895	mg/kg		85.3		50 - 130	
	MSD	2030363-83	ND	2.5348	3.3557	mg/kg	10.2	75.5	30	50 - 130	
4-Nitrophenol	MS	2030363-83	ND	1.5122	1.6447	mg/kg		91.9		30 - 140	
	MSD	2030363-83	ND	1.2694	1.6779	mg/kg	17.5	75.7	30	30 - 140	
Pentachlorophenol	MS	2030363-83	ND	0.94719	1.6447	mg/kg		57.6		30 - 130	
	MSD	2030363-83	ND	0.84682	1.6779	mg/kg	11.2	50.5	30	30 - 130	
Phenol	MS	2030363-83	ND	1.2875	1.6447	mg/kg		78.3		40 - 150	
	MSD	2030363-83	ND	1.1750	1.6779	mg/kg	9.1	70.0	30	40 - 150	
2,4,6-Trichlorophenol	MS	2030363-83	ND	1.3369	1.6447	mg/kg		81.3		50 - 130	
	MSD	2030363-83	ND	1.2046	1.6779	mg/kg	10.4	71.8	30	50 - 130	

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Caltest Analytical
1885 North Kelly Road
Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Base Neutral and Acid Extractables Organic Analysis (EPA Method 8270C)

Quality Control Report - Precision & Accuracy

Constituent	Type	Source Sample ID	Source Result	Result	Spike Added	Units	RPD	Percent Recovery	Control Limits RPD	Control Limits Percent Recovery	Lab Quals
QC Batch ID: B092737	Used client sample: N										
Pyridine	MS	**2030363-83**	**ND**	**0.13562**	**1.6447**	**mg/kg**		**8.2**		**10 - 110**	**J,Q03**
	MSD	**2030363-83**	**ND**	**ND**	**1.6779**	**mg/kg**	**83.9**	**3.3**	**30**	**10 - 110**	**Q03**
2-Fluorophenol (Surrogate)	MS	2030363-83	ND	0.93062	1.3158	mg/kg		70.7		20 - 130	
	MSD	2030363-83	ND	0.86210	1.3423	mg/kg	7.6	64.2		20 - 130	
Phenol-d5 (Surrogate)	MS	2030363-83	ND	1.0584	1.3158	mg/kg		80.4		30 - 130	
	MSD	2030363-83	ND	0.96973	1.3423	mg/kg	8.7	72.2		30 - 130	
Nitrobenzene-d5 (Surrogate)	MS	2030363-83	ND	1.2716	1.3158	mg/kg		96.6		30 - 130	
	MSD	2030363-83	ND	1.1671	1.3423	mg/kg	8.6	86.9		30 - 130	
2-Fluorobiphenyl (Surrogate)	MS	2030363-83	ND	1.0934	1.3158	mg/kg		83.1		30 - 140	
	MSD	2030363-83	ND	1.0119	1.3423	mg/kg	7.7	75.4		30 - 140	
2,4,6-Tribromophenol (Surrogate)	MS	2030363-83	ND	1.5106	1.3158	mg/kg		115		20 - 150	
	MSD	2030363-83	ND	1.3667	1.3423	mg/kg	10.0	102		20 - 150	
p-Terphenyl-d14 (Surrogate)	MS	2030363-83	ND	0.43812	0.65789	mg/kg		66.6		30 - 150	
	MSD	2030363-83	ND	0.41560	0.67114	mg/kg	5.3	61.9		30 - 150	

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Caltest Analytical
1885 North Kelly Road
Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Chemical Analysis

Quality Control Report - Method Blank Analysis

Constituent	QC Sample ID	MB Result	Units	PQL	MDL	Lab Quals
QC Batch ID: B092891						
Ammonia as N	**B092891-BLK1**	**5.0600**	**mg/kg**	**20**	**5.0**	**J**
QC Batch ID: B092977						
Nitrate as N	B092977-BLK1	ND	mg/kg	1.0	0.26	
QC Batch ID: B092978						
Nitrite as N	B092978-BLK1	ND	mg/kg	0.50	0.10	
QC Batch ID: B093433						
Total Kjeldahl Nitrogen	B093433-BLK1	ND	mg/kg	40	13	
QC Batch ID: B093434						
Total Phosphorus	B093434-BLK1	ND	mg/kg	10	3.2	

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Caltest Analytical
1885 North Kelly Road
Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Chemical Analysis

Quality Control Report - Laboratory Control Sample

Constituent	QC Sample ID	Type	Result	Spike Level	Units	Percent Recovery	RPD	Control Limits Percent Recovery	Control Limits RPD	Lab Quals
QC Batch ID: B092891										
Ammonia as N	B092891-BS1	LCS	194.63	196.08	mg/kg	99.3		80 - 120		
QC Batch ID: B092977										
Nitrate as N	B092977-BS1	LCS	5.1260	5.0000	mg/kg	103		90 - 110		
QC Batch ID: B092978										
Nitrite as N	B092978-BS1	LCS	0.53410	0.52632	mg/kg	101		90 - 110		
QC Batch ID: B093433										
Total Kjeldahl Nitrogen	B093433-BS1	LCS	404.16	400.00	mg/kg	101		90 - 110		
QC Batch ID: B093434										
Total Phosphorus	B093434-BS1	LCS	202.86	200.00	mg/kg	101		85 - 115		

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Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Chemical Analysis

Quality Control Report - Precision & Accuracy

Constituent	Type	Source Sample ID	Source Result	Result	Spike Added	Units	RPD	Percent Recovery	Control Limits RPD	Control Limits Percent Recovery	Lab Quals
QC Batch ID: B092891	Used client sample: N										
Ammonia as N	DUP	**2033479-01**	**1756.3**	**1625.6**		**mg/kg**	**7.7**		**20**		
	MS	**2033479-01**	**1756.3**	**1896.4**	**196.08**	**mg/kg**		**71.5**		**80 - 120**	**A03**
	MSD	**2033479-01**	**1756.3**	**1913.5**	**192.31**	**mg/kg**	**0.9**	**81.8**	**20**	**80 - 120**	**A07**
QC Batch ID: B092977	Used client sample: N										
Nitrate as N	DUP	2033480-01	1.5700	1.3700		mg/kg	13.6		20		
	MS	2033480-01	1.5700	52.697	50.505	mg/kg		101		80 - 120	
	MSD	2033480-01	1.5700	52.545	50.505	mg/kg	0.3	101	20	80 - 120	
QC Batch ID: B092978	Used client sample: N										
Nitrite as N	DUP	2033480-01	0.14756	0.15629		mg/kg	5.7		20		J
	MS	2033480-01	0.14756	5.2101	5.2632	mg/kg		96.2		80 - 120	
	MSD	2033480-01	0.14756	5.1711	5.2632	mg/kg	0.8	95.4	20	80 - 120	
QC Batch ID: B093433	Used client sample: N										
Total Kjeldahl Nitrogen	DUP	**2033161-01**	**14167**	**13940**		**mg/kg**	**1.6**		**20**		
	MS	**2033161-01**	**14167**	**16057**	**400.00**	**mg/kg**		**472**		**90 - 110**	**A03**
	MSD	**2033161-01**	**14167**	**15768**	**400.00**	**mg/kg**	**1.8**	**400**	**20**	**90 - 110**	**A03**
QC Batch ID: B093434	Used client sample: N										
Total Phosphorus	DUP	**2033161-01**	**6778.4**	**6570.8**		**mg/kg**	**3.1**		**20**		
	MS	**2033161-01**	**6778.4**	**7396.4**	**200.00**	**mg/kg**		**309**		**80 - 120**	**A03**
	MSD	**2033161-01**	**6778.4**	**7579.6**	**200.00**	**mg/kg**	**2.4**	**401**	**20**	**80 - 120**	**A03**

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Caltest Analytical
1885 North Kelly Road
Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Total Concentrations (TTLC)

Quality Control Report - Method Blank Analysis

Constituent	QC Sample ID	MB Result	Units	PQL	MDL	Lab Quals
QC Batch ID: B092496						
Arsenic	B092496-BLK1	ND	mg/kg	1.0	0.40	
Cadmium	B092496-BLK1	ND	mg/kg	0.50	0.052	
Copper	**B092496-BLK1**	**0.20681**	**mg/kg**	**1.0**	**0.050**	**J**
Lead	B092496-BLK1	ND	mg/kg	2.5	0.41	
Molybdenum	B092496-BLK1	ND	mg/kg	2.5	0.050	
Nickel	B092496-BLK1	ND	mg/kg	0.50	0.15	
Selenium	B092496-BLK1	ND	mg/kg	1.0	0.98	
Zinc	**B092496-BLK1**	**0.36305**	**mg/kg**	**2.5**	**0.087**	**J**
Potassium	B092496-BLK1	ND	mg/kg	50	5.0	

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Napa, CA 94558

Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Total Concentrations (TTLC)

Quality Control Report - Laboratory Control Sample

Constituent	QC Sample ID	Type	Result	Spike Level	Units	Percent Recovery	RPD	Control Limits Percent Recovery	Control Limits RPD	Lab Quals
QC Batch ID: B092496										
Arsenic	B092496-BS1	LCS	18.651	20.000	mg/kg	93.3		75 - 125		
Cadmium	B092496-BS1	LCS	9.9687	10.000	mg/kg	99.7		75 - 125		
Copper	B092496-BS1	LCS	103.41	100.00	mg/kg	103		75 - 125		
Lead	B092496-BS1	LCS	102.67	100.00	mg/kg	103		75 - 125		
Molybdenum	B092496-BS1	LCS	104.60	100.00	mg/kg	105		75 - 125		
Nickel	B092496-BS1	LCS	106.64	100.00	mg/kg	107		75 - 125		
Selenium	B092496-BS1	LCS	19.978	20.000	mg/kg	99.9		75 - 125		
Zinc	B092496-BS1	LCS	103.08	100.00	mg/kg	103		75 - 125		
Potassium	B092496-BS1	LCS	486.81	500.00	mg/kg	97.4		75 - 125		

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Reported: 11/30/2020 13:14
Project: Solid Analysis
Project Number: V110347
Project Manager: Carol Battaglia

Total Concentrations (TTLC)

Quality Control Report - Precision & Accuracy

Constituent	Type	Source Sample ID	Source Result	Result	Spike Added	Units	RPD	Percent Recovery	Control Limits RPD	Control Limits Percent Recovery	Lab Quals
QC Batch ID: B092496	Used client sample: N										
Arsenic	DUP	2033040-01	ND	ND		mg/kg			20		
	MS	2033040-01	ND	17.601	20.000	mg/kg		88.0		75 - 125	
	MSD	2033040-01	ND	17.509	20.000	mg/kg	0.5	87.5	20	75 - 125	
Cadmium	DUP	**2033040-01**	**0.067085**	**0.052039**		**mg/kg**	**25.3**		**20**		**J,A02**
	MS	**2033040-01**	**0.067085**	**10.493**	**10.000**	**mg/kg**		**104**		**75 - 125**	
	MSD	**2033040-01**	**0.067085**	**10.047**	**10.000**	**mg/kg**	**4.3**	**99.8**	**20**	**75 - 125**	
Copper	DUP	2033040-01	0.99458	1.0510		mg/kg	5.5		20		
	MS	2033040-01	0.99458	112.19	100.00	mg/kg		111		75 - 125	
	MSD	2033040-01	0.99458	108.03	100.00	mg/kg	3.8	107	20	75 - 125	
Lead	DUP	2033040-01	0.58127	ND		mg/kg			20		
	MS	2033040-01	0.58127	98.335	100.00	mg/kg		97.8		75 - 125	
	MSD	2033040-01	0.58127	94.971	100.00	mg/kg	3.5	94.4	20	75 - 125	
Molybdenum	DUP	**2033040-01**	**0.21263**	**0.13234**		**mg/kg**	**46.6**		**20**		**J,A02**
	MS	**2033040-01**	**0.21263**	**105.45**	**100.00**	**mg/kg**		**105**		**75 - 125**	
	MSD	**2033040-01**	**0.21263**	**100.70**	**100.00**	**mg/kg**	**4.6**	**100**	**20**	**75 - 125**	
Nickel	DUP	2033040-01	0.31820	0.37360		mg/kg	16.0		20		J
	MS	2033040-01	0.31820	109.18	100.00	mg/kg		109		75 - 125	
	MSD	2033040-01	0.31820	103.06	100.00	mg/kg	5.8	103	20	75 - 125	
Selenium	DUP	2033040-01	ND	ND		mg/kg			20		
	MS	2033040-01	ND	23.168	20.000	mg/kg		116		75 - 125	
	MSD	2033040-01	ND	22.948	20.000	mg/kg	1.0	115	20	75 - 125	
Zinc	DUP	**2033040-01**	**60.827**	**15.998**		**mg/kg**	**117**		**20**		**Q01**
	MS	**2033040-01**	**60.827**	**144.14**	**100.00**	**mg/kg**		**83.3**		**75 - 125**	
	MSD	**2033040-01**	**60.827**	**125.82**	**100.00**	**mg/kg**	**13.6**	**65.0**	**20**	**75 - 125**	**Q03**
Potassium	DUP	2033040-01	ND	ND		mg/kg			20		
	MS	2033040-01	ND	548.27	500.00	mg/kg		110		75 - 125	
	MSD	2033040-01	ND	512.44	500.00	mg/kg	6.8	102	20	75 - 125	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.
All results listed in this report are for the exclusive use of the submitting party. BC Laboratories, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

Caltest Analytical 1885 North Kelly Road Napa, CA 94558	**Reported:** 11/30/2020 13:14 Project: Solid Analysis Project Number: V110347 Project Manager: Carol Battaglia

Notes And Definitions

J	Estimated Value (CLP Flag)
MDL	Method Detection Limit
ND	Analyte Not Detected
PQL	Practical Quantitation Limit
A02	The difference between duplicate readings is less than the quantitation limit.
A03	The sample concentration was more than 4 times the spike level.
A07	Detection and quantitation limits were raised due to sample dilution caused by high analyte concentration or matrix interference.
A10	Detection and quantitation limits were raised due to matrix interference.
A20	Surrogate is low due to matrix interference. Interference verified through second extraction/analysis.
L21	The Laboratory Control Sample Soil (LCSS) recovery is not within laboratory established control limits.
Q01	Sample precision is not within the control limits.
Q03	Matrix spike recovery(s) was(were) not within the control limits.

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.
All results listed in this report are for the exclusive use of the submitting party. BC Laboratories, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

Caltest ANALYTICAL LABORATORY

1885 N. KELLY ROAD NAPA, CA 94558 (707) 258-4000 FAX (707) 226-1001 www.caltestlab.com

SAMPLE CHAIN OF CUSTODY

PAGE 1 OF 1

LAB ORDER # V110347

068

PROJECT NAME / PROJECT NUMBER: BioSolids

P.O. #:

CLIENT: LGVSD

REPORT TO: Sahar Golshani

MAILING ADDRESS: 300 Smith Ranch Rd

STATE: CA

ZIP: 94903

BILLING ADDRESS: same

ATTN: Pam Amatori

PHONE #: 415-472-1734 ext.14

FAX PHONE: 415-472-5014

SAMPLER (PRINT & SIGN NAME): SG

TURN-AROUND TIME: [X] STANDARD [ ] RUSH

DUE DATE:

ANALYSED REQUESTED

CALTEST LAB #	DATE SAMPLED	TIME SAMPLED	SAMPLE MATRIX*	CONTAINER TYPE/ AMOUNT**	PRESERVATIVE	SAMPLE IDENTIFICATION / SITE	CLIENT LAB #	COMP. or GRAB	%TS, pH, Ammonia Nitrogen as N	Arsenic, Cadmium, Copper, Lead, Mercury	Molybdenum, Nickel, Selenium, Zinc	Total Nitrogen, Nitrate Nitrogen as N	Cation Exchange Capacity	Salinity (electrical conductivity)	Total Potassium,Total Phosphorus as P	Fecal Coliform MPN/gram (if applicable)	EPA Method 8270 Semi-Volatile Organics	SW 8461 Method 8080 for PCB Aroclors, Aldrin/Dieldrin	REMARKS
	11/6/2020	10:57	Sludge	2L SJ	4°C	Sludge Pond B		Grab	X	X	X	X	~~X~~ *	X	X	~~X~~ *	X	X	* FECAL COLIFORM NOT NEEDED PER CLIENT
																			* CEC ANALYSIS NOT POSSIBLE DUE TO HIGH % MOISTURE IN SAMPLE -GM 11/12/2020

RELINQUISHED BY	DATE/TIME	RECEIVED BY	RELINQUISHED BY	DATE/TIME	RECEIVED BY
	11/6/20 / 13:35			11/06/20 / 15:30	

FOR LAB USE ONLY

Samples: WC ___ MICRO ___ BIO ___ AA ___ SV ___ VOA ___ pH? Y/N TEMP: SEALED: Y/N INTACT: Y/N

BD: BIO ___ WC ___ AA ___

CC: AA ___ SV ___ VOA ___

SIL: HP ___ PT ___ QT ___ VOA ___

W/HNO_3 ___ H_2SO_4 ___ NaOH ___

PIL: HNO_3 ___ H_2SO_4 ___ NaOH ___ HCL ___

COMMENTS:

TEMP 1.9 / 1.6 C

SEALED: Y / N INTACT: Y / N

ON ICE: Y / N

*MATRIX: AQ = Aqueous Nondrinking Water, Digested Metals; FE = Low R.L.s, Aqueous Nondrinking Water, Digested Metals; DW = Drinking Water; SL = Soil Sludge, Solid ; FP = Free Product

**CONTAINER TYPES: AL = Amber Liter; AHL = 500 ml Amber; PT = Pint (Plastic); QT = Quart (Plastic); HG = Half Gallon (Plastic); SJ = Soil Jar; B4 = 4oz. BACT; BT = Brass Tube; VOA = 40mL VOA; OTC - Other Type Container

R ___ PR ___ M ___ F ___

11/05/20

WHITE - LABORATORY YELLOW - CLIENT COPY TO ACCOMPANY FINAL REPORT PINK - CLIENT COPY AS RECEIPT

Attachment B

Las Gallinas Valley Sanitary District Application Area Soils Constituent Analytical Report

Nutrien Precision Agri Lab
24730 Avenue 13
Madera, CA 93637
Office 559-661-6386

Sample Event Map

2020-01-23

Grower	**Las Gallinas Valley Sanitary District**	Farm	**Las Gallinas Valley Sanitary District**	Lab	**Precision Agri Lab**
Field	**APN 155-01-33**			Area	**61.73 a**

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MOORE TWINING ASSOCIATES, INC.
Since 1888
California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

February 10, 2020

Work Order #: **GA27037**

Chad Bethel
Precision Agri-Lab
24730 Avenue 13
Madera, CA 93637

RE: Analytical Services

Enclosed are the analytical results for samples received by our laboratory on **01/27/20** . For your reference, these analyses have been assigned laboratory work order number **GA27037**.

All analyses have been performed according to our laboratory's quality assurance program. All results are intended to be considered in their entirety, Moore Twining Associates, Inc. (MTA) is not responsible for use of less than complete reports. Results apply only to samples analyzed.

If you have any questions, please feel free to contact us at the number listed above.

Sincerely,

Moore Twining Associates, Inc.

Susan Federico
Client Services Representative

MOORE TWINING ASSOCIATES, INC.
California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary Dist. **Project Manager:** Chad Bethel	**Reported:** 02/10/2020

Analytical Report for the Following Samples

Sample ID	Notes	Laboratory ID	Matrix	Date Sampled	Date Received
APN 155-01-33 (1 at 6")		GA27037-01	Soil	01/23/20 00:00	01/27/20 13:30
APN 155-01-33 (2 at 6")		GA27037-02	Soil	01/23/20 00:00	01/27/20 13:30
APN 155-01-33 (3 at 6")		GA27037-03	Soil	01/23/20 00:00	01/27/20 13:30

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California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary Dist. **Project Manager:** Chad Bethel	**Reported:** 02/10/2020

APN 155-01-33 (1 at 6")

GA27037-01 (Soil) Sampled: 01/23/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Metals (Total)									
Arsenic		**8.3**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Cadmium		ND	0.40	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Chromium		**85**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Copper		**26**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Lead		**21**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Mercury		**0.041**	0.040	mg/kg	1	B0A2808	01/29/20	01/29/20	EPA 7471A
Molybdenum		ND	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Nickel		**72**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Selenium		ND	5.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Zinc		**73**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

APN 155-01-33 (2 at 6")

GA27037-02 (Soil) Sampled: 01/23/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Metals (Total)									
Arsenic		**5.8**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Cadmium		ND	0.40	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Chromium		**76**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Copper		**22**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Lead		**53**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Mercury		**0.055**	0.040	mg/kg	1	B0A2808	01/29/20	01/29/20	EPA 7471A
Molybdenum		ND	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Nickel		**83**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Selenium		ND	5.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Zinc		**51**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

APN 155-01-33 (3 at 6")

GA27037-03 (Soil) Sampled: 01/23/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Metals (Total)									
Arsenic		**7.5**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Cadmium		ND	0.40	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Chromium		**85**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Copper		**26**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Lead		**18**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Mercury		**0.050**	0.040	mg/kg	1	B0A2808	01/29/20	01/29/20	EPA 7471A
Molybdenum		ND	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Nickel		**83**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Selenium		ND	5.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Zinc		**65**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

MOORE TWINING ASSOCIATES, INC.
California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary Dist. **Project Manager:** Chad Bethel	**Reported:** 02/10/2020

Notes and Definitions

MS2	Recovery for this analyte was biased high; associated blank spike recoveries are within range.
µg/L	micrograms per liter (parts per billion concentration units)
mg/L	milligrams per liter (parts per million concentration units)
mg/kg	milligrams per kilogram (parts per million concentration units)
ND	Analyte NOT DETECTED at or above the reporting limit
RPD	Relative Percent Difference
	Analysis of pH, filtration, and residual chlorine is to take place immediately after sampling in the field. If the test was performed in the laboratory, the hold time was exceeded. **(for aqueous matrices only)**

PRECISION AGRI LAB
24730 Avenue 13
Madera CA 93637
Phone: (559) 661-6386
FAX: (559) 661-6135

GA27037

SAMPLE SUBMISSION FORM

Chain of Custody

E

CA/AZ/HI/WA/ID

Log #: 401499

Date Sampled: 1-23-20

Date Received: 1-27-20

Dealer/Branch Name: PRECISION AGRI LAB

Dealer/Branch Fieldman: PRECISION AGRI LAB

Grower/Client Name: LAS GALLINAS VALLEY SANITARY DISTRICT

Crop: _____

Variety: _____

Growth Stage: _____

Crop Monitor Program (CMP) _____ Yes _____ No

Recommendations (Soil Only) _____ Yes _____ No

LPI's NutriScription (Tissue Only) _____ Yes _____ No

Dairy Nutrient Management (NMP) _____ Yes _____ No

Sample Type

Soil X

Plant _____

Other _____

Do not mix sample types on the same form, please.

Event Code

Sample ID	Test ID	Sample Description	Time
1	9999	APN 155-01-33 (1 at 6") 5DHH-BD53	
2	9999	APN 155-01-33 (2 at 6")	
3	9999	APN 155-01-33 (3 at 6")	
4			
5			
6			
7			
8			
9			
10			
11			
12			

Metals Test: Arsenic, Cadmium, Chromium, Copper, Lead, Mercury, Molybdenum, Nickel, Selenium & Zinc.

Client's CPS Account No: _____

Our most popular analysis packages are shown below for your convenience. Please view the complete price list for other analytical needs.

Analysis Type	Test ID	Parameters
Petiole	1151	NO3-N,PO4-P,K
Petiole/Blade	1154	NO3-N,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na
Petiole/Blade	1155	NO3-N,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Petiole/Blade (Grape only)	1161	N,NO3-N,P,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Leaf	1251	N,P,K
Leaf	1253	N,P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na
Leaf	1254	N,P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Almond Hull	1306	B
Soil	2154	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,sCa,sMg,sNa,B
Soil	2155	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,sCa,sMg,sNa,B,SO4-S,Cl
Soil	2555	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,xCa,xMg,xNa,B,%BS,CEC, SO4-S,Cl
Soil	2656	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe,ESP, Free Lime,xCa,xMg,xNa,B,%BS,CEC,SO4-S,Cl, sCa,sMg,sK,sNa
Soil	2657	2656 Test + Organic Matter
Water	3051	pH,ECw,SAR,Ca,Mg,Na,HCO3+CO3,NO3-N,B,Cl, SO4-S,Fe,Mn
Water	3053	pH,Buffer/Titration (specify acid)

	Printed Name	Signature	Date	Time
Sampled By	PAL	OMAR		
Relinquished By	Jenna Overmyer		1/27/2020	1:30pm
Shipping Method	UPS ____ FedEx ____ Drop Off ____ Other (Specify) ____			
Received By	Devon Rivera		1-27-20	13:30

For Lab Use Only Below This Line

	Printed Name	Signature	Date	Time
Relinquished By				
Shipping Method	UPS ____ FedEx ____ Drop Off ____ Other (Specify) ____			
Shipped To				

Please email Results to: Jenna.Overmyer@nutrien.com
Please Bill: Nutrien Precision Agri Lab

Sample Integrity

Page 2 of 2

Moore Twining Associates

WO# GA27037

MTA Bottles: Yes or No

Was temperature within range? Chemistry ≤6°C Micro <10°C Temp_____°C	Yes No N/A	Did all bottle labels agree with COC?	Yes No N/A
If samples were taken today, is there evidence that chilling has begun? Recvd_____C°	Yes No N/A	Was a sufficient amount of sample received?	Yes No N/A
Did all bottles arrive unbroken and intact?	Yes No N/A	Were correct containers and preservatives received for the tests requested?	Yes No N/A
Do samples have a hold time <72 hours?	Yes No N/A		

125ml (A) 250ml (B) 1Liter (C) 40ml VOA (V)	F3															
Bacti $Na_2S_2O_3$																
None (P)																
Cr6 Buffer (P) Borate Carbonate Buffer																
HNO_3 (P)																
H_2SO_4 (P)																
NaOH (P)																
NaOH+ZnAc (P)																
Dissolved Oxygen 300ml (P)																
None (AG)																
None (CG) 500ml																
$Na_2S_2O_3$ 250ml (Brown P) 549																
$Na_2S_2O_3$ (AG)																
$Na_2S_2O_3$ (AG)																
Thio/K Citrate																
NH_4Cl (AG) 552																
HCl (AG)																
None (CG) 500ml																
H_3PO_4 (AG)																
Other:																
Plastic Bag																
Low Level Hg/Metals Double Bag																
Client Own	1															
Glass Jar: 125/ 250/ 500																
Soil Tube: Brass/ Steel/ Plastic																
5 g Encore																

Filter or Split		Container	Preservative	Date/Time/Initials
	S P F			
	S P F			
	S P F			
	S P F			
	S P F			

Labeled by: _____ @ _____

Labels checked by: _____ @ 1530

FL-SC-0003-06

Nutrien Precision Agri Lab
24730 Avenue 13
Madera, CA 93637
Office 559-661-6386

Sample Event Map

2020-01-23

Grower	**Las Gallinas Valley Sanitary District**	Farm	**Las Gallinas Valley Sanitary District**	Lab	**Precision Agri Lab**
Field	**APN 155-01-33**			Area	**61.73 a**

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MOORE TWINING ASSOCIATES, INC.
California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

February 11, 2020

Work Order #: **GA27034**

Jenna Overmyer
Precision Agri-Lab
24730 Avenue 13
Madera, CA 93637

RE: Analytical Services

Enclosed are the analytical results for samples received by our laboratory on **01/27/20** . For your reference, these analyses have been assigned laboratory work order number **GA27034**.

All analyses have been performed according to our laboratory's quality assurance program. All results are intended to be considered in their entirety, Moore Twining Associates, Inc. (MTA) is not responsible for use of less than complete reports. Results apply only to samples analyzed.

If you have any questions, please feel free to contact us at the number listed above.

Sincerely,

Moore Twining Associates, Inc.

Julio Morales
Client Services Supervisor

MOORE TWINING ASSOCIATES, INC.
California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary District **Project Manager:** Jenna Overmyer	**Reported:** 02/11/2020

Analytical Report for the Following Samples

Sample ID	Notes	Laboratory ID	Matrix	Date Sampled	Date Received
APN 155-01-33 (1 at 18")		GA27034-01	Soil	01/23/20 00:00	01/27/20 13:30
APN 155-01-33 (2 at 18")		GA27034-02	Soil	01/23/20 00:00	01/27/20 13:30
APN 155-01-33 (3 at 18")		GA27034-03	Soil	01/23/20 00:00	01/27/20 13:30

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California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary District **Project Manager:** Jenna Overmyer	**Reported:** 02/11/2020

APN 155-01-33 (1 at 18")

GA27034-01 (Soil) Sampled: 01/23/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Ammonia as N		**59**	1.0	mg/kg	1	B0A3111	01/31/20	02/01/20	EPA 350.1
Specific Conductance (EC)		**41**	1.0	µS/cm	1	B0B0708	02/08/20	02/08/20	WREP 125 (S-2.30)
Nitrate + Nitrite as N		**5.0**	0.25	mg/kg	1	B0A2801	01/28/20	01/28/20	SM 4500-NO3 F
Nitrite as N		**0.91**	0.050	mg/kg	1	B0A2802	01/28/20	01/28/20	SM 4500-NO2 B
Phosphorus		**260**	10	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-P
Total Kjeldahl Nitrogen		**1600**	40	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-N
Total Nitrogen		**1600**	40	mg/kg	20	[CALC]	02/03/20	02/03/20	(CALC)
Metals (Total)									
Potassium		**4000**	100	mg/kg	1	B0A2703	01/29/20	01/29/20	EPA 6010B

APN 155-01-33 (2 at 18")

GA27034-02 (Soil) Sampled: 01/23/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Ammonia as N		**43**	1.0	mg/kg	1	B0A3111	01/31/20	02/01/20	EPA 350.1
Specific Conductance (EC)		**19**	1.0	µS/cm	1	B0B0708	02/08/20	02/08/20	WREP 125 (S-2.30)
Nitrate + Nitrite as N		**3.5**	0.25	mg/kg	1	B0A2801	01/28/20	01/28/20	SM 4500-NO3 F
Nitrite as N		**0.34**	0.050	mg/kg	1	B0A2802	01/28/20	01/28/20	SM 4500-NO2 B
Phosphorus		**300**	10	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-P
Total Kjeldahl Nitrogen		**980**	40	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-N
Total Nitrogen		**980**	40	mg/kg	20	[CALC]	02/03/20	02/03/20	(CALC)
Metals (Total)									
Potassium		**2700**	100	mg/kg	1	B0A2703	01/29/20	01/29/20	EPA 6010B

APN 155-01-33 (3 at 18")

GA27034-03 (Soil) Sampled: 01/23/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Ammonia as N		**80**	1.0	mg/kg	1	B0A3111	01/31/20	02/01/20	EPA 350.1
Specific Conductance (EC)		**52**	1.0	µS/cm	1	B0B0708	02/08/20	02/08/20	WREP 125 (S-2.30)
Nitrate + Nitrite as N		**1.9**	0.25	mg/kg	1	B0A2801	01/28/20	01/28/20	SM 4500-NO3 F
Nitrite as N		**0.87**	0.050	mg/kg	1	B0A2802	01/28/20	01/28/20	SM 4500-NO2 B
Phosphorus		**300**	10	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-P
Total Kjeldahl Nitrogen		**2000**	40	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-N
Total Nitrogen		**2000**	40	mg/kg	20	[CALC]	02/03/20	02/03/20	(CALC)
Metals (Total)									
Potassium		**3800**	100	mg/kg	1	B0A2703	01/29/20	01/29/20	EPA 6010B

MOORE TWINING ASSOCIATES, INC.
California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary District **Project Manager:** Jenna Overmyer	**Reported:** 02/11/2020

Notes and Definitions

MS1	Recovery for this analyte was affected by matrix.
MS3	Recovery for this analyte was biased low; associated blank spike recoveries are within range.
µg/L	micrograms per liter (parts per billion concentration units)
mg/L	milligrams per liter (parts per million concentration units)
mg/kg	milligrams per kilogram (parts per million concentration units)
ND	Analyte NOT DETECTED at or above the reporting limit
RPD	Relative Percent Difference
	Analysis of pH, filtration, and residual chlorine is to take place immediately after sampling in the field. If the test was performed in the laboratory, the hold time was exceeded. **(for aqueous matrices only)**

Report of Soil Analysis

DELLAVALLE® Laboratory, Inc.
Chemists and Consultants

1910 W. McKinley, Suite 110, Fresno, CA 93728
FAX (559) 268-8174 - (800) 228-9896 - (559) 233-6129

Moore Twining Associates Inc
2527 Fresno Street
Fresno CA 93721
5187
54

ID:

Lab No. 310545
Sampled Date 1/23/2020
Submitted Date 1/28/2020
Submitted by Julio Morales
Reported Date 2/5/2020
Location/Project
Copy To
Fax (559) 268-0740
E-mail (See Fieldsheet)

No.	Description	Saturated Paste % SP	units pH	lbs/ac-6" Lime	meq/100g CEC
	RL--->	0.50	1.0	500	0.1
	NAPT Methods--->	S1.00	S1.10	S2.50	S10.20 mod
	Handbook 60--->				
1	GA27034-01	89	7.0		31.1
2	GA27034-02	90	6.9		31.4
3	GA27034-03	98	7.3		32.9

PRECISION AGRI LAB
24730 Avenue 13
Madera CA 93637
Phone: (559) 661-6386
FAX: (559) 661-6135

GA 27034

SAMPLE SUBMISSION FORM

Chain of Custody

CA/AZ/HI/WA/ID

Log #: 401500

Date Sampled 1-23-20

Date Received 1-27-20

Dealer/Branch Name PRECISION AGRI LAB

Dealer/Branch Fieldman PRECISION AGRI LAB

Grower/Client Name LAS GALLINAS VALLEY SANITARY DISTRICT

Crop

Variety

Growth Stage

Crop Monitor Program (CMP) ____ Yes ____ No

Recommendations (Soil Only) ____ Yes ____ No

LPI's NutriScription (Tissue Only) ____ Yes ____ No

Dairy Nutrient Management (NMP) ____ Yes ____ No

Sample Type

Soil X

Plant

Other

Do not mix sample types on the same form, please.

Client's CPS Account No:

Our most popular analysis packages are shown below for your convenience. Please view the complete price list for other analytical needs.

Analysis Type	Test ID	Parameters
Petiole	1151	NO3-N,PO4-P,K
Petiole/Blade	1154	NO3-N,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na
Petiole/Blade	1155	NO3-N,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Petiole/Blade (Grape only)	1161	N,NO3-N,P,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Leaf	1251	N,P,K
Leaf	1253	N,P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na
Leaf	1254	N,P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Almond Hull	1306	B
Soil	2154	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,sCa,sMg,sNa,B
Soil	2155	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,sCa,sMg,sNa,B,SO4-S,Cl
Soil	2555	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,xCa,xMg,xNa,B,%BS,CEC, SO4-S,Cl
Soil	2656	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe,ESP, Free Lime,xCa,xMg,xNa,B,%BS,CEC,SO4-S,Cl, sCa,sMg,sK,sNa
Soil	2657	2656 Test + Organic Matter
Water	3051	pH,ECw,SAR,Ca,Mg,Na,HCO3+CO3,NO3-N,B,Cl, SO4-S,Fe,Mn
Water	3053	pH,Buffer/Titration (specify acid)

Sample ID	Test ID	Sample Description	Event Code	Time
1	9999	APN 155-01-33 (1 at 18")	99EL-44KH	
2	9999	APN 155-01-33 (2 at 18")		
3	9999	APN 155-01-33 (3 at 18")		
4				
5				
6				
7				
8				
9				
10				
11				
12				

TEST: Total N, NH3-N, NO3-N, PH, CEC, EC, Total P, Total K

	Printed Name	Signature	Date	Time
Sampled By	PAL	OMAR		
Relinquished By	Jenna Overmyer		1/27/2020	1:30pm
Shipping Method	UPS ____ FedEx ____ Drop Off ____ Other (Specify) ____			
Received By	Devon Rivera		1-27-20	13:30

For Lab Use Only Below This Line

	Printed Name	Signature	Date	Time
Relinquished By				
Shipping Method	UPS ____ FedEx ____ Drop Off ____ Other (Specify) ____			
Shipped To				

Please email Results to: Jenna.Overmyer@nutrien.com
Please Bill: Nutrien Precision Agri Lab

Moore Twining Associates

Sample Integrity — Page 2 of 2 — WO# GA27034 — MTA Bottles: ~~Yes~~ or No

Was temperature within range? Chemistry ≤6°C Micro <10°C Temp_____°C	Yes No N/A	Did all bottle labels agree with COC?	Yes No N/A
If samples were taken today, is there evidence that chilling has begun? Recvd_____C°	Yes No N/A	Was a sufficient amount of sample received?	Yes No N/A
Did all bottles arrive unbroken and intact?	Yes No N/A	Were correct containers and preservatives received for the tests requested?	Yes No N/A
Do samples have a hold time <72 hours?	Yes No N/A		
Were there bubbles in VOA vials? (Volatiles Only)	Yes No N/A		
Was PM notified of discrepancies? PM: By/Time:	Yes No N/A		

125ml (A) 250ml (B) 1Liter (C) 40ml VOA (V)		
Bacti $Na_2S_2O_3$		
None (P)		
Cr6 Buffer (P) Borate Carbonate Buffer		
HNO_3 (P)		
H_2SO_4 (P)		
NaOH (P)		
NaOH+ZnAc (P)		
Dissolved Oxygen 300ml (P)		
None (AG)		
None (CG) 500ml		
$Na_2S_2O_3$ 250ml (Brown P) 549		
$Na_2S_2O_3$ (AG)		
$Na_2S_2O_3$ (AG)		
Thio/K Citrate		
NH_4Cl (AG) 552		
HCl (AG)		
None (CG) 500ml		
H_3PO_4 (AG)		
Other:		
Plastic Bag	1	
Low Level Hg/Metals Double Bag		
Client Own	1	
Glass Jar: 125/ 250/ 500		
Soil Tube: Brass/ Steel/ Plastic		
5 g Encore		

Filter or Split		Container	Preservative	Date/Time/Initials
	S P F			
	S P F			
	S P F			
	S P F			
	S P F			

Labeled by: M @ 1504 Labels checked by: M @ 1505

FL-SC-0003-06

Nutrien Precision Agri Lab
24730 Avenue 13
Madera, CA 93637
Office 559-661-6386

Sample Event Map

2020-01-24

Grower	**Las Gallinas Valley Sanitary District**	Farm	**Las Gallinas Valley Sanitary District**	Lab	**Precision Agri Lab**
Field	**APN 155-01-13**			Area	**104.9 a**

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MOORE TWINING ASSOCIATES, INC.
Since 1888
California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

February 11, 2020

Work Order #: **GA27036**

Jenna Overmyer
Precision Agri-Lab
24730 Avenue 13
Madera, CA 93637

RE: Analytical Services

Enclosed are the analytical results for samples received by our laboratory on **01/27/20** . For your reference, these analyses have been assigned laboratory work order number **GA27036**.

All analyses have been performed according to our laboratory's quality assurance program. All results are intended to be considered in their entirety, Moore Twining Associates, Inc. (MTA) is not responsible for use of less than complete reports. Results apply only to samples analyzed.

If you have any questions, please feel free to contact us at the number listed above.

Sincerely,

Moore Twining Associates, Inc.

Julio Morales
Client Services Supervisor

MOORE TWINING ASSOCIATES, INC.
California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary Dist. **Project Manager:** Jenna Overmyer	**Reported:** 02/11/2020

Analytical Report for the Following Samples

Sample ID	Notes	Laboratory ID	Matrix	Date Sampled	Date Received
APN 155-01-13 (1 at 18")		GA27036-01	Soil	01/24/20 00:00	01/27/20 15:20
APN 155-01-13 (2 at 18")		GA27036-02	Soil	01/24/20 00:00	01/27/20 15:20
APN 155-01-13 (3 at 18")		GA27036-03	Soil	01/24/20 00:00	01/27/20 15:20
APN 155-01-13 (4 at 18")		GA27036-04	Soil	01/24/20 00:00	01/27/20 15:20
APN 155-01-13 (5 at 18")		GA27036-05	Soil	01/24/20 00:00	01/27/20 15:20

MOORE TWINING ASSOCIATES, INC.
California ELAP Certificate #1371

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Fresno, CA 93721
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Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary Dist. **Project Manager:** Jenna Overmyer	**Reported:** 02/11/2020

APN 155-01-13 (1 at 18")

GA27036-01 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Ammonia as N		**40**	1.0	mg/kg	1	B0A3111	01/31/20	02/01/20	EPA 350.1
Specific Conductance (EC)		**280**	1.0	µS/cm	1	B0B0708	02/08/20	02/08/20	WREP 125 (S-2.30)
Nitrate + Nitrite as N		**4.1**	0.25	mg/kg	1	B0A2801	01/28/20	01/28/20	SM 4500-NO3 F
Nitrite as N		**1.6**	0.050	mg/kg	1	B0A2802	01/28/20	01/28/20	SM 4500-NO2 B
Phosphorus		**540**	10	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-P
Total Kjeldahl Nitrogen		**950**	40	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-N
Total Nitrogen		**960**	40	mg/kg	20	[CALC]	02/03/20	02/03/20	(CALC)
Metals (Total)									
Potassium		**2700**	100	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

APN 155-01-13 (2 at 18")

GA27036-02 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Ammonia as N		**65**	1.0	mg/kg	1	B0A3111	01/31/20	02/01/20	EPA 350.1
Specific Conductance (EC)		**110**	1.0	µS/cm	1	B0B0708	02/08/20	02/08/20	WREP 125 (S-2.30)
Nitrate + Nitrite as N		**2.5**	0.25	mg/kg	1	B0A2801	01/28/20	01/28/20	SM 4500-NO3 F
Nitrite as N		**1.3**	0.050	mg/kg	1	B0A2802	01/28/20	01/28/20	SM 4500-NO2 B
Phosphorus		**550**	10	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-P
Total Kjeldahl Nitrogen		**1600**	40	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-N
Total Nitrogen		**1600**	40	mg/kg	20	[CALC]	02/03/20	02/03/20	(CALC)
Metals (Total)									
Potassium		**2900**	100	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

APN 155-01-13 (3 at 18")

GA27036-03 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Ammonia as N		**62**	1.0	mg/kg	1	B0A3111	01/31/20	02/01/20	EPA 350.1
Specific Conductance (EC)		**170**	1.0	µS/cm	1	B0B0708	02/08/20	02/08/20	WREP 125 (S-2.30)
Nitrate + Nitrite as N		**2.9**	0.25	mg/kg	1	B0A2801	01/28/20	01/28/20	SM 4500-NO3 F
Nitrite as N		**1.6**	0.050	mg/kg	1	B0A2802	01/28/20	01/28/20	SM 4500-NO2 B
Phosphorus		**480**	10	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-P
Total Kjeldahl Nitrogen		**1300**	40	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-N
Total Nitrogen		**1300**	40	mg/kg	20	[CALC]	02/03/20	02/03/20	(CALC)
Metals (Total)									
Potassium		**2800**	100	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

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California ELAP Certificate #1371

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Fresno, CA 93721
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(559) 268-0740 Fax

Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary Dist. **Project Manager:** Jenna Overmyer	**Reported:** 02/11/2020

APN 155-01-13 (4 at 18")

GA27036-04 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Ammonia as N		**59**	1.0	mg/kg	1	B0A3111	01/31/20	02/01/20	EPA 350.1
Specific Conductance (EC)		**420**	1.0	µS/cm	1	B0B0708	02/08/20	02/08/20	WREP 125 (S-2.30)
Nitrate + Nitrite as N		**3.0**	0.25	mg/kg	1	B0A2801	01/28/20	01/28/20	SM 4500-NO3 F
Nitrite as N		**1.6**	0.050	mg/kg	1	B0A2802	01/28/20	01/28/20	SM 4500-NO2 B
Phosphorus		**550**	10	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-P
Total Kjeldahl Nitrogen		**1400**	40	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-N
Total Nitrogen		**1400**	40	mg/kg	20	[CALC]	02/03/20	02/03/20	(CALC)
Metals (Total)									
Potassium		**2600**	100	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

APN 155-01-13 (5 at 18")

GA27036-05 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Ammonia as N		**57**	1.0	mg/kg	1	B0A3111	01/31/20	02/01/20	EPA 350.1
Specific Conductance (EC)		**230**	1.0	µS/cm	1	B0B0708	02/08/20	02/08/20	WREP 125 (S-2.30)
Nitrate + Nitrite as N		**2.4**	0.25	mg/kg	1	B0A2801	01/28/20	01/28/20	SM 4500-NO3 F
Nitrite as N		**1.3**	0.050	mg/kg	1	B0A2802	01/28/20	01/28/20	SM 4500-NO2 B
Phosphorus		**390**	10	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-P
Total Kjeldahl Nitrogen		**1200**	40	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-N
Total Nitrogen		**1200**	40	mg/kg	20	[CALC]	02/03/20	02/03/20	(CALC)
Metals (Total)									
Potassium		**2500**	100	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

Notes and Definitions

MS1	Recovery for this analyte was affected by matrix.
MS3	Recovery for this analyte was biased low; associated blank spike recoveries are within range.
µg/L	micrograms per liter (parts per billion concentration units)
mg/L	milligrams per liter (parts per million concentration units)
mg/kg	milligrams per kilogram (parts per million concentration units)
ND	Analyte NOT DETECTED at or above the reporting limit
RPD	Relative Percent Difference

Analysis of pH, filtration, and residual chlorine is to take place immediately after sampling in the field.
If the test was performed in the laboratory, the hold time was exceeded. **(for aqueous matrices only)**

Report of Soil Analysis

DELLAVALLE® Laboratory, Inc.
Chemists and Consultants

1910 W. McKinley, Suite 110, Fresno, CA 93728
FAX (559) 268-8174 - (800) 228-9896 - (559) 233-6129

Moore Twining Associates Inc
2527 Fresno Street
Fresno CA 93721
5187
54

ID:

Lab No. 310545
Sampled Date 1/23/2020
Submitted Date 1/28/2020
Submitted by Julio Morales
Reported Date 2/5/2020
Location/Project
Copy To
Fax (559) 268-0740
E-mail (See Fieldsheet)

No.	Description	Saturated Paste % SP	units pH	lbs/ac-6" Lime	meq/100g CEC
	RL--->	0.50	1.0	500	0.1
	NAPT Methods--->	S1.00	S1.10	S2.50	S10.20 mod
	Handbook 60--->				
4	GA27036-01	91	7.4		31.6
5	GA27036-02	89	7.0		30.6
6	GA27036-03	101	6.9		29.0
7	GA27036-04	95	4.6	19800	26.4
8	GA27036-05	41	5.6	10900	10.0

PRECISION AGRI LAB
24730 Avenue 13
Madera CA 93637
Phone: (559) 661-6386
FAX: (559) 661-6135

GA27036

SAMPLE SUBMISSION FORM

Chain of Custody

CA/AZ/HI/WA/ID

Log #: 401501

Date Sampled: 1-24-20

Date Received: 1-27-20

Dealer/Branch Name: PRECISION AGRI LAB

Dealer/Branch Fieldman: PRECISION AGRI LAB

Grower/Client Name: LAS GALLINAS VALLEY SANITARY DISTRICT

Crop: _____

Variety: _____

Growth Stage: _____

Crop Monitor Program (CMP) _____ Yes _____ No

Recommendations (Soil Only) _____ Yes _____ No

LPI's NutriScription (Tissue Only) _____ Yes _____ No

Dairy Nutrient Management (NMP) _____ Yes _____ No

Sample Type

Soil X

Plant _____

Other _____

Do not mix sample types on the same form, please.

Client's CPS Account No: _____

Sample ID	Test ID	Sample Description	Event Code	Time
1	9999	APN 155-01-13 (1 at 18")	876X-JRBP	
2	9999	APN 155-01-13 (2 at 18")		
3	9999	APN 155-01-13 (3 at 18")		
4	9999	APN 155-01-13 (4 at 18")		
5	9999	APN 155-01-13 (5 at 18")		
6				
7				
8				
9				
10				
11				
12				

Test: Total N, NH3-N, NO3-N, PH, CEC, EC, Total P, Total K

Our most popular analysis packages are shown below for your convenience. Please view the complete price list for other analytical needs.

Analysis Type	Test ID	Parameters
Petiole	1151	NO3-N,PO4-P,K
Petiole/Blade	1154	NO3-N,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na
Petiole/Blade	1155	NO3-N,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Petiole/Blade (Grape only)	1161	N,NO3-N,P,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Leaf	1251	N,P,K
Leaf	1253	N,P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na
Leaf	1254	N,P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Almond Hull	1306	B
Soil	2154	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,sCa,sMg,sNa,B
Soil	2155	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,sCa,sMg,sNa,B,SO4-S,Cl
Soil	2555	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,xCa,xMg,xNa,B,%BS,CEC, SO4-S,Cl
Soil	2656	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe,ESP, Free Lime,xCa,xMg,xNa,B,%BS,CEC,SO4-S,Cl, sCa,sMg,sK,sNa
Soil	2657	2656 Test + Organic Matter
Water	3051	pH,ECw,SAR,Ca,Mg,Na,HCO3+CO3,NO3-N,B,Cl, SO4-S,Fe,Mn
Water	3053	pH,Buffer/Titration (specify acid)

	Printed Name	Signature	Date	Time
Sampled By	PAL	OMAR		
Relinquished By	Jenna	Overmyer	1/27/2020	1:30pm
Shipping Method	UPS ____ FedEx ____ Drop Off ____ Other (Specify) _____			
Received By	Devon Rivas	[illegible]	1-27-20	13:30

For Lab Use Only Below This Line

	Printed Name	Signature	Date	Time
Relinquished By				
Shipping Method	UPS ____ FedEx ____ Drop Off ____ Other (Specify) _____			
Shipped To				

Please email Results to: Jenna.Overmyer@nutrien.com
Please Bill: Nutrien Precision Agri lab

Sample Integrity

Page 2 of 2

Moore Twining Associates

WO# GA27036

MTA Bottles: Yes or No

Was temperature within range? Chemistry ≤6°C Micro <10°C Temp_____°C	Yes No N/A	Did all bottle labels agree with COC?	Yes No N/A
If samples were taken today, is there evidence that chilling has begun? Recvd_____C°	Yes No N/A	Was a sufficient amount of sample received?	Yes No N/A
Did all bottles arrive unbroken and intact?	Yes No N/A	Were correct containers and preservatives received for the tests requested?	Yes No N/A
Do samples have a hold time <72 hours?	Yes No N/A		

Were there bubbles in VOA vials? (Volatiles Only)	Yes No N/A
Was PM notified of discrepancies? PM: By/Time:	Yes No N/A

125ml (A) 250ml (B) 1Liter (C) 40ml VOA (V)	1-5														
Bacti $Na_2S_2O_3$															
None (P)															
Cr6 Buffer (P) Borate Carbonate Buffer															
HNO_3 (P)															
H_2SO_4 (P)															
NaOH (P)															
NaOH+ZnAc (P)															
Dissolved Oxygen 300ml (P)															
None (AG)															
None (CG) 500ml															
$Na_2S_2O_3$ 250ml (Brown P) 549															
$Na_2S_2O_3$ (AG)															
$Na_2S_2O_3$ (AG)															
Thio/K Citrate															
NH_4Cl (AG) 552															
HCl (AG)															
None (CG) 500ml															
H_3PO_4 (AG)															
Other:															
Plastic Bag	1														
Low Level Hg/Metals Double Bag															
Client Own	1														
Glass Jar: 125/ 250/ 500															
Soil Tube: Brass/ Steel/ Plastic															
5 g Encore															

Filter or Split	Container	Preservative	Date/Time/Initials
S P F	Plastic Bag	0	1-27-20 1530 MG
S P F			
S P F			
S P F			
S P F			

Labeled by: MG @_____

Labels checked by: MG @ 1530

FL-SC-0003-06

Nutrien Precision Agri Lab
24730 Avenue 13
Madera, CA 93637
Office 559-661-6386

Sample Event Map

2020-01-24

Grower	**Las Gallinas Valley Sanitary District**	Farm	**Las Gallinas Valley Sanitary District**	Lab	**Precision Agri Lab**
Field	**APN 155-01-13**			Area	**104.9 a**

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MOORE TWINING ASSOCIATES, INC.
California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

February 10, 2020

Work Order #: **GA27035**

Chad Bethel
Precision Agri-Lab
24730 Avenue 13
Madera, CA 93637

RE: Analytical Services

Enclosed are the analytical results for samples received by our laboratory on **01/27/20** . For your reference, these analyses have been assigned laboratory work order number **GA27035**.

All analyses have been performed according to our laboratory's quality assurance program. All results are intended to be considered in their entirety, Moore Twining Associates, Inc. (MTA) is not responsible for use of less than complete reports. Results apply only to samples analyzed.

If you have any questions, please feel free to contact us at the number listed above.

Sincerely,

Moore Twining Associates, Inc.

Susan Federico
Client Services Representative

MOORE TWINING ASSOCIATES, INC.
California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary District **Project Manager:** Chad Bethel	**Reported:** 02/10/2020

Analytical Report for the Following Samples

Sample ID	Notes	Laboratory ID	Matrix	Date Sampled	Date Received
APN 155-01-13 (1 at 6")		GA27035-01	Soil	01/24/20 00:00	01/27/20 13:30
APN 155-01-13 (2 at 6")		GA27035-02	Soil	01/24/20 00:00	01/27/20 13:30
APN 155-01-13 (3 at 6")		GA27035-03	Soil	01/24/20 00:00	01/27/20 13:30
APN 155-01-13 (4 at 6")		GA27035-04	Soil	01/24/20 00:00	01/27/20 13:30
APN 155-01-13 (5 at 6")		GA27035-05	Soil	01/24/20 00:00	01/27/20 13:30

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California ELAP Certificate #1371

2527 Fresno Street
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Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary District **Project Manager:** Chad Bethel	**Reported:** 02/10/2020

APN 155-01-13 (1 at 6")

GA27035-01 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Metals (Total)									
Arsenic		**11**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Cadmium		ND	0.40	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Chromium		**73**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Copper		**32**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Lead		**14**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Mercury		**0.16**	0.040	mg/kg	1	B0A2808	01/29/20	01/29/20	EPA 7471A
Molybdenum		ND	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Nickel		**51**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Selenium		ND	5.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Zinc		**56**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

APN 155-01-13 (2 at 6")

GA27035-02 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Metals (Total)									
Arsenic		**8.4**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Cadmium		ND	0.40	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Chromium		**67**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Copper		**29**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Lead		**12**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Mercury		**0.13**	0.040	mg/kg	1	B0A2808	01/29/20	01/29/20	EPA 7471A
Molybdenum		ND	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Nickel		**46**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Selenium		ND	5.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Zinc		**51**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

APN 155-01-13 (3 at 6")

GA27035-03 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Metals (Total)									
Arsenic		**9.7**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Cadmium		ND	0.40	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Chromium		**69**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Copper		**29**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Lead		**12**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Mercury		**0.13**	0.040	mg/kg	1	B0A2808	01/29/20	01/29/20	EPA 7471A
Molybdenum		ND	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Nickel		**47**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Selenium		ND	5.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Zinc		**52**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

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California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary District **Project Manager:** Chad Bethel	**Reported:** 02/10/2020

APN 155-01-13 (4 at 6")

GA27035-04 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Metals (Total)									
Arsenic		**10**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Cadmium		ND	0.40	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Chromium		**72**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Copper		**32**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Lead		**13**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Mercury		**0.13**	0.040	mg/kg	1	B0A2808	01/29/20	01/29/20	EPA 7471A
Molybdenum		ND	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Nickel		**50**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Selenium		ND	5.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Zinc		**56**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

APN 155-01-13 (5 at 6")

GA27035-05 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Metals (Total)									
Arsenic		**11**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Cadmium		ND	0.40	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Chromium		**67**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Copper		**29**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Lead		**14**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Mercury		**0.12**	0.040	mg/kg	1	B0A2808	01/29/20	01/29/20	EPA 7471A
Molybdenum		ND	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Nickel		**45**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Selenium		ND	5.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Zinc		**51**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

Notes and Definitions

MS2	Recovery for this analyte was biased high; associated blank spike recoveries are within range.
µg/L	micrograms per liter (parts per billion concentration units)
mg/L	milligrams per liter (parts per million concentration units)
mg/kg	milligrams per kilogram (parts per million concentration units)
ND	Analyte NOT DETECTED at or above the reporting limit
RPD	Relative Percent Difference

Analysis of pH, filtration, and residual chlorine is to take place immediately after sampling in the field.
If the test was performed in the laboratory, the hold time was exceeded. **(for aqueous matrices only)**

PRECISION AGRI LAB
24730 Avenue 13
Madera CA 93637
Phone: (559) 661-6386
FAX: (559) 661-6135

GA27035

SAMPLE SUBMISSION FORM

Chain of Custody

CA/AZ/HI/WA/ID

401502

Date Sampled 1-24-20

Date Received 1-27-20

Dealer/Branch Name PRECISION AGRI LAB

Dealer/Branch Fieldman PRECISION AGRI LAB

Grower/Client Name LAS GALLINAS VALLEY SANITARY DISTRICT

Crop Monitor Program (CMP) _____ Yes _____ No

Crop _____

Recommendations (Soil Only) _____ Yes _____ No

Variety _____

LPI's NutriScription (Tissue Only) _____ Yes _____ No

Growth Stage _____

Dairy Nutrient Management (NMP) _____ Yes _____ No

Sample Type

Soil X

Plant _____

Other _____

Do not mix sample types on the same form, please.

Client's CPS Account No: _____

Sample ID	Test ID	Sample Description	Event Code	Time
1	9999	APN 155-01-13 (1 at 6")	5E7W-CY9H	
2	9999	APN 155-01-13 (2 at 6")		
3	9999	APN 155-01-13 (3 at 6")		
4	9999	APN 155-01-13 (4 at 6")		
5	9999	APN 155-01-13 (5 at 6")		
6				
7				
8				
9				
10				
11				
12				

Metals Test: Arsenic, Cadmium, Chromium, Copper, Lead, Mercury, Molybdenum, Nickel, Selenium & Zinc

Our most popular analysis packages are shown below for your convenience. Please view the complete price list for other analytical needs.

Analysis Type	Test ID	Parameters
Petiole	1151	NO3-N,PO4-P,K
Petiole/Blade	1154	NO3-N,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na
Petiole/Blade	1155	NO3-N,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Petiole/Blade (Grape only)	1161	N,NO3-N,P,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Leaf	1251	N,P,K
Leaf	1253	N,P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na
Leaf	1254	N,P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Almond Hull	1306	B
Soil	2154	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,sCa,sMg,sNa,B
Soil	2155	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,sCa,sMg,sNa,B,SO4-S,Cl
Soil	2555	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,xCa,xMg,xNa,B,%BS,CEC, SO4-S,Cl
Soil	2656	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe,ESP, Free Lime,xCa,xMg,xNa,B,%BS,CEC,SO4-S,Cl, sCa,sMg,sK,sNa
Soil	2657	2656 Test + Organic Matter
Water	3051	pH,ECw,SAR,Ca,Mg,Na,HCO3+CO3,NO3-N,B,Cl, SO4-S,Fe,Mn
Water	3053	pH,Buffer/Titration (specify acid)

	Printed Name	Signature	Date	Time
Sampled By	PAL	OMAR		
Relinquished By	Jenna Overmyer		1/27/2020	1:30 pm
Shipping Method	UPS ____ FedEx ____ Drop Off ____ Other (Specify) ____			
Received By	Devon Rivera	[illegible]	1-27-20	13:30

For Lab Use Only Below This Line

	Printed Name	Signature	Date	Time
Relinquished By				
Shipping Method	UPS ____ FedEx ____ Drop Off ____ Other (Specify) ____			
Shipped To				

Please email Results to: Jenna.Overmyer@nutrien.com
Please Bill: Nutrien Precision Agri Lab

Moore Twining Associates

Sample Integrity — Page 2 of 2 — WO# GA27035 — MTA Bottles: Yes or No

Was temperature within range? Chemistry ≤6°C Micro <10°C Temp_____°C	Yes No N/A	Did all bottle labels agree with COC?	Yes No N/A
If samples were taken today, is there evidence that chilling has begun? Recvd_____C°	Yes No N/A	Was a sufficient amount of sample received?	Yes No N/A
Did all bottles arrive unbroken and intact?	Yes No N/A	Were correct containers and preservatives received for the tests requested?	Yes No N/A
Do samples have a hold time <72 hours?	Yes No N/A		

Were there bubbles in VOA vials? (Volatiles Only)	Yes No N/A															
Was PM notified of discrepancies? PM: By/Time:	Yes No N/A															
125ml (A) 250ml (B) 1Liter (C) 40ml VOA (V)																
Bacti $Na_2S_2O_3$																
None (P)																
CrG Buffer (P) Borate Carbonate Buffer																
HNO_3 (P)																
H_2SO_4 (P)																
NaOH (P)																
NaOH+ZnAc (P)																
Dissolved Oxygen 300ml (P)																
None (AG)																
None (CG) 500ml																
$Na_2S_2O_3$ 250ml (Brown P) 549																
$Na_2S_2O_3$ (AG)																
$Na_2S_2O_3$ (AG)																
Thio/K Citrate																
NH_4Cl (AG) 552																
HCl (AG)																
None (CG) 500ml																
H_3PO_4 (AG)																
Other:																
Plastic Bag																
Low Level Hg/Metals Double Bag																
Client Own																
Glass Jar: 125/ 250/ 500																
Soil Tube: Brass/ Steel/ Plastic																
5 g Encore																

Filter or Split		Container	Preservative	Date/Time/Initials
	S P F			
	S P F			
	S P F			
	S P F			
	S P F			

Labeled by: M @_____ Labels checked by: YB @ 1530

FL-SC-0003-06

Nutrien Precision Agri Lab
24730 Avenue 13
Madera, CA 93637
Office 559-661-6386

Sample Event Map

2020-01-24

Grower	**Las Gallinas Valley Sanitary District**	Farm	**Las Gallinas Valley Sanitary District**	Lab	**Precision Agri Lab**
Field	**APN 155-01-14**			Area	**118 a**

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MOORE TWINING ASSOCIATES, INC.
Since 1888
California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

February 10, 2020

Work Order #: **GA27033**

Chad Bethel
Precision Agri-Lab
24730 Avenue 13
Madera, CA 93637

RE: Analytical Services

Enclosed are the analytical results for samples received by our laboratory on **01/27/20** . For your reference, these analyses have been assigned laboratory work order number **GA27033**.

All analyses have been performed according to our laboratory's quality assurance program. All results are intended to be considered in their entirety, Moore Twining Associates, Inc. (MTA) is not responsible for use of less than complete reports. Results apply only to samples analyzed.

If you have any questions, please feel free to contact us at the number listed above.

Sincerely,

Moore Twining Associates, Inc.

Susan Federico
Client Services Representative

MOORE TWINING ASSOCIATES, INC.
California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary District **Project Manager:** Chad Bethel	**Reported:** 02/10/2020

Analytical Report for the Following Samples

Sample ID	Notes	Laboratory ID	Matrix	Date Sampled	Date Received
APN 155-01-14 (1 at 6")		GA27033-01	Soil	01/24/20 00:00	01/27/20 13:30
APN 155-01-14 (2 at 6")		GA27033-02	Soil	01/24/20 00:00	01/27/20 13:30
APN 155-01-14 (3 at 6")		GA27033-03	Soil	01/24/20 00:00	01/27/20 13:30
APN 155-01-14 (4 at 6")		GA27033-04	Soil	01/24/20 00:00	01/27/20 13:30
APN 155-01-14 (5 at 6")		GA27033-05	Soil	01/24/20 00:00	01/27/20 13:30
APN 155-01-14 (6 at 6")		GA27033-06	Soil	01/24/20 00:00	01/27/20 13:30

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2527 Fresno Street
Fresno, CA 93721
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Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary District **Project Manager:** Chad Bethel	**Reported:** 02/10/2020

APN 155-01-14 (1 at 6")

GA27033-01 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Metals (Total)									
Arsenic		**9.3**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Cadmium		ND	0.40	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Chromium		**66**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Copper		**29**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Lead		**14**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Mercury		**0.13**	0.040	mg/kg	1	B0A2808	01/29/20	01/29/20	EPA 7471A
Molybdenum		ND	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Nickel		**48**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Selenium		ND	5.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Zinc		**53**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

APN 155-01-14 (2 at 6")

GA27033-02 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Metals (Total)									
Arsenic		**12**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Cadmium		ND	0.40	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Chromium		**69**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Copper		**31**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Lead		**13**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Mercury		**0.14**	0.040	mg/kg	1	B0A2808	01/29/20	01/29/20	EPA 7471A
Molybdenum		ND	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Nickel		**47**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Selenium		ND	5.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Zinc		**54**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

APN 155-01-14 (3 at 6")

GA27033-03 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Metals (Total)									
Arsenic		**11**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Cadmium		ND	0.40	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Chromium		**71**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Copper		**30**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Lead		**13**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Mercury		**0.11**	0.040	mg/kg	1	B0A2808	01/29/20	01/29/20	EPA 7471A
Molybdenum		ND	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Nickel		**50**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Selenium		ND	5.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Zinc		**56**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

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California ELAP Certificate #1371

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Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary District **Project Manager:** Chad Bethel	**Reported:** 02/10/2020

APN 155-01-14 (4 at 6")

GA27033-04 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Metals (Total)									
Arsenic		**8.9**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Cadmium		ND	0.40	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Chromium		**63**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Copper		**28**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Lead		**11**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Mercury		**0.12**	0.040	mg/kg	1	B0A2808	01/29/20	01/29/20	EPA 7471A
Molybdenum		ND	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Nickel		**43**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Selenium		ND	5.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Zinc		**50**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

APN 155-01-14 (5 at 6")

GA27033-05 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Metals (Total)									
Arsenic		**10**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Cadmium		ND	0.40	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Chromium		**70**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Copper		**30**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Lead		**13**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Mercury		**0.12**	0.040	mg/kg	1	B0A2808	01/29/20	01/29/20	EPA 7471A
Molybdenum		ND	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Nickel		**50**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Selenium		ND	5.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Zinc		**57**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

APN 155-01-14 (6 at 6")

GA27033-06 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Metals (Total)									
Arsenic		**12**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Cadmium		ND	0.40	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Chromium		**78**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Copper		**32**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Lead		**15**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Mercury		**0.11**	0.040	mg/kg	1	B0A2808	01/29/20	01/29/20	EPA 7471A
Molybdenum		ND	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Nickel		**49**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Selenium		ND	5.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B
Zinc		**56**	2.0	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

MOORE TWINING ASSOCIATES, INC.
California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary District **Project Manager:** Chad Bethel	**Reported:** 02/10/2020

Notes and Definitions

MS2	Recovery for this analyte was biased high; associated blank spike recoveries are within range.
µg/L	micrograms per liter (parts per billion concentration units)
mg/L	milligrams per liter (parts per million concentration units)
mg/kg	milligrams per kilogram (parts per million concentration units)
ND	Analyte NOT DETECTED at or above the reporting limit
RPD	Relative Percent Difference
	Analysis of pH, filtration, and residual chlorine is to take place immediately after sampling in the field. If the test was performed in the laboratory, the hold time was exceeded. **(for aqueous matrices only)**

PRECISION AGRI LAB
24730 Avenue 13
Madera CA 93637
Phone: (559) 661-6386
FAX: (559) 661-6135

GAZ7033

SAMPLE SUBMISSION FORM

Chain of Custody

CA/AZ/HI/WA/ID

Log #: 401503

Date Sampled: 1-24-20

Date Received: 1-27-20

Dealer/Branch Name: PRECISION AGRI LAB

Dealer/Branch Fieldman: PRECISION AGRI LAB

Grower/Client Name: LAS GALLINAS VALLEY SANITARY DISTRICT

Crop: _____

Variety: _____

Growth Stage: _____

Crop Monitor Program (CMP) _____ Yes _____ No

Recommendations (Soil Only) _____ Yes _____ No

LPI's NutriScription (Tissue Only) _____ Yes _____ No

Dairy Nutrient Management (NMP) _____ Yes _____ No

Sample Type

Soil X

Plant _____

Other _____

Do not mix sample types on the same form, please.

Client's CPS Account No: _____

Sample ID	Test ID	Sample Description	Event Code	Time
1	9999	APN 155-01-14 (1 at 6")	EJES-CPTA	
2	9999	APN 155-01-14 (2 at 6")		
3	9999	APN 155-01-14 (3 at 6")		
4	9999	APN 155-01-14 (4 at 6")		
5	9999	APN 155-01-14 (5 at 6")		
6	9999	APN 155-01-14 (6 at 6")		
7				
8				
9				
10				
11				
12				

Metals Test: Arsenic, Cadmium, Chromium, Copper, Lead, Mercury, Molybdenum, Nickel, Selenium & Zinc

Our most popular analysis packages are shown below for your convenience. Please view the complete price list for other analytical needs.

Analysis Type	Test ID	Parameters
Petiole	1151	NO3-N,PO4-P,K
Petiole/Blade	1154	NO3-N,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na
Petiole/Blade	1155	NO3-N,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Petiole/Blade (Grape only)	1161	N,NO3-N,P,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Leaf	1251	N,P,K
Leaf	1253	N,P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na
Leaf	1254	N,P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Almond Hull	1306	B
Soil	2154	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,sCa,sMg,sNa,B
Soil	2155	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,sCa,sMg,sNa,B,SO4-S,Cl
Soil	2555	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,xCa,xMg,xNa,B,%BS,CEC, SO4-S,Cl
Soil	2656	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe,ESP, Free Lime,xCa,xMg,xNa,B,%BS,CEC,SO4-S,Cl, sCa,sMg,sK,sNa
Soil	2657	2656 Test + Organic Matter
Water	3051	pH,ECw,SAR,Ca,Mg,Na,HCO3+CO3,NO3-N,B,Cl, SO4-S,Fe,Mn
Water	3053	pH,Buffer/Titration (specify acid)

	Printed Name	Signature	Date	Time
Sampled By	PAL	OMAR		
Relinquished By	Jenna Overmyer		1/27/2020	1:30pm
Shipping Method	UPS ____ FedEx ____ Drop Off ____ Other (Specify) ____			
Received By	Devon Rivera	[illegible]	1-27-20	13:30

For Lab Use Only Below This Line

	Printed Name	Signature	Date	Time
Relinquished By				
Shipping Method	UPS ____ FedEx ____ Drop Off ____ Other (Specify) ____			
Shipped To				

Please email Results to: Jenna.Overmyer@nutrien.com
Please Bill: Nutrien Precision Agri lab

Sample Integrity

Page 2 of 2

Moore Twining Associates

WO# GA27033

MTA Bottles: Yes or No

Was temperature within range? Chemistry ≤6°C Micro <10°C Temp_____°C	Yes No N/A	Did all bottle labels agree with COC?	Yes No N/A
If samples were taken today, is there evidence that chilling has begun? Recvd_____C°	Yes No N/A	Was a sufficient amount of sample received?	Yes No N/A
Did all bottles arrive unbroken and intact?	Yes No N/A	Were correct containers and preservatives received for the tests requested?	Yes No N/A
Do samples have a hold time <72 hours?	Yes No N/A		

Were there bubbles in VOA vials? (Volatiles Only)	Yes No N/A
Was PM notified of discrepancies? PM: By/Time:	Yes No N/A

125ml (A) 250ml (B) 1Liter (C) 40ml VOA (V)		
Bacti $Na_2S_2O_3$		
None (P)		
Cr6 Buffer (P) Borate Carbonate Buffer		
HNO_3 (P)		
H_2SO_4 (P)		
NaOH (P)		
NaOH+ZnAc (P)		
Dissolved Oxygen 300ml (P)		
None (AG)		
None (CG) 500ml		
$Na_2S_2O_3$ 250ml (Brown P) 549		
$Na_2S_2O_3$ (AG)		
$Na_2S_2O_3$ (AG)		
Thio/K Citrate		
NH_4Cl (AG) 552		
HCl (AG)		
None (CG) 500ml		
H_3PO_4 (AG)		
Other:		
Plastic Bag		
Low Level Hg/Metals Double Bag		
Client Own		
Glass Jar: 125/ 250/ 500		
Soil Tube: Brass/ Steel/ Plastic		
5 g Encore		

Filter or Split	Container	Preservative	Date/Time/Initials
S P F			
S P F			
S P F			
S P F			
S P F			

Labeled by: M @1458 Labels checked by: M @1459

FL-SC-0003-06

Nutrien Precision Agri Lab
24730 Avenue 13
Madera, CA 93637
Office 559-661-6386

Sample Event Map

2020-01-24

Grower	**Las Gallinas Valley Sanitary District**	Farm	**Las Gallinas Valley Sanitary District**	Lab	**Precision Agri Lab**
Field	**APN 155-01-14**			Area	**118 a**

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MOORE TWINING ASSOCIATES, INC.
Since 1888
California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

February 11, 2020

Work Order #: **GA27032**

Jenna Overmyer
Precision Agri-Lab
24730 Avenue 13
Madera, CA 93637

RE: Analytical Services

Enclosed are the analytical results for samples received by our laboratory on **01/27/20** . For your reference, these analyses have been assigned laboratory work order number **GA27032**.

All analyses have been performed according to our laboratory's quality assurance program. All results are intended to be considered in their entirety, Moore Twining Associates, Inc. (MTA) is not responsible for use of less than complete reports. Results apply only to samples analyzed.

If you have any questions, please feel free to contact us at the number listed above.

Sincerely,

Moore Twining Associates, Inc.

Julio Morales
Client Services Supervisor

MOORE TWINING ASSOCIATES, INC.
California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
(559) 268-0740 Fax

Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary Dist. **Project Manager:** Jenna Overmyer	**Reported:** 02/11/2020

Analytical Report for the Following Samples

Sample ID	Notes	Laboratory ID	Matrix	Date Sampled	Date Received
APN 155-01-14 (1 at 18")		GA27032-01	Soil	01/24/20 00:00	01/27/20 13:30
APN 155-01-14 (2 at 18")		GA27032-02	Soil	01/24/20 00:00	01/27/20 13:30
APN 155-01-14 (3 at 18")		GA27032-03	Soil	01/24/20 00:00	01/27/20 13:30
APN 155-01-14 (4 at 18")		GA27032-04	Soil	01/24/20 00:00	01/27/20 13:30
APN 155-01-14 (5 at 18")		GA27032-05	Soil	01/24/20 00:00	01/27/20 13:30
APN 155-01-14 (6 at 18")		GA27032-06	Soil	01/24/20 00:00	01/27/20 13:30

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Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary Dist. **Project Manager:** Jenna Overmyer	**Reported:** 02/11/2020

APN 155-01-14 (1 at 18")

GA27032-01 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Ammonia as N		**100**	1.0	mg/kg	1	B0A3111	01/31/20	02/01/20	EPA 350.1
Specific Conductance (EC)		**140**	1.0	µS/cm	1	B0B0708	02/08/20	02/08/20	WREP 125 (S-2.30)
Nitrate + Nitrite as N		**3.0**	0.25	mg/kg	1	B0A2801	01/28/20	01/28/20	SM 4500-NO3 F
Nitrite as N		**1.5**	0.050	mg/kg	1	B0A2802	01/28/20	01/28/20	SM 4500-NO2 B
Phosphorus		**400**	10	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-P
Total Kjeldahl Nitrogen		**1200**	40	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-N
Total Nitrogen		**1200**	40	mg/kg	20	[CALC]	02/03/20	02/03/20	(CALC)
Metals (Total)									
Potassium		**2800**	100	mg/kg	1	B0A2806	01/30/20	01/30/20	EPA 6010B

APN 155-01-14 (2 at 18")

GA27032-02 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Ammonia as N		**74**	1.0	mg/kg	1	B0A3111	01/31/20	02/01/20	EPA 350.1
Specific Conductance (EC)		**170**	1.0	µS/cm	1	B0B0708	02/08/20	02/08/20	WREP 125 (S-2.30)
Nitrate + Nitrite as N		**3.2**	0.25	mg/kg	1	B0A2801	01/28/20	01/28/20	SM 4500-NO3 F
Nitrite as N		**1.7**	0.050	mg/kg	1	B0A2802	01/28/20	01/28/20	SM 4500-NO2 B
Phosphorus		**470**	10	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-P
Total Kjeldahl Nitrogen		**1700**	40	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-N
Total Nitrogen		**1700**	40	mg/kg	20	[CALC]	02/03/20	02/03/20	(CALC)
Metals (Total)									
Potassium		**2700**	100	mg/kg	1	B0A2703	01/29/20	01/29/20	EPA 6010B

APN 155-01-14 (3 at 18")

GA27032-03 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Ammonia as N		**60**	1.0	mg/kg	1	B0A3111	01/31/20	02/01/20	EPA 350.1
Specific Conductance (EC)		**200**	1.0	µS/cm	1	B0B0708	02/08/20	02/08/20	WREP 125 (S-2.30)
Nitrate + Nitrite as N		**3.7**	0.25	mg/kg	1	B0A2801	01/28/20	01/28/20	SM 4500-NO3 F
Nitrite as N		**1.5**	0.050	mg/kg	1	B0A2802	01/28/20	01/28/20	SM 4500-NO2 B
Phosphorus		**480**	10	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-P
Total Kjeldahl Nitrogen		**1500**	40	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-N
Total Nitrogen		**1500**	40	mg/kg	20	[CALC]	02/03/20	02/03/20	(CALC)
Metals (Total)									
Potassium		**2700**	100	mg/kg	1	B0A2703	01/29/20	01/29/20	EPA 6010B

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California ELAP Certificate #1371

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(559) 268-0740 Fax

Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary Dist. **Project Manager:** Jenna Overmyer	**Reported:** 02/11/2020

APN 155-01-14 (4 at 18")

GA27032-04 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Ammonia as N		**51**	1.0	mg/kg	1	B0A3111	01/31/20	02/01/20	EPA 350.1
Specific Conductance (EC)		**230**	1.0	µS/cm	1	B0B0708	02/08/20	02/08/20	WREP 125 (S-2.30)
Nitrate + Nitrite as N		**2.9**	0.25	mg/kg	1	B0A2801	01/28/20	01/28/20	SM 4500-NO3 F
Nitrite as N		**1.4**	0.050	mg/kg	1	B0A2802	01/28/20	01/28/20	SM 4500-NO2 B
Phosphorus		**550**	10	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-P
Total Kjeldahl Nitrogen		**1600**	40	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-N
Total Nitrogen		**1600**	40	mg/kg	20	[CALC]	02/03/20	02/03/20	(CALC)
Metals (Total)									
Potassium		**2800**	100	mg/kg	1	B0A2703	01/29/20	01/29/20	EPA 6010B

APN 155-01-14 (5 at 18")

GA27032-05 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Ammonia as N		**47**	1.0	mg/kg	1	B0A3111	01/31/20	02/01/20	EPA 350.1
Specific Conductance (EC)		**290**	1.0	µS/cm	1	B0B0708	02/08/20	02/08/20	WREP 125 (S-2.30)
Nitrate + Nitrite as N		**4.2**	0.25	mg/kg	1	B0A2801	01/28/20	01/28/20	SM 4500-NO3 F
Nitrite as N		**1.6**	0.050	mg/kg	1	B0A2802	01/28/20	01/28/20	SM 4500-NO2 B
Phosphorus		**510**	10	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-P
Total Kjeldahl Nitrogen		**1500**	40	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-N
Total Nitrogen		**1500**	40	mg/kg	20	[CALC]	02/03/20	02/03/20	(CALC)
Metals (Total)									
Potassium		**2600**	100	mg/kg	1	B0A2703	01/29/20	01/29/20	EPA 6010B

APN 155-01-14 (6 at 18")

GA27032-06 (Soil) Sampled: 01/24/20 00:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Ammonia as N		**56**	1.0	mg/kg	1	B0A3111	01/31/20	02/01/20	EPA 350.1
Specific Conductance (EC)		**140**	1.0	µS/cm	1	B0B0708	02/08/20	02/08/20	WREP 125 (S-2.30)
Nitrate + Nitrite as N		**3.5**	0.25	mg/kg	1	B0A2801	01/28/20	01/28/20	SM 4500-NO3 F
Nitrite as N		**1.9**	0.050	mg/kg	1	B0A2802	01/28/20	01/28/20	SM 4500-NO2 B
Phosphorus		**200**	10	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-P
Total Kjeldahl Nitrogen		**1200**	40	mg/kg	20	B0A3103	02/01/20	02/03/20	SM 4500-N
Total Nitrogen		**1200**	40	mg/kg	20	[CALC]	02/03/20	02/03/20	(CALC)
Metals (Total)									
Potassium		**3000**	100	mg/kg	1	B0A2703	01/29/20	01/29/20	EPA 6010B

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California ELAP Certificate #1371

2527 Fresno Street
Fresno, CA 93721
(559) 268-7021 Phone
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Precision Agri-Lab 24730 Avenue 13 Madera CA, 93637	**Project:** Analytical Services **Project Number:** Las Gallinas Valley Sanitary Dist. **Project Manager:** Jenna Overmyer	**Reported:** 02/11/2020

Notes and Definitions

MS1	Recovery for this analyte was affected by matrix.
MS3	Recovery for this analyte was biased low; associated blank spike recoveries are within range.
µg/L	micrograms per liter (parts per billion concentration units)
mg/L	milligrams per liter (parts per million concentration units)
mg/kg	milligrams per kilogram (parts per million concentration units)
ND	Analyte NOT DETECTED at or above the reporting limit
RPD	Relative Percent Difference
	Analysis of pH, filtration, and residual chlorine is to take place immediately after sampling in the field. If the test was performed in the laboratory, the hold time was exceeded. **(for aqueous matrices only)**

Report of Soil Analysis

DELLAVALLE® Laboratory, Inc.
Chemists and Consultants

1910 W. McKinley, Suite 110, Fresno, CA 93728
FAX (559) 268-8174 - (800) 228-9896 - (559) 233-6129

Moore Twining Associates Inc
2527 Fresno Street
Fresno CA 93721
5187
54

ID:

Lab No. 310545
Sampled Date 1/23/2020
Submitted Date 1/28/2020
Submitted by Julio Morales
Reported Date 2/5/2020
Location/Project
Copy To
Fax (559) 268-0740
E-mail (See Fieldsheet)

No.	Description	Saturated Paste % SP	units pH	lbs/ac-6" Lime	meq/100g CEC
	RL--->	0.50	1.0	500	0.1
	NAPT Methods--->	S1.00	S1.10	S2.50	S10.20 mod
	Handbook 60--->				
9	GA27032-01	86	5.1	19100	24.0
10	GA27032-02	84	6.0		25.6
11	GA27032-03	95	6.9		30.4
12	GA27032-04	86	6.9		29.4
13	GA27032-05	85	7.0		28.1
14	GA27032-06	91	7.1		31.3

PRECISION AGRI LAB
24730 Avenue 13
Madera CA 93637
Phone: (559) 661-6386
FAX: (559) 661-6135

GA27032

SAMPLE SUBMISSION FORM

Chain of Custody

CA/AZ/HI/WA/ID

Log #: 401504

Date Sampled: 1-24-20

Date Received: 1-27-20

Dealer/Branch Name: PRECISION AGRI LAB

Dealer/Branch Fieldman: PRECISION AGRI LAB

Grower/Client Name: LAS GALLINAS VALLEY SANITARY DISTRICT

Crop: _____

Variety: _____

Growth Stage: _____

Crop Monitor Program (CMP) _____ Yes _____ No

Recommendations (Soil Only) _____ Yes _____ No

LPI's NutriScription (Tissue Only) _____ Yes _____ No

Dairy Nutrient Management (NMP) _____ Yes _____ No

Sample Type

Soil X

Plant _____

Other _____

Do not mix sample types on the same form, please.

Client's CPS Account No: _____

Sample ID	Test ID	Sample Description	Event Code	Time
1	9999	APN 155-01-14 (1 at 18")	B3C6-HMYP	
2	9999	APN 155-01-14 (2 at 18")		
3	9999	APN 155-01-14 (3 at 18")		
4	9999	APN 155-01-14 (4 at 18")		
5	9999	APN 155-01-14 (5 at 18")		
6	9999	APN 155-01-14 (6 at 18")		
7				
8				
9				
10				
11				
12				

Test: Total N, NH3-N, NO3-N, PH, CEC, EC, Total P, Total K

Our most popular analysis packages are shown below for your convenience. Please view the complete price list for other analytical needs.

Analysis Type	Test ID	Parameters
Petiole	1151	NO3-N,PO4-P,K
Petiole/Blade	1154	NO3-N,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na
Petiole/Blade	1155	NO3-N,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Petiole/Blade (Grape only)	1161	N,NO3-N,P,PO4-P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Leaf	1251	N,P,K
Leaf	1253	N,P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na
Leaf	1254	N,P,K,Ca,Mg,S,B,Cu,Fe,Mn,Zn,Na,Cl
Almond Hull	1306	B
Soil	2154	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,sCa,sMg,sNa,B
Soil	2155	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,sCa,sMg,sNa,B,SO4-S,Cl
Soil	2555	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe, ESP,Free Lime,xCa,xMg,xNa,B,%BS,CEC, SO4-S,Cl
Soil	2656	NO3-N,Olsen P,K,Zn,Mn,Fe,Cu,SP,pH,ECe,ESP, Free Lime,xCa,xMg,xNa,B,%BS,CEC,SO4-S,Cl, sCa,sMg,sK,sNa
Soil	2657	2656 Test + Organic Matter
Water	3051	pH,ECw,SAR,Ca,Mg,Na,HCO3+CO3,NO3-N,B,Cl, SO4-S,Fe,Mn
Water	3053	pH,Buffer/Titration (specify acid)

	Printed Name	Signature	Date	Time
Sampled By	PAL	OMAR		
Relinquished By	Jenna Overmyer		1/27/2020	1:30pm
Shipping Method	UPS ____ FedEx ____ Drop Off ____ Other (Specify) ____			
Received By	[illegible]	[illegible]	1-27-20	13:30

For Lab Use Only Below This Line

	Printed Name	Signature	Date	Time
Relinquished By				
Shipping Method	UPS ____ FedEx ____ Drop Off ____ Other (Specify) ____			
Shipped To				

Please email Results: Jenna.Overmyer@nutrien.com
Please Bill: Nutrien Precision Agrilab

Moore Twining Associates

Sample Integrity Page 2 of 2 WO# GA27032 MTA Bottles: Yes or No

Was temperature within range? Chemistry ≤6°C Micro <10°C Temp_____°C	Yes No N/A	Did all bottle labels agree with COC?	Yes No N/A	Were there bubbles in VOA vials? (Volatiles Only)	Yes No N/A
If samples were taken today, is there evidence that chilling has begun? Recvd_____C°	Yes No N/A	Was a sufficient amount of sample received?	Yes No N/A	Was PM notified of discrepancies? PM: By/Time:	Yes No N/A
Did all bottles arrive unbroken and intact?	Yes No N/A	Were correct containers and preservatives received for the tests requested?	Yes No N/A		
Do samples have a hold time <72 hours?	Yes No N/A				

125ml (A) 250ml (B) 1Liter (C) 40ml VOA (V)	1														
Bacti $Na_2S_2O_3$															
None (P)															
Cr6 Buffer (P) Borate Carbonate Buffer															
HNO_3 (P)															
H_2SO_4 (P)															
NaOH (P)															
NaOH+ZnAc (P)															
Dissolved Oxygen 300ml (P)															
None (AG)															
None (CG) 500ml															
$Na_2S_2O_3$ 250ml (Brown P) 549															
$Na_2S_2O_3$ (AG)															
$Na_2S_2O_3$ (AG)															
Thio/K Citrate															
NH_4Cl (AG) 552															
HCl (AG)															
None (CG) 500ml															
H_3PO_4 (AG)															
Other:															
Plastic Bag	1														
Low Level Hg/Metals Double Bag															
Client Own	1														
Glass Jar: 125/ 250/ 500															
Soil Tube: Brass/ Steel/ Plastic															
5 g Encore															

Filter or Split	Container	Preservative	Date/Time/Initials
S P F	plastic bag	0	1-27-20 1505 MB
S P F			
S P F			
S P F			
S P F			

Labeled by: [signature] @_____ Labels checked by: [signature] @ 1450

FL-SC-0003-06